

Distributed Beamforming in Wireless Multiuser Relay-Interference Networks With Quantized Feedback

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Abstract—We study fixed data rate communication schemes for wireless relay-interference networks with any number of transmitters, relays, and receivers. The transmitters and the relays have individual short-term power constraints. We analyze both amplify-and-forward (AF) and decode-and-forward (DF) relaying strategies with a two channel use quantized network beamforming protocol. We design the quantizer of the channel state information to minimize the probability that at least one receiver incorrectly decodes its desired symbol(s). Correspondingly, we introduce a generalized diversity measure that encapsulates the conventional one as the *first-order diversity*. Additionally, it incorporates the *second-order diversity*, which is concerned with the transmitter power dependent logarithmic terms that appear in the error rate expression. We first show that for AF relays, the maximal achievable diversity in the presence of interference is strictly less than the transmit diversity bound in terms of the second-order diversity. We then prove that it is possible to achieve the transmit diversity bound using DF relays as if there is no interference and as if coding over an arbitrary number of channel uses is allowed. Relay selection provides the best possible diversity gain for both relaying strategies. Finally, we show that all the aforementioned diversity gains can be achieved using distributed decision making with asymptotically zero feedback rate per receiver. Such a performance is made possible by a special distributed quantizer design method we have called *localization*.

Index Terms—Beamforming, distributed vector quantization, diversity gain, interference, wireless relay network.

I. INTRODUCTION

It has been demonstrated in several studies that cooperation can greatly improve the performance and reliability of wireless network communications [1]–[5]. Most of the emphasis has been on improving the performance of a single-user (point-to-point) communication through cooperation. However, the insights from the single-user setting may not be sufficient to address the challenges of multiuser communication networks. One of the most distinctive features of wireless multiuser networks is the interference phenomenon, dealing with which becomes very important when shifting from the single-user paradigm to a multiple-user paradigm. In such a paradigm shift,

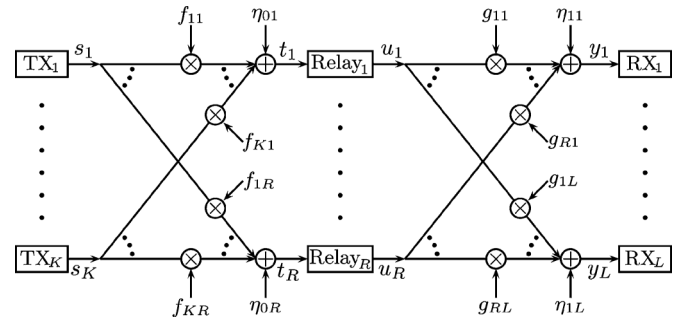


Fig. 1. System block diagram. In the figure, TX_k , Relay_r , and RX_ℓ stand for the k th transmitter, r th relay, and the ℓ th receiver, respectively.

the natural question that arises is the following—what are the benefits of cooperation in a multiuser network setting suffering from interference and how to achieve them? The main goal of this paper is to address this general question in terms of the diversity gain. We consider a particular networking scenario that consists of K transmitters, R relays, and L receivers, as shown in Fig. 1.

A. Assumptions

First we summarize our assumptions concerning the channel and system models. We follow the literature while making the assumptions listed below:

- All the nodes in the network have a single antenna and are half-duplex, i.e., they can either transmit or receive at a given time. Simultaneous transmissions are perfectly synchronized.
- Unless otherwise specified, we assume that there are no direct links between the transmitters and the receivers.
- We assume a quasi-static block fading channel model, in which the channel realizations vary independently from one block to another, while within each block the channels remain constant. Denote the channel from the k th transmitter to the r th relay by f_{kr} and the channel from the r th relay to the ℓ th receiver by $g_{r\ell}$. Let $\mathbf{h} = [f_{11} \cdots f_{KR} \ g_{11} \cdots g_{RL}]$ denote the channel state of the entire network. We assume that the entries of $\mathbf{h}$ are independent and distributed as $f_{kr} \sim \text{CN}(0, \sigma_{f_{kr}}^2)$, $g_{r\ell} \sim \text{CN}(0, \sigma_{g_{r\ell}}^2)$ with non-zero and finite variances $0 < \sigma_{f_{kr}}, \sigma_{g_{r\ell}} < \infty, \forall r, k, \ell$, where $\text{CN}(0, \sigma^2)$ denotes a circularly symmetric complex Gaussian random variable with variance $\frac{\sigma^2}{2}$ per dimension. For brevity, let $\mathbf{f} \triangleq [f_{11} \cdots f_{KR}]$ and

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$\mathbf{g}_\ell \triangleq [g_{1\ell} \ \cdots \ g_{R\ell}]$ represent the channel states from the transmitters to the relays, and from the relays to the ℓ th receiver, respectively.

- We assume that each receiver knows the channels from the transmitters to the relays and its own receiving channels. In other words, the ℓ th receiver knows $\mathbf{f}$ and $\mathbf{g}_\ell$. We assume that the transmitters do not have any CSI, and that each relay knows its own receiving channels, i.e., the r th relay knows f_{kr} , $k = 1, \dots, K$. Each relay and each receiver also has partial CSI in the form of quantized instantaneous CSI provided by feedback from the receivers. The corresponding CSI quantizers will be explained later on. Some possible procedures to reveal the aforementioned channel state information to the relays and the receivers can be found in [6]–[8]. For completeness, we give an outline of one possible way: The ℓ th destination can acquire the knowledge of $g_{r\ell}$ by training from the r th relay. The r th relay can acquire the knowledge of f_{kr} using training sequences from the k th source. It can also amplify and forward its received training signal from the source to the destination, so that the destination can estimate the product of f_{kr} and $g_{r\ell}$. As $g_{r\ell}$ is known by the destination, f_{kr} can be estimated.
- Only the short-term power constraint is considered, which means that for every constant fading block, the average power levels used at the k th transmitter and the r th relay are no larger than P_{S_k} and P_{R_r} , respectively. Let $P_{R_r} = p_{R_r}P$, $r = 1, \dots, R$, $P_{S_k} = p_{S_k}P$, $k = 1, \dots, K$, where $0 < p_{S_k}, p_{R_r} < \infty$ are independent of P . In other words, we allow the power constraint of each transmitting terminal to grow linearly with P .
- Any receiver can be interested in the messages of an arbitrary set of transmitters. Some (but not all) of the transmitters might be “pure” interferers, in which case their messages are not wanted by any of the receivers. We define the network error rate (NER) as the probability that at least one receiver incorrectly decodes its desired message(s).
- We assume that at each block, each transmitter uses a fixed and non-zero rate code for its message; rate adaptation will not be considered in this paper. Furthermore, we assume that the system has very strict decoding-delay requirements: A given realization of transmitter messages are to be transmitted over the minimum possible block length of two channel uses.
- Unlike [9], we do not allow the transmission rates to scale with P , and hence the “multiplexing gain” of our schemes in the sense defined in [9] will be zero. A diversity-multiplexing tradeoff analysis will not be pursued in this paper: Putting a tradeoff analysis aside, even finding the maximum achievable multiplexing gain of the relay-interference network for a general K , R , and L is a challenging open problem [10].

B. A Cut-Set Lower Bound on the NER

The fixed-rate nature of the system, together with the assumption of short-term power constraints, implies that communication with arbitrarily small NERs is not possible even in the “in-

finite block length regime.” Indeed, we have the following general lower bound on the NER.

Theorem 1: For all P sufficiently large, the NER is lower bounded by $\frac{C_0}{P^R}$ for any coding scheme. The factor C_0 is a finite positive constant that depends on the transmission rates but is independent of P . The lower bound holds for an arbitrary block length, arbitrary encoding/decoding mechanisms and relay functionalities, and full CSI at all the nodes.

Proof: The proof relies on (now) standard arguments [9], [11] based on Fano’s inequality [12, Theorem 2.10.1] and the max-flow min-cut theorem [12, Theorem 15.10.1]. We refer the reader to [11, Lemma 1] and [13, Section I.D.c] for a general form of the theorem and its proof. ■

The theorem demonstrates that there is an explicit relationship between the best possible NER and the transmission power P . In this context, the diversity gain determines how fast the NER decays with P . Traditionally, the diversity gain of a scheme is defined as the negative exponent of P in the error rate expression as $P \rightarrow \infty$. According to this traditional definition, the maximum diversity achievable in our network model, by Theorem 1, is R .

C. A Generalized Diversity Measure

In this paper, we use a generalized diversity measure to characterize the asymptotic behavior of the NER. In what follows, we define this measure together with its motivations: Imagine two hypothetical communication schemes S_1 and S_2 with asymptotic NERs $C_1 \frac{\log^R P}{P^R}$ and $C_2 \frac{1}{P^R}$, respectively, with C_1 and C_2 being constants that are independent of P . Both S_1 and S_2 provide a traditional diversity gain of R . On the other hand, when P is sufficiently large, S_2 will always outperform S_1 regardless of C_1 and C_2 are. This is because the $\frac{\log^R P}{P^R}$ decay of the NER of S_1 is much worse than the $\frac{1}{P^R}$ decay of the NER of S_2 . Obviously, the traditional definition of diversity cannot capture the $\log^R P$ term and its effect on the performance. Motivated by this observation, we generalize the definition of diversity to capture the log-term differences. As we explain in the following, this is done by defining higher order diversities that depend on the exponent of the logarithmic terms and are backward compatible with the existing traditional diversity order definition in the literature.

Suppose that the NER can be expressed asymptotically as $C_3(P^\alpha \log^\beta P)^{-1}$, where C_3 is a constant that is independent of P . Then, we call α and β , the first-order and the second-order diversity gains, respectively, and say that the scheme achieves diversity (α, β) . For two diversity gains $d = (d_1, d_2)$, and $d' = (d'_1, d'_2)$, we say that d is higher than d' (or $d > d'$) if either $d_1 > d'_1$ or $d_1 = d'_1, d_2 > d'_2$. According to these definitions, the diversity gains of our hypothetical schemes S_1 and S_2 will be $(R, -R)$ and $(R, 0)$, respectively, and thus we can conclude that S_2 has a higher diversity since $(R, 0) > (R, -R)$.

We shall see later by more concrete examples that the new higher order diversities can capture some of the fundamental performance differences between different schemes that achieve the same traditional diversity order. Note that, according to Theorem 1, the maximum diversity with any

scheme is $(R, 0)$, which we call the “full” diversity order, or the transmit diversity bound.

D. Related Work

One way of achieving high diversity gains through cooperation is network (relay) beamforming [14]–[22], in which the relays adaptively adjust their transmit powers and phases according to the channel information that they have. Indeed, for networks with a single-user networks with $K = L = 1$ and no interference, network beamforming using amplify-and-forward (AF) relays has shown to achieve the full diversity gain, $(R, 0)$ [14], [18]. On the other hand, the optimal beamforming policy requires one or two CSI-dependent real numbers to be made available at the relays. For single-user networks, the only node in the network that can obtain the entire CSI is the receiver. Therefore, the real numbers should be broadcasted from the receiver to the relays as feedback information. In practice, the feedback links have a limited bandwidth which results in a limited number of bits that can be utilized for feedback. One way of capturing such a limited feedback is to model it as a source coding problem (see e.g., [23], [24] and the references therein). Then, one is concerned with the question of how to design the quantizers and the corresponding quantized beamforming codebooks to maximize the reliability of the system. Such a formulation proves to be very effective: It has been shown in [8], [25] that for AF networks with $K = L = 1$ that it is possible to obtain both maximal diversity, as well as high array gain with only a few feedback bits from the receiver.

A special case of quantized feedback for cooperative networks is the relay selection scheme [8], [26]–[38]. In this scheme, only one relay is allowed to transmit given a constant fading block. This requires $\lceil \log_2 R \rceil$ bits of feedback per channel state. It has been analytically shown in [8] that, for single-user AF networks, the relay selection scheme provides full diversity. For single-user networks with DF relays, [39] proves that relay selection achieves full diversity as well. Selection cooperation with DF relays in an interference-free multi-source network with orthogonal channel allocation has been discussed in [40].

There are several other works in the context of cooperative networking with limited feedback apart from the ones we have already mentioned. An iterative phase adjustment scheme for distributed beamforming has been proposed in [41]. In [42], the authors study the optimal outage exponents for DF relay channels with quantized CSI. Cooperative relaying with partial rate and/or power control have been studied in [6], [43]. The idea of Grassmannian beamforming [44], which was originally proposed for point-to-point multiple-input multiple-output (MIMO) channels, has been extended to MIMO relay networks in [45].

A comprehensive overview of the vast literature on MIMO systems with feedback can be found in [23], [24]. In particular, beamforming [44], [46]–[50] and precoded space-time coding [51]–[58] schemes that incorporate limited feedback have been extensively studied. Diversity-multiplexing tradeoffs in MIMO channels with quantized CSI have been determined in [59], [60]. Other works that study the interplay between rate and outage in fading channels with partial CSI include [61]–[64]. An interfer-

ence cancellation scheme for a two-user MIMO multiple access channel with quantized feedback has been introduced in [65].

Quantized feedback schemes have also been studied for non-cooperative multiuser interference networks with several receivers. In [66], the author considers zero-forcing beamforming with finite rate feedback in multiple-input multiple-output (MIMO) broadcast channels. Interference alignment for multiuser interference networks with limited feedback has been studied in [67]. Unlike what we shall study in this work, where we seek to optimize the reliability of the system in terms of the diversity gain, the goal of the above two papers was to optimize the data transmission rate in terms of the multiplexing gain. A common conclusion that we can infer from both studies is that, in order to achieve the same multiplexing gain as a system with perfect CSI, the feedback rate per receiver should be increased at least logarithmically with the transmitter power; any constant feedback rate results in a complete loss of multiplexing gain. This is unlike point-to-point systems where feedback is not even necessary to achieve the maximal multiplexing gain [66], and a few feedback bits is usually sufficient to transmit with rates that are close to the one with perfect CSI [48]. The feedback requirements of interference networks thus appear to be considerably higher than that of interference-free networks. A relevant question in this context is to determine the feedback requirements if instead we would like to ensure maximal reliability in the presence of interference.

E. Objectives, Challenges and Approaches

The main objective of this paper is, as we have mentioned, to study the benefits of cooperation in multiuser relay-interference networks in terms of the generalized diversity gain perspective. We consider this problem with both AF and DF strategies while following the assumptions in Section I-A. For each forwarding strategy, we propose quantized network beamforming schemes. In that sense, this paper is also a generalization of single-user quantized network beamforming to multiuser interference networks with AF or DF relays. Such a generalization is quite challenging though because of the distributed nature of the network. Let us now describe some of these challenges and our approaches to address them.

In interference networks with AF relays, the relays amplify both noise and interference, which results in completely different problem formulations and solutions. Second, in general, there are multiple receivers that have different optimal beam directions. As a result, it is difficult to design a scheme that can provide a reasonable performance to all the users.

Another difficulty is related to acquiring feedback information from several separated receivers. In a multi-receiver relay network with limited feedback, the source coding formulation discussed in Section I-D results in a distributed source coding scenario. This is because none of the receivers can acquire the entire CSI of the network, and thus each of them will have to provide its own feedback message. In other words, according to Fig. 1, the ℓ th receiver has to provide its own message according to the channel vector $[f \ g_\ell] = [f_{11} \dots f_{KR} \ g_{1\ell} \dots g_{R\ell}]$. Designing a good distributed vector quantizer for non-linear distortion functions, such as the ones that will appear in our analysis, is a very challenging non-trivial problem even in the case

of independent and identically distributed sources [68]. Our approach is first to design a *global* quantizer (GQ) that has access to the entire CSI. While such a quantizer is obviously not practical, our key idea is to synthesize, out of the GQ, a practical finite-rate *local* quantizer (LQ) that can provably achieve the same performance as the GQ asymptotically. We call this distributed quantizer design process *localization*. The practicality of our LQs stem from the fact that they only require the ℓ th receiver to know $[\mathbf{f} \ \mathbf{g}_\ell]$, and therefore can be implemented in a distributive manner. We apply the localization method to design such practical LQs for our network model, and compare its performance with that of the GQ.

F. Summary of the Main Results

The main contributions of this paper can be summarized as follows:

- 1) The higher-order diversities can more precisely distinguish between different communication scenarios that achieve the same traditional (first-order) diversity. Indeed, as we shall see later, all the schemes that we propose will achieve the same first-order diversity, what distinguishes them will be their different second-order diversity measures.
- 2) The maximal achievable diversity with AF relays is $(R, -R)$ when $K > 1$, whereas it is $(R, 0)$ when $K = 1$. In other words, with AF relays, the relay-interference network suffers from a second-order diversity loss compared to an interference-free network that can achieve diversity $(R, 0)$. This shows that in interference networks, simple linear forwarding schemes are not sufficient to achieve the transmit diversity bound on the diversity gain $(R, 0)$ provided by Theorem 1.
- 3) It is possible to achieve the full diversity gain $(R, 0)$ using DF relays as if there is no interference and as if coding over an arbitrary block length is allowed. Therefore, although it imposes a considerable amount of complexity at the relays, the DF strategy achieves the transmit diversity bound on the diversity gain $(R, 0)$ for any K , and is thus optimal.
- 4) Relay selection provides the best possible diversity gain for both AF and DF strategies.
- 5) It is possible to achieve full diversity using distributed decision making with asymptotically zero feedback rate per receiver. This provides a very fortunate answer regarding the feedback requirements for maximal reliability: In a relay-interference network, it is possible to achieve maximal reliability using arbitrarily low feedback rates per receiver, when the transmitter powers are sufficiently large. Such a performance is made possible by the localization method described at the end of Section I-E.

G. Organization of the Paper

The rest of the paper is organized as follows: In Section II, we introduce our data transmission schemes with AF and DF relays. In Section III, we provide a formal description of the NER measure and the associated generalized diversity gains. In Section IV, we determine the maximal diversity gains of the AF and DF strategies, and design global quantizers that can achieve these gains. In Section V, we introduce the more practical local quantizers and analyze their performance. Numerical results are

provided in Section VI. In Section VII, we provide extensions of our results to networks with direct links. In Section VIII, we draw our major conclusions. Some technical proofs are provided in the appendices.

H. Notation

We use the following notation throughout the paper. $\|\cdot\|$ indicates the 2-norm, $\|\cdot\|_\infty$ is the infinite norm, $\langle \cdot | \cdot \rangle$ is the inner product. $\mathbb{C}$, $\mathbb{R}$ and $\mathbb{Z}^+$ represent the sets of complex numbers, real numbers, and positive integers, respectively. $\det(\mathbf{A})$ is the determinant of a square matrix $\mathbf{A}$. $\mathbf{A}^T$, $\mathbf{A}^H$ denote the transpose and the Hermitian transpose of $\mathbf{A}$, respectively. $\Re(\mathbf{A}) \triangleq \frac{1}{2}(\mathbf{A} + \mathbf{A}^H)$ and $\Im(\mathbf{A}) \triangleq \frac{1}{2j}(\mathbf{A} - \mathbf{A}^H)$ are the real and imaginary parts of the matrix $\mathbf{A} = \Re(\mathbf{A}) + \sqrt{-1}\Im(\mathbf{A})$. P represents the probability. $f_X(\cdot)$ is the probability density function (PDF), and $F_X(\cdot)$ is the cumulative distribution function (CDF) of a random variable X . $E[X]$ is the expected value of X . $X \sim \Gamma(k, \theta)$ means that X is a Gamma random variable with $f_X(x) = \frac{x^{k-1}e^{-x/\theta}}{\theta^k \Gamma(k)}$ for $x \geq 0$ and $f_X(x) = 0$ for $x < 0$, $k, \theta > 0$. For any sets $\mathcal{A}$ and $\mathcal{B}$, $\mathcal{A} - \mathcal{B}$ is the set of elements in $\mathcal{A}$, but not in $\mathcal{B}$. $|\mathcal{A}|$ is the cardinality of $\mathcal{A}$. $\mathcal{A}^r = \{(a_1, \dots, a_r) : a_1, \dots, a_r \in \mathcal{A}\}$, $r \in \mathbb{Z}^+$, is the Cartesian power. In a similar manner, $\prod_{i=1}^n \mathcal{A}_i$ denotes the Cartesian product of sets $\mathcal{A}_i$, $i = 1, \dots, n$. $\gamma_e = 0.577 \dots$ is the Euler-Mascheroni constant, $e = \exp(1)$ is the base of the natural logarithm, and $\emptyset$ is the empty set. For a real-valued function $f : \mathcal{C} \rightarrow \mathbb{R}$ with $\mathcal{C} \subset \mathbb{C}^K$, let $\mathcal{M} \triangleq \{\mathbf{x} : \mathbf{x} \in \mathcal{C}, f(\mathbf{x}) = \max_{\mathbf{x}' \in \mathcal{C}} f(\mathbf{x}')\}$. Then, $\arg \max_{\mathbf{x} \in \mathcal{C}} f(\mathbf{x})$, if it exists, is the unique vector $\mathbf{x}^*$ with the property that $\mathbf{x}^* \prec \mathbf{x}$, $\forall \mathbf{x} \in \mathcal{M}$, and “ $\prec$ ” represents some partial ordering (e.g., lexicographical ordering) of complex vectors. We define $\arg \min(\cdot)$ in a similar manner. For any logical statement ST, $\mathbf{1}_{\text{ST}} = 1$ if ST is true, and $\mathbf{1}_{\text{ST}} = 0$, otherwise. Finally, $\log(\cdot)$ is the natural logarithm, $\log_2(\cdot)$ is the logarithm to base 2, $\cosh(\cdot)$ is the hyperbolic cosine, $Q(\cdot)$ is the Gaussian tail function, $\Gamma(\cdot)$ is the gamma function, $E_1(x) \triangleq \int_1^\infty t^{-1}e^{-xt}dt$ is the exponential integral, and $K_\nu(\cdot)$ is the modified Bessel function of the second kind of order ν .

II. DATA TRANSMISSION SCHEMES

In this section, we provide the formal descriptions of the AF and DF strategies outlined in Section I-E. We introduce a two-channel use quantized beamforming scheme for each strategy. The key component of these schemes is a channel quantizer, the general description of which will be provided in Section II-A. The first step of the two-channel use protocol is common to both forwarding strategies and will be described in Section II-B. The second step of the protocol with AF and DF relays will be provided in Sections II-C and II-D, respectively.

A. A General Description of Channel Quantization

At the heart of our data transmission schemes is a channel quantizer $\mathbf{Q} : \mathbb{C}^{R(K+L)} \rightarrow \mathcal{C}$ that maps the entire channel state of the network to a quantized beamforming codebook $\mathcal{C} = \{\mathbf{x}_1, \dots, \mathbf{x}_{|\mathcal{C}|}\}$ with $\mathcal{C} \subset \mathbb{C}^R$. For each $\mathbf{h}$, the quantized beamforming vector $\mathbf{Q}(\mathbf{h})$ will be made available at the relays and the receivers. The relays will use the beamforming vector $\mathbf{Q}(\mathbf{h})$

to adjust their transmit power and phase. The purpose of providing $\mathbf{Q}(\mathbf{h})$ to each receiver is to ensure that they will know which quantized beamforming vector is used by the relays. The receivers will then be able to do coherent decoding.

For the purposes of this section, it is sufficient for us to imagine the quantization process in the general abstract manner described above. We shall later relate this general description to the more specific global and local quantization strategies outlined in Section I-E.

B. First Step of Data Transmission

The first step of our two-channel use data transmission protocol is the same for both AF and DF strategies. In this step, the k th transmitter selects a symbol s_k from a constellation $\mathcal{S}_k$, where $|\mathcal{S}_k| < \infty$, $\mathbf{P}(s_k) = |\mathcal{S}_k|^{-1}$, $\forall s_k \in \mathcal{S}_k$, and sends $\sqrt{P_{S_k}} s_k$. We normalize s_k as $\mathbf{E}_{s_k}[|s_k|^2] = 1$.¹ Thus, the average power used at the k th transmitter is P_{S_k} . During the first step, there is no reception at the receivers, but the r th relay receives

$$t_r = \sum_{k=1}^K f_{kr} s_k \sqrt{P_{S_k}} + \eta_{0r}, \quad (1)$$

where $\eta_{0r} \sim \mathcal{CN}(0, 1)$.

Suppose that a quantizer $\mathbf{Q} : \mathbb{C}^{R(K+L)} \rightarrow \mathcal{C}$ is employed in the network, and $\mathbf{Q}(\mathbf{h}) = \mathbf{x}$, for some $\mathbf{x} \in \mathcal{C}$. Then, the relays use the beamforming vector $\mathbf{x}$ to adjust their transmit power and transmit phase. Without loss of generality, the r th relay uses the r th component x_r of the vector $\mathbf{x} = [x_1 \cdots x_R]$. Regardless of the forwarding strategy, during the second step, the transmitters remain silent. We now introduce the relay operation for the AF and DF cases separately.

C. Second Step of Data Transmission With AF Relays

For AF relays, the r th relay transmits

$$u_r = x_r \sqrt{\rho_r} t_r, \quad (2)$$

where the relay normalization factor ρ_r is given by

$$\rho_r \triangleq \frac{P_{R_r}}{1 + \mathbf{E}_{s_1, \dots, s_K} \left[\left| \sum_{k=1}^K f_{kr} s_k \sqrt{P_{S_k}} \right|^2 \right]}. \quad (3)$$

In particular, if $K = 1$ or $\mathbf{E}[s_k] = 0$, $\forall k$, we have

$$\rho_r \triangleq \frac{P_{R_r}}{1 + \sum_{k=1}^K |f_{kr}|^2 P_{S_k}}. \quad (4)$$

In either case, the average power used at the r th relay can be calculated to be $\mathbf{E}_{s_1, \dots, s_K, \eta_{0r}}[|u_r|^2] = |x_r|^2 P_{R_r}$, $\forall \mathbf{h}$. We require $0 \leq |x_r| \leq 1$ as a result of the short-term power constraint. The channel state dependent normalization factors ρ_r ensure that the

instantaneous transmit power of each relay remains within its power constraint with high probability.²

Also, note that within the restriction of $0 \leq |x_r| \leq 1$, ρ_r is the maximal normalization factor that we can use. In other words, if a factor ρ'_r satisfies $\rho'_r > \rho_r$ for some $\mathbf{h}$, then it violates the short-term power constraint. Still, one can employ another factor ρ'_r with $\rho'_r \leq \rho_r$, $\forall \mathbf{h}$. We shall discuss later in Section IV whether or not such a different choice of the normalization factor can improve the network performance for AF relaying.

After the two steps of transmission, the received signal at the ℓ th receiver for AF relays can be expressed as:

$$y_\ell = \sum_{k=1}^K \sum_{r=1}^R x_r \sqrt{\rho_r} f_{kr} g_{r\ell} \sqrt{P_{S_k}} s_k + \sum_{r=1}^R x_r g_{r\ell} \sqrt{\rho_r} \eta_{0r} + \eta_{1\ell}, \quad (5)$$

where $\eta_{1\ell} \sim \mathcal{CN}(0, 1)$ is the noise at the ℓ th receiver. We assume that the noises η_{0r} , $r = 1, \dots, R$, and $\eta_{1\ell}$, $\ell = 1, \dots, L$ are independent.

D. Second Step of Data Transmission With DF Relays

For DF relays, the r th relay first decodes all the symbols from all the transmitters using a joint maximum-likelihood (ML) decoder. More formally, let $\mathbf{s} \triangleq [s_1 \cdots s_K]$ denote the transmitted “super-symbol,” and $\mathcal{S} = \mathcal{S}_1 \times \cdots \times \mathcal{S}_K$ be the corresponding alphabet. Also, let $\mathbf{f}_r \triangleq [f_{1r} \cdots f_{Kr}]$ denote all the channels from the transmitters to the r th relay. We have

$$\text{DEC}_r(t_r, \mathbf{f}_r) \triangleq \arg \max_{\mathbf{s}' \in \mathcal{S}} \mathbf{P}(\mathbf{s}' | t_r, \mathbf{f}_r) \quad (6)$$

as the ML estimate of the super-symbol $\mathbf{s}$ at the r th relay. Note that $\text{DEC}_r(t_r, \mathbf{f}_r) \in \mathcal{S} \subset \mathbb{C}^K$. The next step for the r th relay is to reencode its decoded super-symbol to a new alphabet $\mathcal{S}_R \subset \mathbb{C}$ using an arbitrary but deterministic injective function $\text{REENC} : \mathcal{S} \rightarrow \mathcal{S}_R$. Note that with these definitions, we have implicitly assumed that the reencoding function is the same for all relays and each relay uses the same output alphabet. Also, throughout the paper, we assume that

$$\mathcal{S}_R \triangleq \left\{ -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{\varsigma - 1}i + \sqrt{-1} \left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{\varsigma - 1}j \right) : i, j = 0, \dots, \varsigma - 1 \right\} \quad (7)$$

is a peak-power constrained square ς^2 -QAM constellation with $\varsigma = 2^{\lceil \log_4 |\mathcal{A}| \rceil}$. Note that $|\mathcal{S}_R| = \varsigma^2 = 4^{\lceil \log_4 |\mathcal{A}| \rceil} \geq |\mathcal{A}|$.

¹In order to avoid unnecessary technical complications, we also assume that there exists $\epsilon_s > 0$ such that the signal constellations satisfy $|\mathbf{E}[s_k]|^2 \leq 1 - \epsilon_s$, $\forall k$, $\forall P$. Note that the case $|\mathbf{E}[s_k]| = 1$ corresponds to a degenerate constellation consisting of a single signal point.

²Because of the noise at its received signal, a relay can exceed its transmit power constraint at some instants. The phrase “short-term” comes from the observation that, regardless of the channel states, the relay always obeys its power constraint when its transmit power is averaged over the transmitted symbols and the noise.

Hence, $\mathcal{S}_R$ is large enough to represent all $|\mathcal{A}|$ possible transmitted super-symbols.³

Hence, for DF relays, the r th relay transmits⁴

$$u_r = x_r \sqrt{P_{R_r}} \text{REENC}(\text{DEC}_r(t_r, \mathbf{f}_r)), \quad (8)$$

where x_r is, as we have already mentioned, the r th component of the quantized beamforming vector $\mathbf{x} = \mathbf{Q}(\mathbf{h})$.

Note that regardless of the input super-symbol $\mathbf{s}$, channel state $\mathbf{f}_r$, and the additive noise η_{0r} , we have

$$\begin{aligned} |u_r| &= |x_r| \sqrt{P_{R_r}} |\text{REENC}(\text{DEC}_r(t_r, \mathbf{f}_r))| \\ &\leq \sqrt{P_{R_r}} |\text{REENC}(\text{DEC}_r(t_r, \mathbf{f}_r))| \\ &\leq \sqrt{P_{R_r}} \max_{s_R \in \mathcal{S}_R} |s_R| \\ &= \sqrt{P_{R_r}}, \end{aligned}$$

and therefore $\mathbb{E}_{\mathbf{s}, \eta_{0r}}[|u_r|^2] \leq P_{R_r}$, satisfying the short-term power constraint.

After the two steps of transmission, the received signal at the ℓ th receiver for DF relays can be expressed as

$$y_\ell = \sum_{r=1}^R x_r \sqrt{P_{R_r}} \text{REENC}(\text{DEC}_r(t_r, \mathbf{f}_r)) g_{r\ell} + \eta_{1\ell}, \quad (9)$$

where $\eta_{1\ell} \sim \text{CN}(0, 1)$ is the noise at the ℓ th receiver as before.

III. PERFORMANCE MEASURES: NETWORK ERROR RATE AND GENERALIZED DIVERSITY

In the previous section, we have introduced the AF and DF transmission schemes. In this section, we define the corresponding performance measures. Clearly, many different measures can be envisioned. In this paper, we work with the NER quantity that can represent the overall error probability performance of the network. The formal definition of the NER will be given in Section III-A. We shall then introduce in Section III-B our generalized diversity measure that determines how fast the NER decays with the transmission power.

A. Network Error Rate (NER)

In this paper, we work with a general scenario in which the ℓ th receiver attempts to decode the symbols of the transmitters with indices given by an arbitrary but fixed set $\mathcal{D}_\ell \subset \{1, \dots, K\}$, $\mathcal{D}_\ell \neq \emptyset$. As an example, for a network with $K = 4$ and $L = 2$, let $\mathcal{D}_1 = \{1, 2\}$ and $\mathcal{D}_2 = \{2, 3\}$. Then, the first receiver is interested only in the symbols of the first and the second transmitters, while the second receiver is interested only in the symbols of the second and the third transmitters. The fourth transmitter is a pure interferer, and its symbols are not wanted by any of the receivers. In general, not all of the transmitters can be pure interferers since $\mathcal{D}_\ell \neq \emptyset, \forall \ell$ by the definition above.

³Therefore, unlike the AF strategy, for the DF strategy, we make some choices that will result in a loss of optimality in terms of the error probability. We make these choices in order to make the presentation simple. Also, as we shall demonstrate later on, our DF scheme will still be able to achieve the transmit diversity bound $(R, 0)$. Since the best possible diversity gain achievable by any scheme is $(R, 0)$, our DF scheme will not suffer from any loss of optimality in terms of the diversity gain.

⁴For brevity, we omit the dependence of the relay's transmitted signal to the particular forwarding strategy as it is clear from the context.

Let us call the vector of transmitted symbols $\mathbf{s}_\ell = [s_k]_{k \in \mathcal{D}_\ell}$ as the super-symbol relevant to the ℓ th receiver, and $\tilde{\mathbf{s}}_\ell$ be its decoded version. We say that an error event occurs at a receiver if it incorrectly decodes its desired super-symbol. In this case, the optimal decoder at the ℓ th receiver is an individual ML decoder⁵ given by $\tilde{\mathbf{s}}_\ell = \arg \max_{\mathbf{s}'_\ell \in \mathcal{S}_\ell} P(\mathbf{s}'_\ell | y_\ell, \mathbf{x}, \mathbf{h})$, where $\mathcal{S}_\ell = \prod_{k \in \mathcal{D}_\ell} \mathcal{S}_k$ is the relevant super-symbol alphabet. Note that the individual ML decoder at the ℓ th receiver knows the symbol alphabets of all the transmitters, not just the alphabets of the transmitters in its decoding set $\mathcal{D}_\ell$.

Let us now define a single quantity that represents the super-symbol error rate (SER) performance of all the receivers. We define the *conditional* network error rate (conditional NER, or CNER) as the probability that at least one receiver incorrectly decodes its desired super-symbol. For a fixed channel state $\mathbf{h}$, and beamforming vector $\mathbf{x}$, we denote the CNER as $\text{CNER}(\mathbf{x}, \mathbf{h})$. We would like to emphasize that, the CNER, and many of the other measures that we will define later on, depend on the particular forwarding strategy that is employed in the network. For brevity, we omit the indication of this dependence as it will be clear from the context.

Our performance measure, the NER, is the expected value of the CNER. Given any quantizer $\mathbf{Q}$, the NER can thus be expressed as

$$\text{NER}(\mathbf{Q}) \triangleq \mathbb{E}_{\mathbf{h}}[\text{CNER}(\mathbf{Q}(\mathbf{h}), \mathbf{h})]. \quad (10)$$

B. A Generalized Diversity Measure

Let us also define a unique diversity measure for our network. We recall from Section I-A that $P_{R_r} = p_{R_r} P$, $r = 1, \dots, R$, $P_{S_k} = p_{S_k} P$, $k = 1, \dots, K$, where $0 < p_{S_k}, p_{R_r} < \infty$ are independent of P . In other words, we allow the power constraint of each transmitting terminal to grow linearly with P . Then, the *first-order* diversity achieved by a quantizer $\mathbf{Q}$ is given by

$$d_1(\mathbf{Q}) \triangleq \lim_{P \rightarrow \infty} -\frac{\log \text{NER}(\mathbf{Q})}{\log P}. \quad (11)$$

As we have already exemplified in Section I-C, one problem with this conventional definition of diversity is that it fails to characterize the asymptotic effect of possible sub-linear P -dependent terms (e.g., logarithmic terms) in the error rate expression. In order to properly handle such cases, we define the *second-order* diversity as

$$d_2(\mathbf{Q}) \triangleq \lim_{P \rightarrow \infty} -\frac{\log \text{NER}(\mathbf{Q}) + d_1(\mathbf{Q}) \log P}{\log \log P}. \quad (12)$$

The *diversity* (gain) achieved by a quantizer $\mathbf{Q}$ is given by $d(\mathbf{Q}) \triangleq (d_1(\mathbf{Q}), d_2(\mathbf{Q}))$. Note that the first-order diversity is always positive, while the second-order diversity can be negative.

With these definitions, the asymptotic performance with a quantizer $\mathbf{Q}$, as P grows to infinity, can be expressed as

$$\text{NER}(\mathbf{Q}) \cong G_{\mathbf{A}}(\mathbf{Q}) (\log P)^{-d_2(\mathbf{Q})} P^{-d_1(\mathbf{Q})}, \quad (13)$$

⁵In the literature, the phrase “individual” usually refers to the cases in which the a posteriori probability is maximized over a single transmitter alphabet. Note that, in our case, the maximization is over the product alphabet $\mathcal{S}_\ell$ that represents the set of all super-symbols that the ℓ th receiver is interested in.

where the factor $G_A(Q)$ is the *array gain* with the following properties:

- For any arbitrary $\alpha > 0$, $G_A(Q) \leq \log^\alpha P$ for sufficiently large P .
- For any arbitrary $\beta > 0$, $G_A(Q) \geq \log^{-\beta} P$ for sufficiently large P .

We use the array gain only when we compare the performance of two quantizers that provide the same diversity gain.

Finally, for completeness, we would like to recall that for two diversity gains $d = (d_1, d_2)$, and $d' = (d'_1, d'_2)$, we say that d is higher than d' (or $d > d'$) if either $d_1 > d'_1$ or $d_1 = d'_1, d_2 > d'_2$. It is straightforward to show using (13) that such a comparison of the diversity gains of any two systems is consistent with their asymptotic NER performances: The NER of a system with a higher diversity gain will be lower than the one with the lower diversity gain for all P sufficiently large.

IV. OPTIMAL DIVERSITY GAINS AND THEIR ACHIEVABILITY USING GLOBAL QUANTIZERS

Sections II and III have been of an introductory nature on our main objectives described in Section I-E. In Section II, we have introduced a general description of channel quantization, and the AF and DF cooperative schemes for our relay-interference network model. In Section III, we have defined the NER and the generalized diversity gain quantities to measure the performance of these schemes. Equipped with the necessary tools, we can now begin working on the following aforementioned main questions that we intend to study in this paper:

- 1) What are the best possible (generalized) diversity gains with AF and DF strategies?, and
- 2) How to achieve these diversity gains using a practical distributed feedback scenario?

In this section, we provide an answer to the first question and address the second one in part by relaxing the constraint of a “practical distributed feedback scenario.” We will assume that there is a global quantizer (GQ) that knows the channel state of the entire network $\mathbf{h}$ and provides feedback information to the relays and the receivers via global feedback bits. While such a quantizer cannot exist in practice, this GQ-based approach has two main advantages: First, the simple structure of GQs allows us to find achievable diversity gains without worrying about the analytical complications that would otherwise arise while dealing with a distributed quantizer. (We have mentioned some of these complications in Section I-E.) Second and more importantly, we can use a GQ to synthesize practical distributed quantizers that can operate without any global feedback information. The details of how to do such a synthesis and the performance of the resulting quantizers will be discussed later in Section V.

We introduce our global quantization scheme in Section IV-A. In Section IV-B, we determine the optimal GQs for the AF and DF strategies. As it turns out, these optimal GQs have complicated structures. We therefore design suboptimal GQs that can perform as well as the optimal GQs in terms of the diversity gain. In Section IV-C, we determine the diversity gains provided by our GQs using the relay selection codebook. We show that with AF relays, our GQs can achieve diversity $(R, 0)$ if $K = 1$, and $(R, -R)$ if $K > 1$. Conversely, we prove

that for any two-channel use AF scheme, even with full CSI at the relays, the best possible diversity gain is the same as the one achievable by our GQs. This part of our analysis thus provides a refinement of the transmit diversity bound on the diversity gain $(R, 0)$ provided by Theorem 1. It also shows that in the presence of interference, simple linear forwarding schemes are not sufficient to achieve the full diversity gain $(R, 0)$. Then, we show that with DF relays, we can achieve full diversity for any K . In other words, although it significantly complicates the relay operation, the DF strategy achieves the transmit diversity bound, and is thus optimal.

A. Global Quantization

Our global quantization scheme operates as follows: For each constant fading block, the channel state $\mathbf{h}$ is quantized using a global quantizer GQ defined by the encoder and the decoder mappings $\text{GQE} : \mathbb{C}^{R(K+L)} \rightarrow \mathcal{I}^g$ and $\text{GQD} : \mathcal{I}^g \rightarrow \mathcal{C}^g$, where $\mathcal{I}^g \triangleq \{1, \dots, |\mathcal{C}^g|\}$ represents the index set for the codebook elements. The (global) feedback bits represent the binary description of the encoder output, and they are received by each relay and each receiver without any errors or delays. Therefore, there are a fixed number of $\lceil \log_2 |\mathcal{C}^g| \rceil$ global feedback bits per channel state. Each relay and each receiver decodes the index that corresponds to the feedback bits using GQD, to find the corresponding codebook element. Without loss of generality, for $\text{GQE}(\mathbf{h}) = m$, we set $\text{GQD}(m) = \mathbf{x}_m \in \mathcal{C}^g$. For the rest of the paper, we use the well-known notation $\text{GQ}(\mathbf{h}) \triangleq (\text{GQD} \circ \text{GQE})(\mathbf{h})$. Therefore, $\text{GQ} : \mathbb{C}^{R(K+L)} \rightarrow \mathcal{C}^g$, and $\text{GQ}(\mathbf{h}) = \mathbf{x}$, for some $\mathbf{x} \in \mathcal{C}^g$.⁶

B. Efficient Suboptimal Approaches to Optimal Global Quantization

In order to determine a GQ that can achieve maximal diversity, let us first determine the optimal GQ given a fixed codebook with finite cardinality.

Proposition 1: Given a fixed codebook $\mathcal{C}$ with $|\mathcal{C}| < \infty$, the optimal global quantizer $\text{GQ}_\mathcal{C}^* : \mathbb{C}^{R(K+L)} \rightarrow \mathcal{C}$ that minimizes the NER is given by

$$\text{GQ}_\mathcal{C}^*(\mathbf{h}) = \arg \min_{\mathbf{x} \in \mathcal{C}} \text{CNER}(\mathbf{x}, \mathbf{h}). \quad (14)$$

Proof: For any global quantizer $\text{GQ}' : \mathbb{C}^{R(K+L)} \rightarrow \mathcal{C}$, we have $\text{CNER}(\text{GQ}_\mathcal{C}^*(\mathbf{h}), \mathbf{h}) \leq \text{CNER}(\text{GQ}'(\mathbf{h}), \mathbf{h}) \implies \text{NER}(\text{GQ}_\mathcal{C}^*) \leq \text{NER}(\text{GQ}')$. Thus, $\text{GQ}_\mathcal{C}^*$ performs at least as well as any GQ with codebook $\mathcal{C}$. ■

We would like to note that the proposition holds for both AF and DF strategies due to the inherent dependence of the CNER on the particular forwarding method.

Given that we employ an optimal GQ encoder given by Proposition 1, the GQ codebook thus uniquely determines the system performance. But, there is one complication: If we ever want to implement the optimal GQ encoder, we should be able to evaluate $\text{CNER}(\mathbf{x}, \mathbf{h})$, for any given $\mathbf{x}$ and $\mathbf{h}$. Unfortunately, for both AF and DF strategies, a closed form characterization

⁶The GQ is thus a special case of the general quantizer described in Section II-A.

of the CNER is very difficult, if not impossible. For that reason, we design a suboptimal quantizer that, instead of the actual CNER, uses an upper bound on the CNER. Fortunately, this suboptimal quantizer will be powerful enough to achieve the best possible diversity that the particular forwarding strategy can provide.

Moreover, we shall see that in order to achieve the best possible diversity gain, it is sufficient to evaluate the CNER with only one of a finite number of beamforming vectors. In fact, for our achievability results, we work with only R beamforming vectors, each of which corresponds to a relay selection vector that allows only one relay to cooperate given a constant fading block. Formally, let $\mathbf{e}_r \triangleq [e_{r1} \cdots e_{rR}]$ with $e_{rq} = 1, q = r$ and $e_{rq} = 0, q \neq r$ denote the r th relay selection vector. Then, the useful upper bounds on $\text{CNER}(\mathbf{e}_r, \mathbf{h})$, $r = 1, \dots, R$, which will form the basis of our GQs, are given by the following proposition:⁷

Proposition 2: For a given $r \in \{1, \dots, R\}$, let

$$\gamma^L(\mathbf{e}_r, \mathbf{h}) \triangleq \min_{\ell} \gamma_{\ell}(\mathbf{e}_r, \mathbf{h}) \quad (15)$$

with

$$\gamma_{\ell}(\mathbf{e}_r, \mathbf{h}) = \min_{\ell} \min_{\substack{\mathbf{s}, \hat{\mathbf{s}} \in \mathcal{S} \\ \mathbf{s} \neq \hat{\mathbf{s}}}} \frac{\left| \sum_{k=1}^K (s_k - \hat{s}_k) \sqrt{P_{S_k}} f_{kr} \right|^2 |g_{r\ell}|^2 P_{R_r}}{4\bar{C}(1 + \sum_{k=1}^K |f_{kr}|^2 P_{S_k} + |g_{r\ell}|^2 P_{R_r})}, \quad (16)$$

$$\bar{C} \triangleq \begin{cases} 1, & K = 1 \text{ or } \mathbb{E}[s_k] = 0, \forall k. \\ K, & \text{otherwise} \end{cases} \quad (17)$$

for AF relays, and

$$\gamma_{\ell}(\mathbf{e}_r, \mathbf{h}) = \min \left\{ \frac{\min_{\ell} |g_{r\ell}|^2 P_{R_r}}{2(\varsigma - 1)^2}, \frac{1}{4} \min_{\substack{\mathbf{s}, \hat{\mathbf{s}} \in \mathcal{S} \\ \mathbf{s} \neq \hat{\mathbf{s}}}} \left| \sum_{k=1}^K (s_k - \hat{s}_k) \sqrt{P_{S_k}} f_{kr} \right|^2 \right\} \quad (18)$$

for DF relays.

Then, for any $r \in \{1, \dots, R\}$ and $\mathbf{h}$, we have

$$\text{CNER}(\mathbf{e}_r, \mathbf{h}) \leq C_4 \exp(-\gamma^L(\mathbf{e}_r, \mathbf{h})), \quad (19)$$

for a constant $0 < C_4 < \infty$ that is independent of P and the forwarding strategy.

Proof: The proof is provided in Appendix A. ■

We would like to note the similarity of (19) to the conventional error rate expressions for single user wireless communication systems. Actually, we interpret the term $\gamma^L(\mathbf{e}_r, \mathbf{h})$ as a network signal-to-noise ratio (NSNR) measure that characterizes the overall performance of the network. Also, we call $\gamma_{\ell}(\mathbf{e}_r, \mathbf{h})$ as the *local* NSNR at the ℓ th receiver given that the r th relay is selected.

⁷As a byproduct of our analysis, we also derive general simple upper bounds on the CNER that hold for an arbitrary beamforming vector. We do not present these upper bounds in the statement of the proposition for brevity and since we will not employ them in this paper. For the interested reader, we refer to the proof of the proposition.

We can now verify the claims regarding the performance of our GQs that we have made earlier in the beginning of this section.

C. Achieving Maximum Diversity With Relay Selection

For $K = L = 1$, it has been shown in [8] and [39] that a feedback scheme based on relay selection can achieve diversity $(R, 0)$ with AF and DF strategies, respectively. Here, we generalize this result to any L .

For $K > 1$, due to multiuser interference at both relays and receivers, finding the best possible diversity gain and the question of how to achieve it both become non-trivial problems. In this section, we show that simple relay selection based GQs that maximize the NSNR can achieve diversity $(R, -R)$ and $(R, 0)$ for the AF and DF strategies, respectively. We also show by a converse that the best possible diversity gain with AF relays and any quantizer is indeed $(R, -R)$ whenever $K > 1$. Note that the transmit diversity bound $(R, 0)$ established by Theorem 1 provides a converse for the remaining scenarios that can already achieve diversity $(R, 0)$.

For any K and L , and regardless of the forwarding scheme, we define our GQ as

$$\text{GQ}_{C_s}(\mathbf{h}) = \arg \max_{\mathbf{e}_r \in C_s} \gamma^L(\mathbf{e}_r, \mathbf{h}), \quad (20)$$

where $C_s \triangleq \{\mathbf{e}_r : r = 1, \dots, R\}$ represents the relay selection codebook, and $\gamma^L(\mathbf{e}_r, \mathbf{h})$ is given by (16) and (18) for the AF and DF strategies, respectively. Note that in any case, GQ_{C_s} chooses the relay selection vector that maximizes the NSNR provided by the forwarding strategy.

The following theorem determines the best possible diversity gain for our two-channel use AF protocol, and shows that the relay selection based GQ GQ_{C_s} can achieve this diversity.

Theorem 2: For the AF strategy, there are constants $0 < C_5$, $C_6 < \infty$ that are independent of P such that for all P sufficiently large,

$$\text{NER}(\text{GQ}_{C_s}) \leq C_5 \frac{1}{P^R}, \quad K = 1, \quad (21)$$

$$\text{NER}(\text{GQ}_{C_s}) \leq C_6 \frac{\log^R P}{P^R}, \quad K > 1. \quad (22)$$

Conversely, for any quantizer $\mathbf{Q}$ for the AF strategy, there are constants $0 < C_7, C_8 < \infty$ that are independent of $\mathbf{Q}$ and P such that for all P sufficiently large,

$$\text{NER}(\mathbf{Q}) \geq C_7 \frac{1}{P^R}, \quad K = 1, \quad (23)$$

$$\text{NER}(\mathbf{Q}) \geq C_8 \frac{\log^R P}{P^R}, \quad K > 1. \quad (24)$$

Moreover, (23) and (24) hold for any relay normalization factor ρ'_r that satisfies $\rho'_r \leq \rho_r, \forall \mathbf{h}$.

Proof: The proof of (21) and (22) is provided in Appendix B. We prove (24) in Appendix C. Note that (23) is an immediate consequence of Theorem 1 with $C_7 = C_0$. ■

In other words, for any K , the relay selection scheme with a GQ achieves the maximal diversity gain that the AF scheme

can provide. It is remarkable that the maximal diversity gain is achieved regardless of the number of transmitters and receivers.

By definition, our selection scheme requires $\lceil \log_2 R \rceil$ global feedback bits. With $\lceil \log_2 R_0 \rceil$ global feedback bits, where $R_0 \in \{1, \dots, R-1\}$, diversity orders $(R_0, 0)$ and $(R_0, -R_0)$ are achievable for $K = 1$, and $K > 1$, respectively, simply by considering the selection scheme for any fixed R_0 of the relays and disregarding the others. Note that using more than $\lceil \log_2 R \rceil$ feedback bits can only improve the array gain performance, since the maximal diversity gain can already be achieved using $\lceil \log_2 R \rceil$ bits.

As we have mentioned in the statement of the theorem, the lower bound on the NER in (24) holds for *any* quantizer $\mathbf{Q} : \mathbb{C}^{R(K+L)} \rightarrow \mathcal{C}$. This means that we can use the “quantizer” $\mathbf{Q}_{\mathcal{X}}^*(\mathbf{h}) = \arg \max_{\mathbf{x} \in \mathcal{X}} \text{CNER}(\mathbf{x}, \mathbf{h})$ that chooses the best possible beamforming vector over the set of all feasible beamforming vectors $\mathcal{X} \triangleq \{\mathbf{x} \in \mathbb{C}^R : \|\mathbf{x}\|_\infty \leq 1\}$. The employment of the quantizer $\mathbf{Q}_{\mathcal{X}}^*$ is equivalent to the assumption of having full CSI at the relays. On the other hand, for a two-channel use protocol, the most general operation of an AF relay can only be a CSI-dependent linear scaling of its signal received in the first step. Therefore, we can interpret (24) as a lower bound that indeed holds for any two-channel use AF (linear) scheme.⁸

Theorem 2 also shows that a different relay normalization factor ρ_r' *cannot* improve the diversity upper bounds, provided that it satisfies the short-term power constraint, and a codebook $\mathcal{C} \subset \mathcal{X}$ is employed. Thus, for the rest of this paper, we will only consider ρ_r as our AF relay normalization factor.

One of the main arguments of Theorem 2 is that for the AF strategy, the maximum achievable diversity is $(R, -R)$ when $K > 1$. In other words, there is always a second-order diversity loss of R compared to an interference-free network that can achieve diversity $(R, 0)$. This result implies that for sufficiently large P , orthogonalizing the transmissions of different transmitters will outperform the non-orthogonal transmission studied in this paper. Another advantage of having orthogonal transmissions will be a much simpler decoding structure at the receivers. On the other hand, in practical networks, such a “perfect” orthogonalization may not be feasible due to time and bandwidth limitations. For example, in cellular networks, co-channel interference is usually unavoidable because of multiple simultaneous uses of the same time and frequency

channels [69]. In this perspective, Theorem 2 provides valuable information on the performance limits of such systems suffering from unwanted interference. It also demonstrates that it is possible to achieve these limits using a GQ. We shall later see that these limits can as well be achieved by a practical distributed quantization strategy.

We now consider DF relaying. The following theorem shows that for the DF strategy, our relay selection based GQ $\text{GQ}_{\mathcal{C}_s}$ can achieve the full diversity gain of $(R, 0)$ for any K .

Theorem 3: For the DF strategy, there is a constant $0 < \mathcal{C}_9 < \infty$ that is independent of P such that for all P sufficiently large,

$$\text{NER}(\text{GQ}_{\mathcal{C}_s}) \leq \mathcal{C}_9 \frac{1}{PR}. \quad (25)$$

Conversely, for any quantizer $\mathbf{Q}$ for the DF strategy, there is a constant $0 < \mathcal{C}_{10} < \infty$ that is independent of $\mathbf{Q}$ and P such that for all P sufficiently large,

$$\text{NER}(\mathbf{Q}) \geq \mathcal{C}_{10} \frac{1}{PR}. \quad (26)$$

Proof: We prove (25) in Appendix D. The converse follows from Theorem 1 with $\mathcal{C}_{10} = \mathcal{C}_0$. ■

Theorem 3 shows that the DF strategy with a GQ achieves the transmit diversity bound $(R, 0)$ as if there is no interference and it is thus optimal. Interestingly, according to Theorem 1 the full diversity gain of $(R, 0)$ is already the best possible for an arbitrary block length and full CSI. The DF scheme can achieve it with the minimum possible delay of two channel uses and only $\lceil \log_2 R \rceil$ global feedback bits. Similarly, with $\lceil \log_2 R_0 \rceil$ global feedback bits, where $R_0 \in \{1, \dots, R-1\}$, diversity order $(R_0, 0)$ is achievable by considering the selection scheme for any fixed R_0 of the relays and disregarding the others. Compared to the AF strategy, the only disadvantage of the DF strategy is the extra ML decoding burden imposed upon the relays.

V. DIVERSITY WITH LOCAL QUANTIZERS

In the previous section, we have found the maximum achievable diversity gains with AF and DF strategies. We have also designed GQs that can achieve these gains. On the other hand, our goal of designing a *practical* feedback scenario that can provide maximum diversity is far from over: None of the terminals can acquire the CSI $\mathbf{h}$ of the entire network, yet the GQ encoder needs $\mathbf{h}$ to operate. This makes the GQ structure impractical. In this section, we design distributed (local) quantizers (LQs) that can operate without any global feedback information, and surprisingly, can achieve the same diversity gains as the GQ. In our local quantization scheme, each receiver provides its own feedback information according to the channel states $[\mathbf{f} \ \mathbf{g}_\ell]$ that it can acquire via the training methods outlined in Section I-A.

We describe our local quantization scheme in Section V-A. In Section V-B, we introduce our *localization* method to synthesize practical LQs out of an existing GQ. We shall thus see that even though the GQ structure by itself is impractical, it serves the important purpose of providing the basis for our practical LQs. The

⁸Obviously, the truth of this statement depends on how one defines amplifying and forwarding. In this context, the literature follows the assumption that the amplification factor of an AF relay is a deterministic function of the CSI that does not depend on the relay's instantaneous received signal. Note that this paper, and in particular our statement, both follow this assumption. Nevertheless, if one allows the amplification factor to be a function of the relay's received signal, then any viable forwarding strategy can be interpreted as a special case of AF relaying. As an example, suppose that for a particular forwarding strategy, the relay receives t and sends u . Then, we can set the relay operation to be $f(t) = (\frac{u}{t})t$. This way we can call this forwarding strategy an AF scheme with an amplification factor $x = \frac{u}{t}$. Let us also briefly sketch how a non-deterministic AF relay can be handled and would indeed be suboptimal. Suppose that the amplification factor x is allowed to vary according to some density $f(x|\mathbf{h})$ that depends on the CSI but is independent of the relay's received signal, the additive noises in the network, and the transmitted symbols. Then, for a given $\mathbf{h}$, one can pick the best factor, say $x^*(\mathbf{h})$, that yields the best CNER averaged over all the noises and the transmitted symbols. We can then replace this non-deterministic AF relay with a deterministic one that uses a “fixed” CSI-dependent factor $x^*(\mathbf{h})$. This deterministic relay will outperform its non-deterministic counterpart.

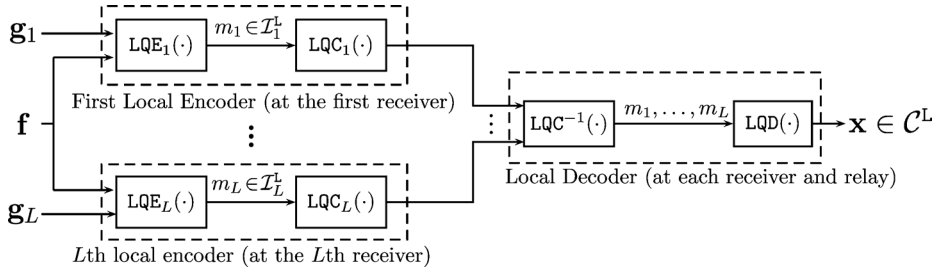


Fig. 2. Local quantizer operation.

diversity performance of our LQs with the AF and DF strategies will be provided in the remaining Sections V-C and V-D.

A. Local Quantization

We define our local quantizer LQ by L local encoders, with the ℓ th encoder at the ℓ th receiver, and a unique local decoder at each receiver and relay, as described in Fig. 2. The ℓ th local encoder comprises of two parts: First, an LQ encoder $\text{LQE}_\ell : \mathbb{C}^{R(K+1)} \rightarrow \mathcal{I}_\ell^L$ maps the part $[\mathbf{f} \ \mathbf{g}_\ell]$ of the entire channel realization $\mathbf{h}$ to an index $m_\ell \in \mathcal{I}_\ell^L$. Then, a lossless local compressor $\text{LQC}_\ell : \mathcal{I}_\ell^L \rightarrow \mathcal{J}_\ell^L$ maps this index to a binary description. Note that the domain of each LQ encoder is different from the domain of the GQ encoder. For the ℓ th encoder, which is at the ℓ th receiver, the domain corresponds to the channel states known by the ℓ th receiver. In other words, for the ℓ th encoder, the domain corresponds to the channel states from the transmitters to the relays and from the relays to the ℓ th receiver, represented by $[\mathbf{f} \ \mathbf{g}_\ell]$. Note that since $\mathbf{f}$ is common among all local encoders, the sources $[\mathbf{f} \ \mathbf{g}_\ell]$, $\ell = 1, \dots, L$ to be quantized are correlated despite the fact that the channels themselves are independent.

Let $l(j)$ denote the length of a binary description j . The ℓ th receiver feeds back $\text{LQC}_\ell(\text{LQE}_\ell([\mathbf{f} \ \mathbf{g}_\ell]))$, using $l(\text{LQC}_\ell(\text{LQE}_\ell([\mathbf{f} \ \mathbf{g}_\ell])))$ bits. We call LQ a fixed-length LQ (fLQ) if the number of feedback bits per receiver remains fixed for each constant fading block, i.e., if $l(\text{LQC}_\ell(\text{LQE}_\ell([\mathbf{f} \ \mathbf{g}_\ell]))) = \lceil \log_2 |\mathcal{I}_\ell^L| \rceil$, $\forall \mathbf{h}$, $\forall \ell$.⁹ Otherwise, we call it a variable-length LQ (vLQ). For the latter case, the feedback rate of the ℓ th receiver can be expressed as

$$R_\ell(\text{LQ}) \triangleq \mathbb{E} \left[l \left(\text{LQC}_\ell(\text{LQE}_\ell([\mathbf{f} \ \mathbf{g}_\ell])) \right) \right]. \quad (27)$$

After all the L feedback messages are exchanged between the receivers and the relays without any errors or delays, each of them decodes the feedback bits using a unique local decoder. The local decoder is the composition of a lossless decompressor $\text{LQC}^{-1} : \prod_\ell \mathcal{J}_\ell^L \rightarrow \prod_\ell \mathcal{I}_\ell^L$ and an LQ decoder $\text{LQD} : \prod_\ell \mathcal{I}_\ell^L \rightarrow \mathcal{C}^L$. The decompressor reconstructs the quantization indices m_ℓ , $\ell = 1, \dots, L$ respectively from the received binary descriptions $\text{LQC}_\ell(\text{LQE}_\ell([\mathbf{f} \ \mathbf{g}_\ell]))$, $\ell = 1, \dots, L$, and the LQ decoder maps the quantization indices to a codebook element.

Overall,

$$\text{LQ}(\mathbf{h}) \triangleq \text{LQD} \left(\text{LQC}^{-1} \left(\text{LQC}_1(\text{LQE}_1([\mathbf{f} \ \mathbf{g}_1])), \dots, \right. \right.$$

$$\left. \left. \text{LQC}_L(\text{LQE}_L([\mathbf{f} \ \mathbf{g}_L])) \right) \right) \\ = \text{LQD} \left(\text{LQE}_1([\mathbf{f} \ \mathbf{g}_1]), \dots, \text{LQE}_L([\mathbf{f} \ \mathbf{g}_L]) \right).$$

Thus, $\text{LQ} : \mathbb{C}^{R(K+L)} \rightarrow \mathcal{C}^L$, and $\text{LQ}(\mathbf{h}) = \mathbf{x}$, for some $\mathbf{x} \in \mathcal{C}^L$.

B. Localization

In Section IV, we showed that a GQ using relay selection can achieve the maximal diversity gain for both AF and DF strategies. Motivated by this result, we expect that a relay selection based LQ will achieve high diversity orders for both forwarding schemes. Later in this section, we will design two such LQs: An fLQ that achieves maximal first-order diversity, and a vLQ that achieves maximal diversity. Both quantizers will have similar structures. As we have mentioned, we construct them using the *localization* method [70], in which we synthesize an LQ out of an existing GQ. The synthesized LQ and the GQ share the same codebook. For our particular quantization scheme, we use the global quantizer GQ_{C_s} in (20) as the basis of our LQs. Since GQ_{C_s} is based on relay selection, all of our LQs will be based on relay selection as well.

Let $\text{LQ}_{\xi, N}^{(\mathbf{f}|\mathbf{v})}$ denote a generic localization of GQ_{C_s} . For the synthesized local quantizer $\text{LQ}_{\xi, N}^{(\mathbf{f}|\mathbf{v})}$, the superscript indicates whether it is fixed-length (f) or variable-length (v); and ξ, N are design parameters that we shall specify later on. For a fixed power level P , and a particular channel state $\mathbf{h}$, the components of the synthesized quantizer operate as follows:

1) *LQ Encoders*: For notational convenience, let $\omega_{r\ell} = \gamma_\ell^L(\mathbf{e}_r, \mathbf{h})$ denote the local NSNR at the ℓ th receiver given that the r th relay is selected.¹⁰ For future reference, let us also recall from Proposition 2 that $\min_\ell \omega_{r\ell}$ is thus the NSNR given that the r th relay is selected.

The ℓ th LQ encoder calculates $\omega_{r\ell}$, $r = 1, \dots, R$. In other words, it calculates its own local NSNR for all possible relay selection vectors. Then, it quantizes each of the local NSNRs using a scalar quantizer $\mathcal{N} : \mathbb{R} \rightarrow \{0, \dots, N-1\}$ with

$$\mathcal{N}(x) = \begin{cases} n, & \exists n \in \{0, \dots, N-2\}, x \in [n\xi, (n+1)\xi), \\ N-1, & \text{otherwise.} \end{cases}$$

Its output message is the concatenation of R sub-messages $\mathcal{N}(\omega_{r\ell})$, $r = 1, \dots, R$.

2) *An Illustration of the LQ Encoders*: Let us now illustrate the operation of the LQ encoders with a simple example with

¹⁰Note that, as shown in Proposition 2, $\gamma_\ell^L(\mathbf{e}_r, \mathbf{h})$ depends on the particular forwarding strategy that is employed in the network. Therefore, the discussion that will follow holds for both AF and DF strategies.

⁹Therefore, the GQ studied in Section IV is indeed a fixed-length quantizer.

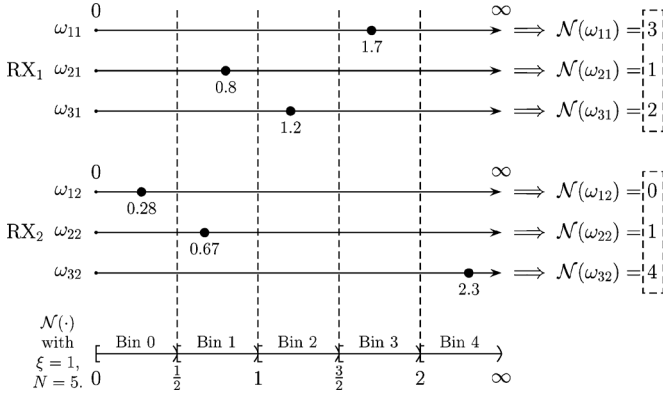


Fig. 3. An illustration of the LQ encoders.

$R = 3$, and $L = 2$, as shown in Fig. 3. For some fixed channel variances, power constraints, and channel state $\mathbf{h}'$, suppose that $w_{11} = 1.7$, $w_{21} = 0.8$, $w_{31} = 1.2$, $w_{12} = 0.28$, $w_{22} = 0.67$, and $w_{32} = 2.3$. In the figure, each of these local NSNR values are represented by a disk ($\bullet$) on the real axis. Since we are using an LQ, w_{r1} , $r = 1, 2, 3$ can be calculated only by the first receiver, and similarly, w_{r2} , $r = 1, 2, 3$ can be calculated only by the second receiver. Note that the GQ has access to all the local SNRs and in this example, selects the relay with index $\arg \max_{r \in \{1,2,3\}} \min_{\ell} w_{r\ell} = 3$.

After the LQ encoder calculates its local NSNR values, it quantizes them using a scalar quantizer $\mathcal{N}$ that is uniquely determined by the parameters ξ and N . In our example, we use $N = 5$ bins and set $\xi = \frac{1}{2}$. Each bin is represented by a half open interval ($\left[\text{---} \right)$) on the real axis. The output message of the LQ encoder is the concatenation of its quantized local NSNR values (submessages), shown as frames with a dashed outline, on the right hand side of the figure.

3) *Compressors*: In general, there are R sub-messages, each with N possible values. Therefore, for a fixed-length synthesis $\text{LQ}_{\xi, N}^f$, at each channel state, each receiver feeds back $\lceil R \log_2 N \rceil$ bits without any compression.

For a variable-length synthesis $\text{LQ}_{\xi, N}^v$, we use a lossless compressor that produces an empty codeword (of length 0) whenever $\mathcal{N}(w_{r\ell}) = N - 1$, $\forall r$, and otherwise a codeword of length $\lceil \log_2(N^R - 1) \rceil$ bits that can uniquely represent each $\mathcal{N}(w_{r\ell})$. In other words, for a given channel state, the number of feedback bits produced by any receiver is either 0 bits or $\lceil \log_2(N^R - 1) \rceil$ bits¹¹.

We should re-emphasize that in the above description of our fLQs and vLQs, for clarity of exposition, we have assumed that the power level P is fixed. In general, as we shall discuss later in Sections V-C and V-D in more detail, we allow one or both of the parameters ξ and N to vary with P , depending on whether the

¹¹If the empty codeword is not allowed, one can use a “0” (a codeword of length 1 bit) instead of the empty codeword, and append a “1” to each remaining codeword of length $\lceil \log_2(N^R - 1) \rceil$ bits. The resulting codewords are uniquely decodable as well. Then, all of the results in this paper will hold for the case where the empty codeword is forbidden, given that the required feedback rates are increased by 1 bit. Also, note that one can achieve a better compression by using entropy encoders instead of the *suboptimal* compressors that we employ. Even though the localization method was introduced originally with entropy encoders, the compressors that we use in this paper will be good enough for our purposes.

synthesized quantizer is fixed-length or variable-length. More specifically, for our fLQs, we set ξ to be a function of P , while N remains a fixed positive constant that is independent of P . In other words, regardless of P or the channel state $\mathbf{h}$, the fLQ will always operate with a fixed $\lceil R \log_2 N \rceil$ feedback bits per receiver. With such a design, despite the fact that the structure of the fLQ varies with P (since ξ is a function of P), the rate of the fLQ remains fixed. In contrast, for our vLQs, we allow both ξ and N to be a function of P . In this case, the number of feedback bits of a receiver depends on both $\mathbf{h}$ and P , and thus the rate of the vLQ varies with P .

Regardless of the type of the quantizer, after all the L feedback messages of the receivers are exchanged between the receivers and the relays, each of them decodes the feedback bits using the local decoder. The decoder operation is the same for each receiver and relay.

4) *Decompressor*: First, a decompressor perfectly recovers all the submessages from all the receivers, $\mathcal{N}(w_{r\ell})$, $r = 1, \dots, R$, $\ell = 1, \dots, L$. All of these submessages are passed to the LQ decoder.

5) *An Illustration of the LQ Decoder*: For clarity of exposition, let us first present the LQ decoder for the example scenario in Section V-B-II and the same channel state $\mathbf{h}'$. A more formal description of the general LQ decoder operation will be presented afterwards.

In general, the main goal of the LQ decoder is to imitate the GQ as well as possible. For our particular example, the GQ selects the relay with index $\arg \max_{r \in \{1,2,3\}} w_r$, where $w_r = \min\{w_{r1}, w_{r2}\}$ is the NSNR given that the r th relay is selected. Then, the first goal of the LQ decoder should be to determine w_r . However, the LQ decoder only knows the quantized local NSNR values, $\mathcal{N}(w_{r\ell})$, $r = 1, 2, 3$, $\ell = 1, 2$, as shown in Fig. 3. Therefore, it cannot determine the exact value of w_r . However, as we shall describe in what follows, it can perfectly determine a subset of $\mathbb{R}$ where w_r resides.

For any $\omega \in \mathbb{R}$, $\mathcal{N}(\omega) = n \implies \omega \in [\frac{n}{2}, \frac{n+1}{2})$, $n = 0, \dots, 3$, and $\mathcal{N}(\omega) = 4 \implies \omega \in [2, \infty)$. We can use these facts to determine the possible locations of the local NSNR values, as represented in Fig. 4 by half-open intervals ($\left[\text{---} \right)$) of $\mathbb{R}$.

Since $\omega_1 = \min\{w_{11}, w_{12}\}$, and we know for sure that $w_{11} \in [\frac{3}{2}, 2)$ and $w_{12} \in [0, \frac{1}{2})$, we should have $\omega_1 \in [0, \frac{1}{2})$. Using the same arguments for all r , we can obtain $w_2 \in [\frac{1}{2}, 1)$, and $w_3 \in [1, \frac{3}{2})$. We have thus determined the possible locations of w_r , as shown in Fig. 5, by having access only to the quantized versions of w_r .

The LQ decoder's main goal was to find $\arg \max_{r \in \{1,2,3\}} w_r$. Using the possible locations of w_r that we have found, it is now clear that the third relay should provide the best NSNR. The LQ decoder's output will be $\mathbf{e}_3$. Note that this is the same output as the GQ output. Therefore, for this particular channel state, the LQ operates in the same manner as the GQ.

However, the LQ decoder will not be this lucky in general. As an example, another channel state might result in $\omega_1 \in [0, \frac{1}{2})$ and $\omega_2, \omega_3 \in [1, \frac{3}{2})$. In this case, the LQ decoder will know for sure that both the second relay and the third relay provide larger NSNRs compared to the first relay. On the other hand, it cannot

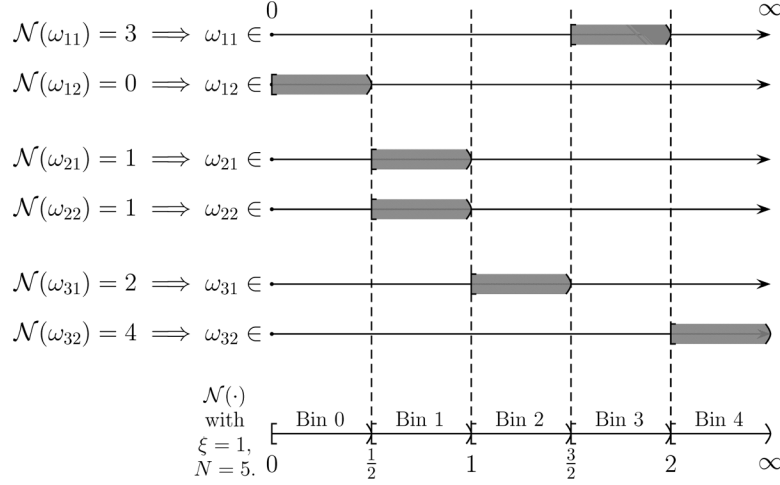


Fig. 4. Possible locations of the local NSNRs according to the LQ Decoder.

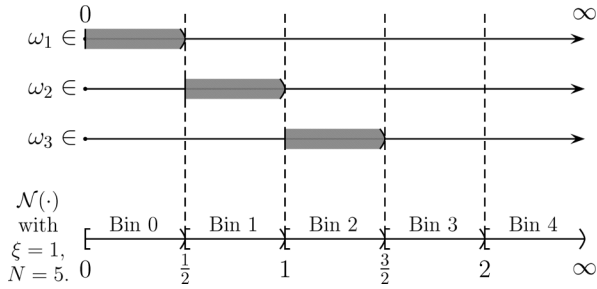


Fig. 5. Possible locations of the NSNRs according to the LQ Decoder.

determine which one of the second and the third relays provides the best NSNR. Therefore, it chooses one of them, and its decision may not be the optimal one that would instead be provided by the GQ. We shall quantify the effect of such suboptimal decisions later on.

6) *LQ Decoder*: We now give the general and formal description of the LQ decoder.

Let $\mathcal{R}_g \triangleq \{q : \min_{\ell} \omega_{q\ell} = \max_r \min_{\ell} \omega_{r\ell}\}$ denote the set of indices from which our GQ in (20) produces its output.¹² In other words, $\mathcal{R}_g$ is the set of indices of relays that provide the maximal NSNR. Also, let $\mathcal{R}_l \triangleq \{q : \mathcal{N}(\min_{\ell} \omega_{q\ell}) = \mathcal{N}(\max_r \min_{\ell} \omega_{r\ell})\}$. Note that $\mathcal{R}_g \subset \mathcal{R}_l$. Moreover, due to the structure of $\mathcal{N}$, not only

$$\mathcal{N}(\min_{\ell} \omega_{q\ell}) = \min_{\ell} \mathcal{N}(\omega_{q\ell}),$$

but also

$$\begin{aligned} \mathcal{N}(\max_r \min_{\ell} \omega_{r\ell}) &= \max_r \mathcal{N}(\min_{\ell} \omega_{r\ell}) \\ &= \max_r \min_{\ell} \mathcal{N}(\omega_{r\ell}). \end{aligned}$$

Therefore, $\mathcal{R}_l = \{q : \min_{\ell} \mathcal{N}(\omega_{q\ell}) = \max_r \min_{\ell} \mathcal{N}(\omega_{r\ell})\}$, and $\mathcal{R}_l$ can be easily calculated by the LQ decoder.

Since $\mathcal{R}_g \subset \mathcal{R}_l$, the LQ decoder can determine which relay selection vector(s) can possibly provide the maximal NSNR. In

¹² $\mathcal{R}_g$ is not necessarily a singleton, but our definition of the $\arg \max$ guarantees that the GQ output is unique.

general, it can choose any one of the relay selection vectors that are indicated by $\mathcal{R}_\ell$. But, to be more precise, we define

$$\text{LQ}_{\xi, N}^{(\mathbf{f}|\mathbf{v})}(\mathbf{h}) \triangleq \arg \max_{\mathbf{e}_r \in \mathcal{C}_s} \min_{\ell} \mathcal{N}(\omega_{r\ell}).$$

7) *Localization Distortion*: Let us now study two possible cases of interest regarding the LQ output: If $\mathcal{R}_g = \mathcal{R}_l$, then the LQ output provides the same NSNR as the GQ output. Otherwise, the LQ might make a suboptimal decision. This results in what we call the *localization distortion*, given by

$$\text{LD}(\xi, N) \triangleq \text{NER}(\text{LQ}_{\xi, N}^{(\mathbf{f}|\mathbf{v})}) - \text{NER}(\text{GQ}_{\mathcal{C}_s}). \quad (28)$$

It is very difficult to find a closed-form expression for the exact value of the localization distortion. We shall later see that instead, upper bounds on the localization distortion are good enough for our purposes.

We can now analyze the performance of our LQs that are synthesized using the localization method. First, we consider the simpler case of fLQs where the number of feedback bits per receiver per channel state remains fixed.

C. Maximal First-Order Diversity With an fLQ

Regardless of the forwarding strategy, for our fLQ, we set $\xi_{\mathbf{f}} = R \log P$ and $N_{\mathbf{f}} = 2$. This means that the corresponding fixed-length local quantizer $\text{LQ}_{\xi_{\mathbf{f}}, N_{\mathbf{f}}}^{\mathbf{f}}$ uses a fixed R feedback bits per receiver per channel state. We first analyze the performance of such an fLQ. Later in this section, we will discuss the reasoning behind our choice of the parameters $\xi_{\mathbf{f}}$ and $N_{\mathbf{f}}$.

The performance of our fLQ for AF relays is given by the following theorem:

Theorem 4: For AF relays, the NER with $\text{LQ}_{\xi_{\mathbf{f}}, N_{\mathbf{f}}}^{\mathbf{f}}$ is upper bounded by

$$\text{NER}(\text{LQ}_{\xi_{\mathbf{f}}, N_{\mathbf{f}}}^{\mathbf{f}}) \leq \mathcal{C}_{11} \frac{\log^R P}{P^R}, \quad K = 1, \quad (29)$$

$$\text{NER}(\text{LQ}_{\xi_{\mathbf{f}}, N_{\mathbf{f}}}^{\mathbf{f}}) \leq \mathcal{C}_{12} \frac{\log^{2R} P}{P^R}, \quad K > 1. \quad (30)$$

for all P sufficiently large. The constants $0 < \mathcal{C}_{11}, \mathcal{C}_{12} < \infty$ are independent of P .

Proof: See Appendix E. ■

In other words, using a fixed R feedback bits per receiver per channel state, we can achieve diversity $(R, -R)$ for $K = 1$, and diversity $(R, -2R)$ for $K > 1$. Comparing with Theorem 2, since $(R, -R) < (R, 0)$ for $K = 1$, and $(R, -2R) < (R, -R)$ for $K > 1$, our fLQ has a second-order diversity loss compared to the optimal performance for any K . Also, it is straightforward to show that, using R_0 bits, where $R_0 \in \{1, \dots, R\}$, we can achieve diversity gains $(R_0, -R_0)$ and $(R_0, -2R_0)$ for $K = 1$ and for $K > 1$, respectively.

With DF relays, the performance of our fLQ is analogous to the case with AF relays:

Theorem 5: For DF relays, the NER with LQ_{ξ_f, N_f}^f is upper bounded by

$$\text{NER}(\text{LQ}_{\xi_f, N_f}^f) \leq C_{13} \frac{\log^R P}{P^R}, \quad (31)$$

for all P sufficiently large. The constant $0 < C_{13} < \infty$ is independent of P .

Proof: The proof relies on the same steps as the proof of Theorem 4 in Appendix E. We thus skip it for brevity. ■

Therefore, for any K , our fLQ can achieve a diversity of $(R, -R)$ with DF relays, which means a second-order diversity loss of R compared to the one achievable by the GQ in Theorem 3. Similarly, using R_0 bits, where $R_0 \in \{1, \dots, R\}$, we can achieve a diversity gain of $(R_0, -R_0)$.

For both AF and DF strategies, we have thus constructed an fLQ that uses a fixed R feedback bits per receiver per channel state. Our fLQ achieves maximal first-order diversity, but it incurs a second-order diversity loss of R compared to the best possible diversity gain that the forwarding strategy can provide.

We now give an intuitive reasoning that is behind our choice of the parameters $\xi_f = R \log P$ and $N_f = 2$. Note that in general, by the description in Section V-B-III, an fLQ uses a fixed $\lceil R \log_2 N_f \rceil$ feedback bits per receiver per channel state. Since the case $N_f = 1$ corresponds to an open loop system with no feedback (and thus no diversity gain), we need at least $N_f = 2$ in order to benefit from the feedback information.

In order to motivate the choice of ξ_f , let us consider an AF network with $K = 1$ and $R = 2$. We later generalize the same intuition to any R and K . The number of receivers is irrelevant as far as our discussion here is concerned.

For an AF network with $K = 1$ and $R = 2$, Theorem 2 tells us that the GQ can achieve the maximum possible diversity gain of $(2, 0)$. Such a performance is possible as the GQ can always select the best relay that maximizes the NSNR. The fLQ however, as we have explained in greater detail in Sections V-B-II through V-B-VI, can only determine whether the NSNR provided by a particular relay is in $[0, \xi_f)$ or in $[\xi_f, \infty)$. In other words, letting γ_1 and γ_2 to respectively denote the NSNR provided by the first and the second relay, the fLQ can only distinguish between the following events:

$$E_1 : \gamma_1 \in [0, \xi_f), \gamma_2 \in [\xi_f, \infty),$$

$$E_2 : \gamma_1 \in [\xi_f, \infty), \gamma_2 \in [0, \xi_f),$$

$$E_3 : \gamma_1 \in [\xi_f, \infty), \gamma_2 \in [\xi_f, \infty)$$

$$E_4 : \gamma_1 \in [0, \xi_f), \gamma_2 \in [0, \xi_f).$$

Now, let NER denote the NER with the fLQ. Also, for $i \in \{1, \dots, 4\}$, let NER_i denote the contribution of event E_i to NER. We have

$$\text{NER} = \text{NER}_1 + \dots + \text{NER}_4. \quad (32)$$

If either E_1 or E_2 occurs, the fLQ can perfectly determine which relay provides the maximum NSNR and operates in the same manner as the GQ. The contribution of these events to the NER cannot be more than the overall NER of the GQ, and thus, according to (21), we have

$$\text{NER}_1 + \text{NER}_2 \leq \frac{C_5}{P^2}. \quad (33)$$

The proper choice of ξ_f is crucial rather in the event of E_3 or E_4 . Here, the main intuition behind setting $\xi_f = R \log P$ is to avoid the worst case scenarios. To see how this is done, let us first consider the CNER given E_3 . In this case, the fLQ cannot distinguish which one of the relays provides the best NSNR, and therefore the NSNR can be as low as $\min\{\gamma_1, \gamma_2\} \geq \xi_f$. According to Proposition 2, the CNER given E_3 is at most $C_4 \exp(-\min\{\gamma_1, \gamma_2\}) \leq C_4 \exp(-\xi_f)$. Therefore,

$$\begin{aligned} \text{NER}_3 &\leq C_4 \exp(-\xi_f) P(E_3) \\ &\leq C_4 \exp(-\xi_f). \end{aligned} \quad (34)$$

The final upper bound thus represents the worst case scenario regarding the event E_3 .

For event E_4 , let us assume for the sake of simplicity that the source-to-relay links are noiseless, in which case γ_1 and γ_2 follow an exponential distribution, i.e., $f(\gamma_i) = \frac{1}{P} \exp(-\frac{\gamma_i}{P})$, $i = 1, 2$.¹³ Then, $P(E_4) = (1 - \exp(-\frac{\xi_f}{P}))^2 \leq \frac{\xi_f^2}{P^2}$ as $1 - e^{-x} \leq x$, $\forall x \geq 0$, and the NSNR provided by the fLQ is at least $\min\{\gamma_1, \gamma_2\} \geq 0$. Therefore,

$$\begin{aligned} \text{NER}_4 &\leq C_4 \exp(-0) P(E_4) \\ &\leq C_4 \frac{\xi_f^2}{P^2}. \end{aligned} \quad (35)$$

Substituting our worst case upper bounds (33), (34), and (35) to (32), we obtain

$$\text{NER} \leq \frac{C_5}{P^2} + C_4 \left(\exp(-\xi_f) + \frac{\xi_f^2}{P^2} \right). \quad (36)$$

We now need to optimize (36) with an appropriate choice of ξ_f . Note that the term inside the parenthesis is a concave function of ξ_f and thus has a unique global minimum ξ_f^* given by

$$\frac{\partial}{\partial \xi_f} \left(\exp(-\xi_f) + \frac{\xi_f^2}{P^2} \right) \Big|_{\xi_f^*} = 0 \implies \xi_f^* \exp(\xi_f^*) = \frac{P^2}{2}.$$

¹³Indeed, Proposition 3 in Appendix B shows that the actual PDFs also follow “roughly” an exponential distribution.

The solution is $\xi_f^* = 2 \log P - o(\log P)$, where $o(\log P)$ represents a function with $\lim_{P \rightarrow \infty} \frac{o(\log P)}{\log P} = 0$.¹⁴ Our choice of $\xi_f = 2 \log P$ is thus an “asymptotically optimal” solution that minimizes the worst case upper bound on the NER given by (36). Substituting $\xi_f = 2 \log P$ to (36), we can observe that NER decays no slower than $\frac{\log^2 P}{P^2}$, in agreement with Theorem 4.

The proof of Theorem 4 is a formal generalization of the above arguments to any $R > 2$ and $K > 1$. Similarly, there are two dominating error events:

- The NSNR provided by at least two of the relays is in the interval $[\xi_f, \infty)$.
- The NSNR provided by all of the relays is in the interval $[0, \xi_f)$.

The only difference is that we do not resort to any approximations regarding the PDFs of the NSNRs. Still, we first calculate a reasonably tight upper bound on the NER with the fLQ. In the process of such a calculation, we take into account specifically the worst case scenarios that result in high NERs. In the final result, we observe that for any R, K , and L , and for both AF and DF forwarding strategies, the choice of $\xi_f = R \log P$ provides a high diversity performance.

As an obvious alternative approach for a better performance, one may instead calculate the exact NER as a function of ξ_f (e.g., $\text{NER}(\xi_f)$) and optimize ξ_f accordingly. However, putting the optimization of ξ_f aside, finding an analytical expression for the exact $\text{NER}(\xi_f)$ is by itself very difficult, if not impossible. Nevertheless, an interesting open problem in this context is to determine whether a better choice of ξ_f could improve the second order diversity performance of our fLQs. More generally, given an arbitrary but fixed number of feedback bits per receiver per channel state, determining the best possible diversity gain and the question of how to achieve it both remain challenging open problems. Our fLQ design solves these problems, in part, by providing the maximal first order diversity gain of R with only R bits.

Fortunately, as we shall demonstrate in what follows, we can provably achieve the maximal diversity gain using instead a vLQ. Such a performance is made possible by appropriately increasing the scalar quantizer resolution with P instead of constraining ourselves to the fixed $\log_2 N_f = 1$ bit per local NSNR resolution of the fLQ. Despite the fact that the resolution of the vLQ thus grows without bound, surprisingly, the compressors make sure that the feedback rate remains bounded.

D. Maximal Diversity With a vLQ

Before showing that we can achieve the maximal diversity gain with a vLQ, we need to specify the parameters of the corresponding scalar quantizers. Let $\Lambda(\cdot)$ be an arbitrary positive real-valued function that satisfies the following properties for all P sufficiently large:

- 1) $\exists \epsilon > 0$ such that $\Lambda(P) \geq \epsilon$.
- 2) $\exists \Lambda_0 > 0, \Lambda_0 < \infty$ such that $\Lambda(P) \leq \Lambda_0 P$.

¹⁴For a formal proof, note that we seek to find the real number x that satisfies of $f(x) = xe^x - \frac{P^2}{2} = 0$. For sufficiently large P , we observe that $f(2 \log P) > 0$ and $f(2 \log(\frac{P}{\log P})) < 0$. Since $f(\cdot)$ is a monotonically increasing function, it follows that $2 \log(\frac{P}{\log P}) < x < 2 \log P$, or equivalently $x = 2 \log P - o(\log P)$.

As an example, for an arbitrary $\lambda > 0$, we can choose $\Lambda(P) = \lambda, \forall P$. Similarly, for an arbitrary $\lambda' > 0$, we can choose $\Lambda(P) = \lambda' P$. Both choices satisfy the properties listed above.

We now set

$$\xi_v = \frac{1}{\Lambda(P)}, \quad (37)$$

and

$$N_v = \lceil \Lambda(P) \log \Lambda(P) + R \Lambda(P) \log P \rceil. \quad (38)$$

Note that, since $\Lambda(P) \geq \epsilon$ and N_v is an increasing function of P , we have $N_v \geq 1$ when P is sufficiently large. This makes sure that the scalar quantizers have at least one bin and thus they are well-defined.

With the above choices of the scalar quantizer resolution and cell width, the performance of our vLQ with AF relays is given by the following theorem:

Theorem 6: For AF relays and P sufficiently large, we have

$$\text{LD}(\xi_v, N_v) \leq C_{14} \frac{1}{\Lambda(P) P^R}, \quad K = 1, \quad (39)$$

$$\text{LD}(\xi_v, N_v) \leq C_{15} \frac{\log^R P}{\Lambda(P) P^R}, \quad K > 1, \quad (40)$$

and, in addition, the feedback rate of the ℓ th receiver satisfies

$$R_\ell(\text{LQ}_{\xi_v, N_v}^v) \leq C_{16} \frac{\log^2 P}{P}, \quad K = 1, \quad (41)$$

$$R_\ell(\text{LQ}_{\xi_v, N_v}^v) \leq C_{17} \frac{\log^3 P}{P}, \quad K > 1, \quad (42)$$

where $0 < C_{14}, C_{15}, C_{16}, C_{17} < \infty$ are constants that are independent of P and $\Lambda(P)$.

Proof: See Appendix F. ■

We now describe several consequences of this theorem for $K > 1$. The consequences for $K = 1$ will be analogous.

Let us first recall from (28) that $\text{NER}(\text{LQ}_{\xi_v, N_v}^v) = \text{NER}(\text{GQ}_{C_s}) + \text{LD}(\xi_v, N_v)$. We have found an upper bound for $\text{NER}(\text{GQ}_{C_s})$ in Theorem 2. An upper bound for $\text{LD}(\xi_v, N_v)$ is given by Theorem 6. Combining the two bounds and setting $\Lambda(P) = P$, we have

$$\text{NER}(\text{LQ}_{\xi_v, N_v}^v) \leq \left(C_6 + \frac{C_{15}}{P} \right) \frac{\log^R P}{P^R} \quad (43)$$

$$\leq (C_6 + C_{15}) \frac{\log^R P}{P^R}, \quad (44)$$

where (44) follows from (43) whenever $P \geq 1$. The upper bound on the NER given by (44) implies that $d(\text{LQ}_{\xi_v, N_v}^v) \geq (R, -R)$. On the other hand, according to Theorem 2, the diversity gain with any quantizer for the AF strategy is upper bounded by $(R, -R)$. Therefore, $d(\text{LQ}_{\xi_v, N_v}^v) = (R, -R)$, and our vLQ indeed achieves the maximal diversity gain that the AF scheme can provide.

Moreover, using the same arguments as in the previous paragraph, we have

$$\text{NER}(\text{LQ}_{\xi_v, N_v}^v) \leq \text{NER}(\text{GQ}_{C_s}) + C_{15} \frac{\log^R P}{P^{R+1}}. \quad (45)$$

Note that the $\frac{\log^R P}{P^{R+1}}$ decay is much faster than the $\frac{\log^R P}{P^R}$ decay provided by GQ_{C_s} . Thus, the array gain of our vLQ is the same as the one provided by GQ_{C_s} .

What is more interesting is the behavior of the upper bound on the feedback rate given by (41) and (42). As P grows to infinity, the required feedback rate decays to zero. In other words, both the diversity and array gain benefits of $\text{NER}(\text{GQ}_{C_s})$ can be achieved using arbitrarily low feedback rates, when P is sufficiently large.

Let us now determine the performance of our vLQs for the DF strategy by the following theorem:

Theorem 7: For DF relays and P sufficiently large, we have

$$\text{LD}(\xi_v, N_v) \leq C_{18} \frac{1}{\Lambda(P)P^R}, \quad (46)$$

and, in addition, the feedback rate of the ℓ th receiver satisfies

$$R_\ell(\text{LQ}_{\xi_v, N_v}^v) \leq C_{19} \frac{\log^2 P}{P}, \quad (47)$$

where $0 < C_{18}, C_{19} < \infty$ are constants that are independent of P and $\Lambda(P)$.

Proof: The proof relies on the same steps as the proof of Theorem 6 in Appendix F. We thus skip it for brevity. ■

The interpretation of Theorem 7 is analogous to the case with AF relays. The full diversity gain of $(R, 0)$ can be achieved using DF relays with arbitrarily low feedback rates per receiver for sufficiently large P .

The best possible diversity gain for any scheme with an arbitrary block length and full CSI at the relays is $(R, 0)$ according to Theorem 1. We have thus shown by Theorem 7 that we can achieve this diversity gain with a two-channel use DF protocol using distributed decision making with asymptotically zero feedback rate per receiver.

VI. SIMULATION RESULTS

In this section, we present numerical evidence that verifies our analytical results. We assume that each receiver attempts to decode all the symbols from all the transmitters. In other words, $\mathcal{D}_\ell = \{1, \dots, K\}$, $\forall \ell$. In the graphs, “GQ” represents GQ_{C_s} in (20), “fLQ” denotes LQ_{ξ_f, N_f}^f with $\xi_f = R \log P$ and $N_f = 2$ as defined in the beginning of Section V-C. Also, “vLQ, $\Lambda(P) = \Lambda'(P)$ ” represents LQ_{ξ_v, N_v}^v with $\xi_v = \frac{1}{\Lambda'(P)}$ and $N_v = \lceil \Lambda'(P) \log \Lambda'(P) + R \Lambda'(P) \log P \rceil$ as defined in the beginning of Section V-D. We consider the choices $\Lambda'(P) = 1$, $\forall P$, and $\Lambda'(P) = P$.

A. Networks With Equal Parameters

In Fig. 6, we show the performance results for a network with $K = R = L = 2$, $\sigma_{f_{r,k}}^2 = \sigma_{g_{r,\ell}}^2 = p_{R_r} = p_{S_k} = 1$, $\forall r, k, \ell$, and $\mathcal{S}_1 = \mathcal{S}_2 = \{+1, -1\}$. For this network, the NERs with the GQ, fLQ, and vLQs for $\Lambda(P) = 1$, P is presented in Fig. 6(a). The horizontal and the vertical axes represent P in decibels (dBs), and the NER, respectively.

We can observe that for AF relays both our GQ and vLQs achieve the maximal diversity gain $(2, -2)$ that the AF scheme can provide, while the fLQ achieves diversity $(2, -4)$. For DF relays, both our GQ and vLQs achieve the full diversity gain

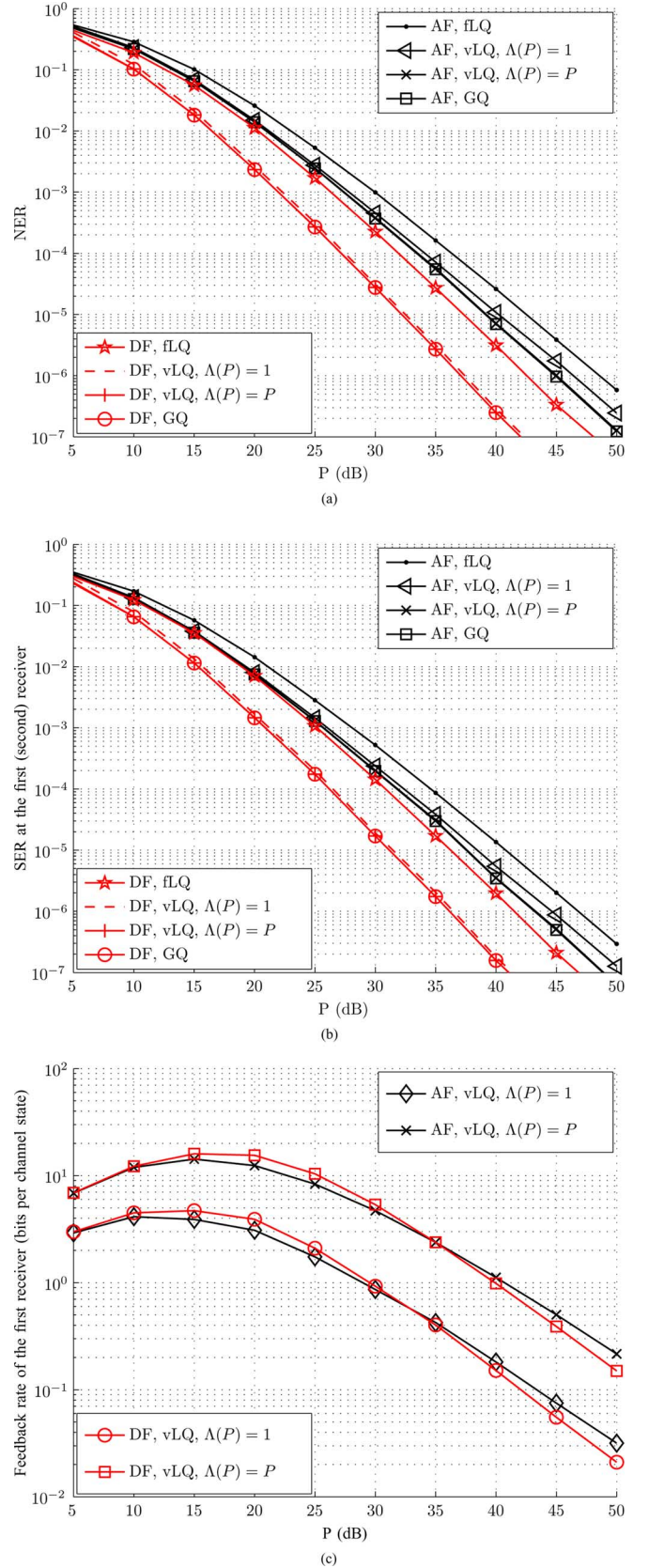


Fig. 6. Performance results for a network with $K = R = L = 2$ (a) NERs (b) SERs (c) Feedback rates.

$(2, 0)$, while the fLQ achieves diversity $(2, -2)$. For the choice of $\Lambda(P) = P$, we expect, as a result of Theorems 6 and 7,

that our vLQs achieve the same array gain as the GQ. Indeed, even for the finite values of P , the performance of the vLQ with $\Lambda(P) = P$ is very close to that of the GQ. For AF relays, the difference is less than 0.1 dB at an NER of 10^{-7} . For DF relays, within the numerical limits of the simulations, we have observed that the two achieve the same performance for $P = 30, 35, 40$, and 45 dB. Even for the choice of $\Lambda(P) = 1, \forall P$, the performance of the vLQ is close to that of the GQ; at an NER of 10^{-6} , the difference is around 1 dB and 0.2 dB, for the AF and DF strategies, respectively.

In Fig. 6(b), we show the SERs at the first receiver for the same network. The horizontal axis represents P in decibels, while the vertical axis represents the SER at the first (second) receiver. As a result of our choice of network parameters, the SERs of different receivers are the same. Also, a particular quantizer achieves the same diversity as in Fig. 6(a). On the other hand, since the SER is upper bounded by the NER, any quantizer in Fig. 6(b) provides more array gain than it does in Fig. 6(a). Indeed, due to the symmetry of the network parameters, the SER performance is around 1.5 dB better than the NER performance for all quantizers.

The corresponding feedback rates of our vLQs are shown in Fig. 6(c). The horizontal axis represents P in decibels, while the vertical axis represents the feedback rate of the first receiver in bits per channel state. Similarly, due to our choice of the network parameters, the feedback rates of each receiver will be the same. We can observe the validity of Theorem 6, as for both choices of $\Lambda(P) = 1, P$, the required feedback rate decays to zero at high P . As expected, the choice of $\Lambda(P) = P$, which will ensure that the vLQ achieves the same array gain as the GQ, requires a higher feedback rate compared to the choice of $\Lambda(P) = 1$. In other words, there is a tradeoff between performance and feedback rate. An important question is to determine the required amount of feedback rate for a certain NER requirement.

Suppose that for DF relays at 35 dB, we would like to achieve an NER of 10^{-5} . According to Fig. 6(a) and (c), at 35 dB and for the choice of $\Lambda(P) = 1$, our vLQ can achieve an NER of 3.18×10^{-6} with a feedback rate of 0.40 bits per receiver per channel state. Clearly, the NER of 3.18×10^{-6} is much less than the target 10^{-5} . One can achieve the target NER of 10^{-5} using a lesser amount of feedback by an adjustment $\Lambda(35 \text{ dB}) = \lambda_0$ for some $0 < \lambda_0 < 1$. We have numerically determined that a choice of $\Lambda(35 \text{ dB}) = 0.17241$ yields an NER of 9.92×10^{-6} , which satisfies the target 10^{-5} and requires 0.11 bits per receiver per channel state. Similarly, for DF relays at 30 dB, if our goal is to achieve an NER of 10^{-4} instead of the NER of 3.11×10^{-5} provided by the choice $\Lambda(30 \text{ dB}) = 1$ with 0.93 bits per receiver per channel state, we can choose $\Lambda(30 \text{ dB}) = 0.16807$ which yields an NER of 9.98×10^{-5} with 0.34 bits per receiver per channel state. These examples demonstrate the flexibility of our vLQ design.

The (very) difficult question in this context is to determine the exact value of $\Lambda(P)$ that will yield a given target NER at power level P . We do not have a complete answer to this question other than proposing the sheer experimentation with different values of $\Lambda(P)$ as explained above. On the other hand, we would like also to note that numerical optimization is typical for variable-length quantizers. In other words, our method of choosing $\Lambda(P)$

for a target NER through numerical search is similar to any other variable-length quantizer design. A typical practice is to design off-line quantizers for different values of design parameters (in our case, the design parameter is $\Lambda(P)$) and pick the one that fits the requirement [71]. Note that the time-consuming design process is done off-line and after the quantizer is designed, the complexity of the encoding/decoding operation is the same for different quantizers.

B. Networks With Unequal Parameters

Our results also hold for networks with unequal power constraints and/or channel variances. To demonstrate that, we consider a network with $K = R = 3$ and $L = 4$. The parameters of the network are assumed to be $p_{S_1} = 1, p_{S_2} = 1.3, p_{S_3} = 0.7, p_{R_1} = 0.6, p_{R_2} = 2, p_{R_3} = 0.7, \mathcal{S}_1 = \mathcal{S}_3 = \{+1, -1\}$, and $\mathcal{S}_2 = \{e^{j\frac{\pi}{4}}, e^{j\frac{3\pi}{4}}, e^{j\frac{5\pi}{4}}, e^{j\frac{7\pi}{4}}\}$. Also, we assume that $\sigma_{f_{kr}}^2 = F_{kr}, \sigma_{g_{r\ell}}^2 = G_{r\ell}, k = 1, \dots, K, r = 1, \dots, R, \ell = 1, \dots, L$, where

$$\mathbf{F} = \begin{bmatrix} 2 & 1 & 0.7 \\ 1.5 & 0.9 & 3 \\ 1 & 4 & 0.5 \end{bmatrix}, \quad (48)$$

and

$$\mathbf{G} = \begin{bmatrix} 7 & 1.2 & 2.5 & 0.9 \\ 0.4 & 1.3 & 3 & 2 \\ 1.3 & 0.9 & 1.6 & 5 \end{bmatrix}. \quad (49)$$

In Fig. 7(a), we show the NERs of the GQ, fLQ, and vLQs for $\Lambda(P) = 1, P$. The results are analogous to what we have observed in Fig. 6(a). For AF relays, both the GQ and the vLQs achieve the maximal diversity gain $(3, -3)$ that the AF scheme can provide, while the fLQ achieves diversity $(3, -6)$. For DF relays, both the GQ and the vLQs achieve the full diversity gain $(3, 0)$, while the fLQ achieves diversity $(3, -3)$. For $\Lambda(P) = P$ and both forwarding strategies, the array gain performance of our vLQ and the GQ is the same.

The SERs at the second and the fourth receiver are shown in Fig. 7(b) and (c), respectively. We can observe that, unlike the previous network with equal parameters, the SERs at different receivers are different for this network with unequal parameters.

For the vLQs, the feedback rates of the second and the fourth receivers are shown in Fig. 7(d) and (e). For both figures, the feedback rates decay to zero as P grows to infinity, verifying Theorem 6.

VII. EXTENSIONS TO NETWORKS WITH DIRECT LINKS

Our results in the previous sections were based on the assumption that there are no direct links between the transmitters and the receivers. In general, direct links may exist between one or more of the transmitter-receiver pairs. These direct links can be exploited to improve the achievable diversity gain.

In this section, we consider a scenario where a direct link exists between each transmitter-receiver pair, and provide the corresponding extensions of our results. Such extensions are indeed straightforward using the tools that we have already developed throughout the paper.

As a first result for networks with a direct link, it can be shown that the NER with any coding scheme is lower bounded

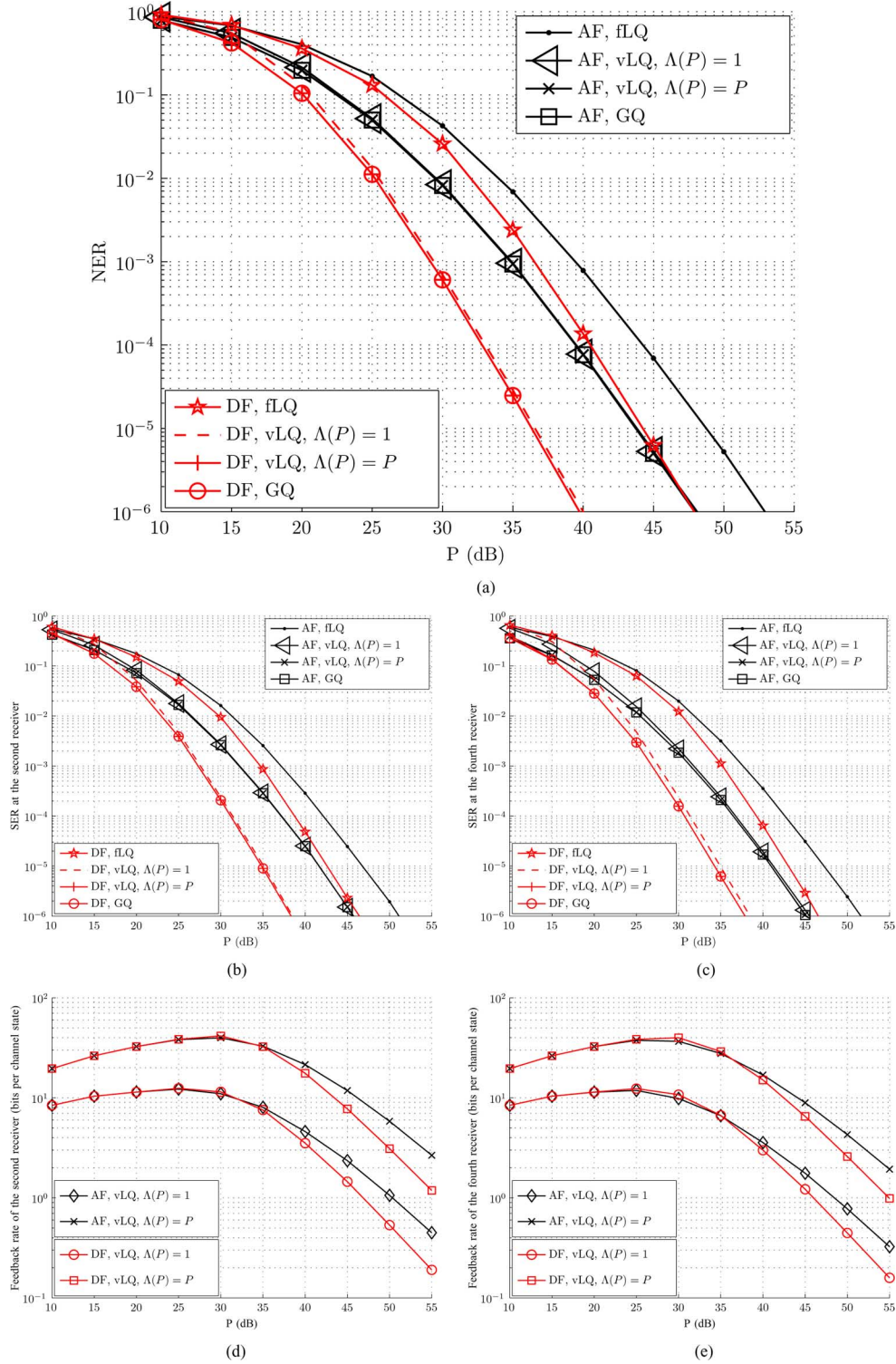


Fig. 7. Performance results for a network with $K = R = 3$, $L = 4$ (a) NERs (b) SERs at the second receiver (c) SERs at the fourth receiver (d) Feedback rates at the second receiver (e) Feedback rates at the fourth receiver.

by $C_{20}P^{-R-1}$, for P sufficiently large and a constant C_{20} that is independent of P . This argument holds under the same general assumptions in the statement of Theorem 1. The proof relies on the same standard arguments that we have discussed in the proof of Theorem 1. Therefore, the maximal diversity gain for networks with direct links is $(R + 1, 0)$. In order to achieve this diversity gain, we first need to generalize the data transmission schemes in Section II so as to make use of the direct links.

A. Data Transmission Schemes for Networks With Direct Links

Let us recall that for both AF and DF strategies and networks without a direct link, we have employed a two-step data transmission protocol. These two step protocols can easily be extended to networks with direct links. Let $d_{k\ell} \sim \text{CN}(0, \sigma_{d_{k\ell}}^2)$, $0 < \sigma_{d_{k\ell}}^2 < \infty$ denote the direct link channel from the k th transmitter to the ℓ th receiver. We assume that the components

of the channel state of the network $\bar{\mathbf{h}} \triangleq [d_{11} \cdots d_{KL} \mathbf{h}]$ are independent.

In general, the transmitters can use the direct links for both transmission steps. To be more precise, if the k th transmitter wishes to convey a symbol $s_k \in \mathcal{S}_k$ to one or more of the receivers, in the most general case and regardless of the forwarding strategy, it can send $\alpha_k s_k \sqrt{P_{S_k}}$ and $\beta_k s_k \sqrt{P_{S_k}}$ in the first and the second steps, respectively. We require $|\alpha_k|^2 + |\beta_k|^2 \leq 1, \forall k$ due to the short-term power constraint of the transmitters. In practice, the factors $\alpha_k, \beta_k, k = 1, \dots, K$ need to be optimized to minimize the NER with a specific quantizer employed in the network.

The first step of the transmission scheme is the same for both AF and DF strategies. As we have mentioned, in the first step, the k th transmitter sends $\alpha_k s_k \sqrt{P_{S_k}}$. The r th relay and the ℓ th destination receives

$$\bar{t}_r = \sum_{k=1}^K \alpha_k f_{kr} s_k \sqrt{P_{S_k}} + \eta_{0r}, \quad (50)$$

and

$$\bar{y}_{0\ell} = \sum_{k=1}^K \alpha_k d_{k\ell} s_k \sqrt{P_{S_k}} + \eta_{1\ell}, \quad (51)$$

respectively. The terms η_{0r} and $\eta_{1\ell}$ represent the noise at the r th relay and the ℓ th receiver, respectively.

For the second step, the k th transmitter sends $\beta_k s_k \sqrt{P_{S_k}}$. For a quantizer $\mathbf{Q} : \mathbb{C}^{R(K+L)} \rightarrow \mathcal{C}$ in the network, suppose that $\mathbf{Q}(\mathbf{h}) = \mathbf{x}$ for some $\mathbf{x} \in \mathcal{C}$. Then, with AF relays, the r th relay transmits $\bar{u}_r = x_r \sqrt{\bar{\rho}_r} \bar{t}_r$, where the relay normalization factor $\bar{\rho}_r$ is now given by

$$\bar{\rho}_r \triangleq \frac{P_{R_r}}{1 + \mathbb{E}_{s_1, \dots, s_K} \left[\left| \sum_{k=1}^K f_{kr} \alpha_k s_k \sqrt{P_{S_k}} \right|^2 \right]}. \quad (52)$$

The received signal at the ℓ th receiver is thus

$$\bar{y}_{1\ell} = \sum_{k=1}^K \left(\beta_k d_{k\ell} + \sum_{r=1}^R x_r \sqrt{\bar{\rho}_r} \alpha_k f_{kr} g_{r\ell} \right) \sqrt{P_{S_k}} s_k + \sum_{r=1}^R x_r g_{r\ell} \sqrt{\bar{\rho}_r} \eta_{0r} + \eta_{2\ell}, \quad (53)$$

where $\eta_{2\ell} \sim \text{CN}(0, 1)$. We assume that all the noise terms and the channels are mutually independent as usual.

For DF relays, the r th relay transmits

$$\bar{u}_r = x_r \sqrt{P_{R_r}} \text{RENC}(\text{DEC}_r(\bar{t}_r, \mathbf{f}_r)), \quad (54)$$

where the reencoding and decoding operations $\text{RENC}(\cdot)$ and $\text{DEC}_r(\cdot, \cdot)$ are as defined in Section II-D. The received signal at the ℓ th receiver can be expressed as

$$\bar{y}_{1\ell} = \sum_{k=1}^K \beta_k d_{k\ell} \sqrt{P_{S_k}} s_k + \sum_{r=1}^R x_r \sqrt{P_{R_r}} \text{RENC}(\text{DEC}_r(\bar{t}_r, \mathbf{f}_r)) g_{r\ell} + \eta_{2\ell}. \quad (55)$$

The NER and the generalized diversity measure definitions can easily be extended to direct-linked networks without any

modification. What remains is to determine the maximal diversity gains for each forwarding scheme and show how to achieve these gains.

B. Maximal Diversity Gains in Networks With Direct Links and Their Achievability Using Global or Local Quantizers

We first determine the maximal diversity gain for both AF and DF strategies. Compared to networks with no direct link, there exists one extra independently fading path between each transmitter-receiver pair in a network with direct links. Intuitively, this extra path adds at most a diversity gain of $(1, 0)$ to the maximal diversity gains indicated by Theorems 2 and 3. We can therefore conclude that the maximal diversity gain with AF relays is $(R+1, 0)$ if $K = 1$, and $(R+1, -R)$ if $K > 1$. Similarly, for DF relays, the maximal diversity gain is $(R+1, 0)$ for any K . We provide a formal proof of these bounds in Appendix G-A.

We now describe how to achieve these maximal diversity gains. Consider now any one of the relay-selection based quantizers $\mathbf{Q}_{\text{RS}} : \mathbb{C}^{R(K+L)} \rightarrow \mathcal{C}_{\text{RS}}$ that we have discussed in the previous sections for networks without a direct link. In particular, for quantizers that can provide the maximal first-order diversity gain, we have proved that $d(\mathbf{Q}_{\text{RS}}) = (R, d_2)$ for some $d_2 \in \{0, -R, -2R\}$. The exact value of d_2 depends on the specific quantizer that is used and the forwarding strategy (e.g., for an fLQ with the AF strategy, we have $d_2 = -2R$).

We now show that for networks with a direct link, we can indeed achieve a diversity gain of $(R+1, d_2)$ with the same quantizer $\mathbf{Q}_{\text{RS}}$. We consider the employment of the direct link in the first step only, and therefore set $\alpha_k = 1, \beta_k = 0, \forall k$. Imagine now two scenarios: In the first one, the receivers disregard their received signals from the direct links in the first step. Such a scenario is equivalent to a network without the direct links, and therefore we can obviously achieve a diversity gain of $d(\mathbf{Q}_{\text{RS}}) = (R, d_2)$. In the second scenario, the receivers disregard the relay signals in the second step and attempt to decode their desired symbols using what they receive in the first step only. In Appendix G-B, we show that in this case, the conditional NER is upper bounded by

$$\mathcal{C}_{21} \max_{\ell} \exp(-\gamma_{\ell, \text{DL}}(\mathbf{d}_{\ell})), \quad (56)$$

where $0 < \mathcal{C}_{21} < \infty$ is a constant, $\mathbf{d}_{\ell} \triangleq [d_{1\ell} \cdots d_{K\ell}]$, and

$$\gamma_{\ell, \text{DL}}(\mathbf{d}_{\ell}) \triangleq \frac{1}{4} \min_{\mathbf{s} \neq \hat{\mathbf{s}}} \left| \sum_{k=1}^K (s_k - \hat{s}_k) \sqrt{P_{S_k}} d_{k\ell} \right|^2 \quad (57)$$

represents the NSNR of the direct link to the ℓ th receiver. We also show in Appendix G-B that this yields the upper bound $\mathcal{C}_{22} P^{-1}$ on the NER, where $0 < \mathcal{C}_{22} < \infty$ is a constant that is independent of P . Hence, using only the direct links provides a diversity gain of $(1, 0)$.

In order to achieve a diversity gain of $(R+1, d_2)$, we fuse the two aforementioned scenarios together simply by selection combining at the receivers. Let $\gamma_{\ell}(\mathbf{Q}_{\text{RS}}(\mathbf{h}), \mathbf{h})$ denote the NSNR at the ℓ th receiver for the first scenario as defined in Proposition 2. Note that the ℓ th receiver can calculate both $\gamma_{\ell}(\mathbf{Q}_{\text{RS}}(\mathbf{h}), \mathbf{h})$ and $\gamma_{\ell, \text{DL}}(\mathbf{d}_{\ell})$ perfectly. If $\gamma_{\ell}(\mathbf{Q}_{\text{RS}}(\mathbf{h}), \mathbf{h}) < \gamma_{\ell, \text{DL}}(\mathbf{d}_{\ell})$, the ℓ th receiver uses only the received signals in the first step in order

to decode its desired symbols. Otherwise, it uses only the second step signals from the relays. The NSNR at the ℓ th receiver is thus $\max\{\gamma_\ell(Q_{RS}(\mathbf{h}), \mathbf{h}), \gamma_\ell(Q_{DL}(\mathbf{d}_\ell))\}$. Since the NSNR terms in the maximization are independent random variables, the resulting diversity gain is the sum of the diversity gains provided by each NSNR term. In other words, the diversity gain representing the asymptotic behavior of the SER at the ℓ th receiver is $(R+1, d_2)$. By a simple union bound, the NER is upper bounded by the sum of the SERs of the receivers, and therefore, a diversity gain of $(R+1, d_2)$ is achievable. We skip the specific calculations for each quantization and forwarding scheme since they are straightforward.

To summarize, all the main achievability and converse results in Theorems 1 through 6 remain valid for networks with direct links if the term P^R appearing in the denominator of the upper or lower bounds on the NER is replaced with the term P^{R+1} , or equivalently if the implied first-order diversity gains are increased by 1.

VIII. CONCLUSIONS AND DISCUSSIONS

We have studied fixed data rate communication schemes for wireless relay-interference networks with any number of transmitters, relays, and receivers. The transmitters and the relays had individual short-term power constraints. We have analyzed both amplify-and-forward (AF) and decode-and-forward (DF) relaying strategies with a two channel use quantized network beamforming protocol.

A. Generalized Diversity Analysis

We have introduced a generalized diversity measure to characterize the asymptotic performance of the network. For a communication scheme that achieves an asymptotic error rate of $C \frac{1}{P^{\alpha \log^{\beta} P}}$ with C being a constant that is independent of the transmission power P , we have called α and β the first-order and the second-order diversity gains, respectively.

We have demonstrated that the higher order diversities can more precisely distinguish between different communication scenarios that achieve the same traditional (first-order) diversity. Indeed, as we shall summarize in the following and shown throughout the paper, all the schemes that we have proposed achieved the same first-order diversity, what distinguished them was their different second-order diversity gains. Applications of the generalized diversity measures to other networking scenarios would provide better insights in comparing different communication schemes.

We would also like to note that one can similarly generalize the concept of the “multiplexing gain” to capture the sublogarithmic behaviors of the data transmission rate. One possible extension of this work would be a generalized diversity/multiplexing tradeoff analysis of different schemes for the relay-interference network. Currently, the limited knowledge regarding the available degrees of freedom of multihop networks makes this problem difficult to attack.

B. Linear vs. Non-Linear Forwarding in Relay-Interference Networks

We have shown that the maximal achievable diversity with AF relays is $(R, -R)$ when $K > 1$, whereas it is $(R, 0)$ when

$K = 1$. In other words, with AF relays, we have shown that the relay-interference network suffers from a second-order diversity loss compared to an interference-free network that can achieve diversity $(R, 0)$. This has demonstrated that in interference networks, simple linear forwarding schemes are not sufficient to achieve the best possible full diversity gain $(R, 0)$.

We have shown that it is possible to achieve the full diversity gain $(R, 0)$ using DF relays as if there is no interference and as if coding over an arbitrary block length is allowed. Therefore, although it imposes a considerable amount of complexity at the relays, the DF strategy has achieved the full diversity gain $(R, 0)$ for any number of transmitters.

For both AF and DF strategies, we have shown that relay selection provides the best possible diversity gain.

C. Very-Low Rate Channel State Information Quantization

One key contribution of our work that distinguished itself from the earlier literature on limited feedback has been to demonstrate that the full diversity gain can be achieved using distributed decision making with asymptotically zero feedback rate per receiver. Such a performance is made possible by a special distributed (local) quantizer (LQ) design method that we have called “localization.” In this method, we have first designed a global quantizer (GQ) that has access to the entire channel state information. While such a quantizer is obviously not practical, we have shown that it is possible to synthesize, out of the GQ, a practical finite-rate local quantizer (LQ) that can achieve the same performance as the GQ asymptotically. The performance of our LQs has provided a very fortunate answer regarding the feedback requirements for maximal reliability: In a relay-interference network, it is possible to achieve maximal reliability using variable-length LQs with arbitrarily low feedback rates per receiver, when the transmitter powers are sufficiently large.

Still, regarding the LQs, there are many open problems that we have not addressed in this paper. One important problem is to determine whether there exists a fixed-length LQ that can achieve maximal diversity. Another goal might be to generalize our relay-selection based localization result to show that *any* GQ can be localized to synthesize an LQ that can achieve the same array gain as the GQ. Due to the complicated nature of our distortion functions, the latter goal seems difficult to accomplish, even though we have observed its validity by simulations.

APPENDIX A

PROOF OF PROPOSITION 2

For a fixed channel state $\mathbf{h}$, and an arbitrary beamforming vector $\mathbf{x} \in \mathcal{X}$, let $\text{SER}_\ell^{\text{IML}}(\mathbf{x}, \mathbf{h}) \triangleq \mathbb{P}(\tilde{\mathbf{s}}_\ell \neq \mathbf{s}_\ell | \mathbf{x}, \mathbf{h})$ denote the conditional super-symbol error rate (SER) of the ℓ th receiver with the individual ML decoder described in Section III-A.

For the ℓ th receiver, instead of the individual ML decoder, suppose that we employ a joint ML decoder $\hat{\mathbf{s}}_\ell \triangleq \arg \max_{\mathbf{s}' \in \mathcal{S}} \mathbb{P}(\mathbf{s}' | y_\ell, \mathbf{x}, \mathbf{h})$, where $\mathcal{S} = \prod_k \mathcal{S}_k$. Recall that, for the individual ML decoder at the ℓ th receiver, the a posteriori probability was maximized over $\prod_{i \in \mathcal{D}_\ell} \mathcal{S}_i$. For the joint ML decoder, the maximization is over $\prod_k \mathcal{S}_k$ at all the receivers.

Let $\text{SER}_\ell^{\text{JML}}(\mathbf{x}, \mathbf{h}) \triangleq \mathbb{P}(\hat{s}_\ell \neq s | \mathbf{x}, \mathbf{h})$ denote the error rate of the joint ML decoder. Then, we have $\text{SER}_\ell^{\text{JML}}(\mathbf{x}, \mathbf{h}) \leq \text{SER}_\ell^{\text{JML}}(\mathbf{x}, \mathbf{h}), \forall \ell$. Since the CNER is the probability that at least one receiver incorrectly decodes its desired super-symbol, we can use a union bound over all the receivers to obtain

$$\begin{aligned} \text{CNER}(\mathbf{x}, \mathbf{h}) &\leq \sum_{\ell=1}^L \text{SER}_\ell^{\text{JML}}(\mathbf{x}, \mathbf{h}) \\ &\leq \sum_{\ell=1}^L \text{SER}_\ell^{\text{JML}}(\mathbf{x}, \mathbf{h}) \\ &\leq L \max_{\ell} \text{SER}_\ell^{\text{JML}}(\mathbf{x}, \mathbf{h}) \end{aligned} \quad (58)$$

We now prove the upper bounds in the statement of the proposition with AF and DF relays separately.

A. AF Relays

For AF relays, let

$$\tilde{\gamma}_{\ell, s, \hat{s}}(\mathbf{x}, \mathbf{h}) = \frac{|\sum_{k=1}^K (s_k - \hat{s}_k) \sqrt{P_{S_k}} \sum_{r=1}^R f_{kr} \sqrt{\rho_r} g_{r\ell} x_r|^2}{4(1 + \sum_{r=1}^R \rho_r |g_{r\ell}|^2 |x_r|^2)} \quad (59)$$

It is then straightforward to show by inspection of (5) that

$$\begin{aligned} \text{SER}_\ell^{\text{JML}}(\mathbf{x}, \mathbf{h}) &\leq \frac{1}{|\mathcal{S}|} \sum_{s \neq \hat{s}} \mathbb{Q}(\sqrt{2\tilde{\gamma}_{\ell, s, \hat{s}}(\mathbf{x}, \mathbf{h})}) \\ &\leq \frac{|\mathcal{S}| - 1}{2} \max_{s \neq \hat{s}} \mathbb{Q}(\sqrt{2\tilde{\gamma}_{\ell, s, \hat{s}}(\mathbf{x}, \mathbf{h})}) \\ &\leq \frac{|\mathcal{S}| - 1}{4} \max_{s \neq \hat{s}} \exp(-\tilde{\gamma}_{\ell, s, \hat{s}}(\mathbf{x}, \mathbf{h})) \\ &= \frac{|\mathcal{S}| - 1}{4} \exp(-\tilde{\gamma}_\ell(\mathbf{x}, \mathbf{h})), \end{aligned} \quad (60)$$

where $\tilde{\gamma}_\ell(\mathbf{x}, \mathbf{h}) = \min_{s \neq \hat{s}} \tilde{\gamma}_{\ell, s, \hat{s}}(\mathbf{x}, \mathbf{h})$. The first inequality is the well-known union bound for probability of detection error in Gaussian noise. For the third inequality, we have used the Chernoff bound on the $\mathbb{Q}(\cdot)$ function, $\mathbb{Q}(x) \leq \frac{1}{2} \exp(-\frac{x^2}{2})$, $x \geq 0$. As a general remark for the derivation above, we would like to note that the decoded symbol vector for each receiver is obviously different, i.e., $\hat{s}_\ell$, though we have omitted the dependence on ℓ for brevity. We have also omitted the condition $s, \hat{s} \in \mathcal{S}$ in the summation and the maximizations as it is clear from the context.

Substituting (60) to (58), we obtain

$$\begin{aligned} \text{CNER}(\mathbf{x}, \mathbf{h}) &\leq \underbrace{\frac{L|\mathcal{S}| - 1}{4}}_{\triangleq c_{23}} \max_{\ell} \exp(-\tilde{\gamma}_\ell(\mathbf{x}, \mathbf{h})) \\ &= c_{23} \exp(-\tilde{\gamma}^L(\mathbf{x}, \mathbf{h})), \end{aligned} \quad (61)$$

where

$$\tilde{\gamma}^L(\mathbf{x}, \mathbf{h}) = \min_{\ell} \tilde{\gamma}_\ell(\mathbf{x}, \mathbf{h}). \quad (62)$$

Note that (61) holds for an arbitrary beamforming vector $\mathbf{x} \in \mathcal{X}$. where $\mathbf{E}^c$ denotes the complement of the event $\mathbf{E}$.

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In particular, for $\mathbf{x} \in \mathcal{C}_s = \{\mathbf{e}_1, \dots, \mathbf{e}_R\}$, we have

$$\tilde{\gamma}_\ell(\mathbf{e}_r, \mathbf{h}) = \min_{\substack{s, \hat{s} \in \mathcal{S} \\ s \neq \hat{s}}} \frac{|\sum_{k=1}^K (s_k - \hat{s}_k) \sqrt{P_{S_k}} f_{kr}|^2 |g_{r\ell}|^2 \rho_r}{4(1 + |g_{r\ell}|^2 \rho_r)}. \quad (63)$$

Now, if $K = 1$ or $\mathbb{E}[s_k] = 0, \forall k$, then ρ_r is given by (4). Otherwise, for the denominator of (3), we have

$$\begin{aligned} 1 + \mathbb{E}_{s_1, \dots, s_K} \left[\left| \sum_{k=1}^K f_{kr} s_k \sqrt{P_{S_k}} \right|^2 \right] \\ \leq 1 + \mathbb{E}_{s_1, \dots, s_K} \left[K \sum_{k=1}^K |f_{kr} s_k \sqrt{P_{S_k}}|^2 \right] \\ = 1 + K \sum_{k=1}^K |f_{kr}|^2 P_{S_k} \\ \leq K \left(1 + \sum_{k=1}^K |f_{kr}|^2 P_{S_k} \right), \end{aligned}$$

where the second inequality follows from Hölder's inequality. In either case, we have

$$\rho_r \geq \frac{P_{R_r}}{\bar{c} \left(1 + \sum_{k=1}^K |f_{kr}|^2 P_{S_k} \right)}, \quad (64)$$

where $\bar{c}$ is as defined in the statement of the proposition. Substituting (64) to (63),

$$\begin{aligned} \tilde{\gamma}_\ell(\mathbf{e}_r, \mathbf{h}) &\geq \min_{\substack{s, \hat{s} \in \mathcal{S} \\ s \neq \hat{s}}} \frac{|\sum_{k=1}^K (s_k - \hat{s}_k) \sqrt{P_{S_k}} f_{kr}|^2 |g_{r\ell}|^2 P_{R_r}}{4 \left[\bar{c} \left(1 + \sum_{k=1}^K |f_{kr}|^2 P_{S_k} \right) + |g_{r\ell}|^2 P_{R_r} \right]} \\ &\geq \gamma_\ell(\mathbf{e}_r, \mathbf{h}). \end{aligned} \quad (65)$$

Now, let $\gamma^L(\mathbf{e}_r, \mathbf{h}) = \min_{\ell} \gamma_\ell(\mathbf{e}_r, \mathbf{h})$. According to (62) and (65), we have $\tilde{\gamma}^L(\mathbf{e}_r, \mathbf{h}) \geq \gamma^L(\mathbf{e}_r, \mathbf{h})$. Substituting to (61), we obtain

$$\text{CNER}(\mathbf{e}_r, \mathbf{h}) \leq c_{23} \exp(-\gamma^L(\mathbf{e}_r, \mathbf{h})), \quad (66)$$

which is the same upper bound (19) in the statement of the proposition apart from the constant multiplier.

B. DF Relays

Let $\mathbf{E}$ denote the event that in the first step of data transmission, each relay with $x_r \neq 0$ has successfully decoded all the symbols of all the transmitters without any error. In order to find an upper bound on the CNER with DF relays, at each receiver, we use a suboptimal joint ML decoder that assumes that the event $\mathbf{E}$ has occurred. Let $\hat{\mathbf{s}}_\ell = \arg \max_{s \in \mathcal{S}} \mathbb{P}(s | y_\ell, \mathbf{g}_\ell, \mathbf{E})$ denote the transmitted super-symbol's estimate of the suboptimal joint ML decoder at the ℓ th receiver. Also, let $\text{SER}_\ell^{\text{SOJML}}(\mathbf{x}, \mathbf{h}) \triangleq \mathbb{P}(\hat{\mathbf{s}}_\ell \neq s | \mathbf{x}, \mathbf{h})$ denote the corresponding error probability. We have

$$\begin{aligned} \text{SER}_\ell^{\text{JML}}(\mathbf{x}, \mathbf{h}) &\leq \text{SER}_\ell^{\text{SOJML}}(\mathbf{x}, \mathbf{h}) \\ &= \mathbb{P}(\hat{\mathbf{s}}_\ell \neq s | \mathbf{E}) \mathbb{P}(\mathbf{E}) + \mathbb{P}(\hat{\mathbf{s}}_\ell \neq s | \mathbf{E}^c) \mathbb{P}(\mathbf{E}^c) \\ &\leq \mathbb{P}(\hat{\mathbf{s}}_\ell \neq s | \mathbf{E}) + \mathbb{P}(\mathbf{E}^c) \end{aligned} \quad (67)$$

We now observe that given $\mathbf{E}$, the probability of incorrect detection is upper bounded by the symbol error probability of the peak-power constrained ζ^2 -QAM constellation $\mathcal{S}_R$ defined in (7). Furthermore, each signal point in $\mathcal{S}_R$ has at most 4 nearest neighbors with Euclidean distance $\frac{\sqrt{2}}{\zeta-1}$. It follows that

$$\begin{aligned} P(\hat{\mathbf{s}}_\ell \neq \mathbf{s} | \mathbf{E}) &\leq 4Q \left(\sqrt{\frac{1}{(\zeta-1)^2} \left| \sum_{r=1}^R x_r g_{r\ell} \sqrt{P_{R_r}} \right|^2} \right) \\ &\leq 2 \exp \left(-\frac{1}{2(\zeta-1)^2} \left| \sum_{r=1}^R x_r g_{r\ell} \sqrt{P_{R_r}} \right|^2 \right) \end{aligned} \quad (68)$$

For $\mathbf{E}^c$, we use a union bound over all the relays (with $x_r \neq 0$) and all possible transmitted symbols to obtain

$$\begin{aligned} P(\mathbf{E}^c) &\leq \sum_{r: x_r \neq 0} \frac{1}{|\mathcal{A}|} \sum_{\mathbf{s} \neq \hat{\mathbf{s}}} Q \left(\sqrt{\frac{1}{2} \left| \sum_{k=1}^K (s_k - \hat{s}_k) \sqrt{P_{S_k}} f_{kr} \right|^2} \right) \\ &\leq \frac{R(|\mathcal{A}|-1)}{4} \exp \left[-\frac{1}{4} \min_{r: x_r \neq 0} \min_{\mathbf{s} \neq \hat{\mathbf{s}}} \left| \sum_{k=1}^K (s_k - \hat{s}_k) \sqrt{P_{S_k}} f_{kr} \right|^2 \right] \end{aligned}$$

Combining the last inequality with (68), (67) and (58),

$$\text{CNER}(\mathbf{x}, \mathbf{h}) \leq \underbrace{L \max \left\{ 2, \frac{R(|\mathcal{A}|-1)}{4} \right\}}_{\triangleq C_{24}} \exp(-\gamma^L(\mathbf{x}, \mathbf{h})), \quad (69)$$

where

$$\begin{aligned} \gamma^L(\mathbf{x}, \mathbf{h}) &= \min \left\{ \frac{1}{2(\zeta-1)^2} \left| \sum_{r=1}^R x_r g_{r\ell} \sqrt{P_{R_r}} \right|^2, \right. \\ &\quad \left. \frac{1}{4} \min_{r: x_r \neq 0} \min_{\substack{\mathbf{s}, \hat{\mathbf{s}} \in \mathcal{S} \\ \mathbf{s} \neq \hat{\mathbf{s}}}} \left| \sum_{k=1}^K (s_k - \hat{s}_k) \sqrt{P_{S_k}} f_{kr} \right|^2 \right\}. \end{aligned}$$

In particular, if $\mathbf{x} \in \mathcal{C}_S$, we obtain $\gamma^L(\mathbf{e}_r, \mathbf{h})$ as defined in the statement of the proposition. By inspection of (66) and (69), we set $C_4 = \max\{C_{23}, C_{24}\}$. This concludes the proof. ■

APPENDIX B

PROOF OF THE UPPER BOUNDS IN THEOREM 2

In this appendix, we prove the upper bounds (21) and (22) in the statement of Theorem 2. The proof is organized as follows: In Appendix B-A, we provide some useful lemmas. The main idea of the proof is to find upper bounds on the PDF and the CDF of the random variable

$$\Omega_r \triangleq \gamma^L(\mathbf{e}_r, \mathbf{h}) \quad (70)$$

that represents the NSNR given that the r th relay is selected. These upper bounds will be provided in Appendix B-B. Finally, we prove (21) and (22) in Appendix B-C using the results in Appendices B-A and B-B.

A. Some Useful Lemmas

We start with some useful lemmas that we use throughout the proof.

Lemma 1: For $K > 1$, let $\tilde{\mathbf{v}}_c$ and $\tilde{\mathbf{v}}_s$ be zero-mean real Gaussian $K \times 1$ random vectors, with equal diagonal covariance matrices $\mathbf{E}[\tilde{\mathbf{v}}_c \tilde{\mathbf{v}}_c^T] = \mathbf{E}[\tilde{\mathbf{v}}_s \tilde{\mathbf{v}}_s^T] = \mathbf{K}$, $K_{ii} > 0$, $\forall i$, $K_{ij} = 0$, $\forall i \neq j$, and zero cross-covariance $\mathbf{E}[\tilde{\mathbf{v}}_c \tilde{\mathbf{v}}_s^T] = \mathbf{0}_{K \times K}$. Let $\tilde{\mathbf{v}} \triangleq \tilde{\mathbf{v}}_c + j\tilde{\mathbf{v}}_s$ denote the complex Gaussian random vector with real and imaginary parts given by $\tilde{\mathbf{v}}_c$ and $\tilde{\mathbf{v}}_s$, respectively. Also, let $X \triangleq |\langle \mathbf{t}, \tilde{\mathbf{v}} \rangle|^2$, where $\mathbf{t} \in \mathbb{C}^K$ with $\|\mathbf{t}\| = 1$ is a fixed vector, and $W \triangleq \|\tilde{\mathbf{v}}\|^2 - X$. Then, there are constants $0 < \lambda', \lambda'' < \infty$, such that for any x and w , we have

$$f_{X,W}(x, w) \leq \mathbf{1}_{x, w \geq 0} \frac{\lambda'}{\Gamma(K-1)} w^{K-2} \exp[-\lambda''(x+w)].$$

Proof: Let $\mathbf{t}_1, \dots, \mathbf{t}_K$ with $\mathbf{t}_1 = \mathbf{t}$ be an orthonormal basis for $\mathbb{C}^K$. Also, let $\mathbf{U}$ be a $K \times K$ matrix whose k th row is equal to $\mathbf{t}_k^H$. By definition, $\mathbf{U}$ is a unitary matrix, and

$$\mathbf{U}\mathbf{t} = \mathbf{U}\mathbf{t}_1 = \mathbf{e}_1, \quad (71)$$

where $\mathbf{e}_1 = [1 \ 0 \ \dots \ 0]^T$ is as defined in Section IV-C.

Now, let

$$\mathbf{v} \triangleq \mathbf{U}\tilde{\mathbf{v}}, \quad (72)$$

$$X_k \triangleq |v_k|^2, \quad k = 1, \dots, K, \quad (73)$$

and $\mathbf{X} \triangleq [X_1 \dots X_K]^T$. We have

$$X = |\langle \mathbf{t}, \tilde{\mathbf{v}} \rangle|^2 = |\langle \mathbf{U}\mathbf{t}, \mathbf{U}\tilde{\mathbf{v}} \rangle|^2 = |\langle \mathbf{e}_1, \mathbf{v} \rangle|^2 = |v_1|^2 = X_1, \quad (74)$$

where the first equality is as in the statement of the lemma, and the second equality follows since $\mathbf{U}$ is a unitary matrix. For the third equality, we have used (71) and (72). The last equality is by the definition in (73).

Moreover,

$$\begin{aligned} W &= \|\tilde{\mathbf{v}}\|^2 - X = \|\mathbf{U}\tilde{\mathbf{v}}\|^2 - X \\ &= \|\mathbf{v}\|^2 - X_1 = \left(\sum_{i=1}^K X_i \right) - X_1 = \sum_{i=2}^K X_i, \end{aligned} \quad (75)$$

where the first equality is as in the statement of the lemma, and the second equality follows since $\mathbf{U}$ is norm-preserving. The third equality follows from (73) and (74). For the fourth equality, we have used the fact that $\|\mathbf{v}\|^2 = \sum_{i=1}^K |v_i|^2 = \sum_{i=1}^K X_i$.

We have thus achieved $X = X_1$ and $W = X_2 + \dots + X_K$ by (74) and (75), respectively.

Let us now consider the cases $K = 2$ and $K \geq 3$ separately. We first work on the case $K \geq 3$, for which the joint PDF of $X, W, X_3, \dots, X_K$ can be expressed in terms of the joint PDF of $X_1, \dots, X_K$ as [72]

$$f_{X,W,X_3,\dots,X_K}(x, w, x_3, \dots, x_K) = f_{\mathbf{X}}(\mathbf{x}) |\det(\mathbf{J})|, \quad (76)$$

where $\mathbf{x} = [x_1 \ \dots \ x_K]^T$, and

$$\mathbf{J} = \begin{bmatrix} \frac{\partial x_1}{\partial x} & \frac{\partial x_1}{\partial w} & \frac{\partial x_1}{\partial x_3} & \dots & \frac{\partial x_1}{\partial x_K} \\ \frac{\partial x_2}{\partial x} & \frac{\partial x_2}{\partial w} & \frac{\partial x_2}{\partial x_3} & \dots & \frac{\partial x_2}{\partial x_K} \\ \frac{\partial x_3}{\partial x} & \frac{\partial x_3}{\partial w} & \frac{\partial x_3}{\partial x_3} & \dots & \frac{\partial x_3}{\partial x_K} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_K}{\partial x} & \frac{\partial x_K}{\partial w} & \frac{\partial x_K}{\partial x_3} & \dots & \frac{\partial x_K}{\partial x_K} \end{bmatrix}.$$

We now evaluate $\det(\mathbf{J})$. By the definition of the transformation, we have $x_1 = x$, $x_2 = w - \sum_{k=3}^K x_k$, and hence

$$\det(\mathbf{J}) = \det \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & -1 & -1 & \cdots & -1 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \end{pmatrix} = 1. \quad (77)$$

The last equality follows since the determinant of a triangular matrix is equal to the product of its diagonal elements.

Substituting (77) to (76), and rewriting $x_1, \dots, x_K$ in terms of $x, w, x_3, \dots, x_K$, we have

$$f_{X,W,X_3,\dots,X_K}(x, w, x_3, \dots, x_K) = f_{\mathbf{X}} \left(x, w - \sum_{k=3}^K x_k, x_3, \dots, x_K \right). \quad (78)$$

Since we seek to find the joint PDF of X and W , we average out (78) over all possible $x_3, \dots, x_K$ to obtain

$$f_{X,W}(x, w) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{\mathbf{X}} \left(x, w - \sum_{k=3}^K x_k, x_3, \dots, x_K \right) \times dx_K \cdots dx_3 \quad (79)$$

The ranges of integrals in (79) can be restricted as follows: Since $f_{\mathbf{X}}(\mathbf{x}) = 0$ whenever $\exists k \in \{1, \dots, K\}$ with $x_k < 0$, we need to evaluate (79) only for the values of $x, w, x_3, \dots, x_K$ that satisfy $x, w - \sum_{k=3}^K x_k, x_3, \dots, x_K \geq 0$. It is straightforward to show that these inequalities are satisfied if and only if the following hold: $x, w, x_3, \dots, x_K \geq 0, \sum_{k=3}^K x_k \leq w$. Therefore,

$$f_{X,W}(x, w) = \mathbf{1}_{x,w \geq 0} \int_{\substack{x_3, \dots, x_K \geq 0 \\ x_3 + \cdots + x_K \leq w}} f_{\mathbf{X}} \left(x, w - \sum_{k=3}^K x_k, x_3, \dots, x_K \right) \times dx_K \cdots dx_3. \quad (80)$$

For $K = 2$, note that $X_1 = X$ and $X_2 = W$. As a result,

$$f_{X,W}(x, w) = \mathbf{1}_{x,w \geq 0} f_{\mathbf{X}}(x, w). \quad (81)$$

We now find an upper bound for $f_{\mathbf{X}}(\mathbf{x})$ for any K and any $\mathbf{x} = [x_1 \cdots x_K]^T$ with $x_i \geq 0, \forall i$.

Let $\mathbf{U}_c \triangleq \Re(\mathbf{U})$, and $\mathbf{U}_s \triangleq \Im(\mathbf{U})$. The real and imaginary parts of $\mathbf{v}$ defined in (72) can be calculated to be $\mathbf{v}_c \triangleq \Re(\mathbf{v}) = \mathbf{U}_c \tilde{\mathbf{v}}_c - \mathbf{U}_s \tilde{\mathbf{v}}_s$, and $\mathbf{v}_s \triangleq \Im(\mathbf{v}) = \mathbf{U}_c \tilde{\mathbf{v}}_s + \mathbf{U}_s \tilde{\mathbf{v}}_c$. Then,

$$\begin{aligned} \mathbf{K}_{cc} &\triangleq \mathbb{E}[\mathbf{v}_c \mathbf{v}_c^T] \\ &= \mathbb{E}[(\mathbf{U}_c \tilde{\mathbf{v}}_c - \mathbf{U}_s \tilde{\mathbf{v}}_s)(\mathbf{U}_c \tilde{\mathbf{v}}_c - \mathbf{U}_s \tilde{\mathbf{v}}_s)^T] \\ &= \mathbb{E}[(\mathbf{U}_c \tilde{\mathbf{v}}_c - \mathbf{U}_s \tilde{\mathbf{v}}_s)(\tilde{\mathbf{v}}_c^T \mathbf{U}_c^T - \tilde{\mathbf{v}}_s^T \mathbf{U}_s^T)] \\ &= \mathbb{E}[\mathbf{U}_c \tilde{\mathbf{v}}_c \tilde{\mathbf{v}}_c^T \mathbf{U}_c^T - \mathbf{U}_s \tilde{\mathbf{v}}_s \tilde{\mathbf{v}}_c^T \mathbf{U}_c^T - \\ &\quad \mathbf{U}_c \tilde{\mathbf{v}}_c \tilde{\mathbf{v}}_s^T \mathbf{U}_s^T + \mathbf{U}_s \tilde{\mathbf{v}}_s \tilde{\mathbf{v}}_s^T \mathbf{U}_s^T] \\ &= \mathbf{U}_c \mathbb{E}[\tilde{\mathbf{v}}_c \tilde{\mathbf{v}}_c^T] \mathbf{U}_c^T - \mathbf{U}_s \mathbb{E}[\tilde{\mathbf{v}}_s \tilde{\mathbf{v}}_c^T] \mathbf{U}_c^T - \\ &\quad \mathbf{U}_c \mathbb{E}[\tilde{\mathbf{v}}_c \tilde{\mathbf{v}}_s^T] \mathbf{U}_s^T + \mathbf{U}_s \mathbb{E}[\tilde{\mathbf{v}}_s \tilde{\mathbf{v}}_s^T] \mathbf{U}_s^T \\ &= \mathbf{U}_c \mathbf{K} \mathbf{U}_c^T + \mathbf{U}_s \mathbf{K} \mathbf{U}_s^T, \end{aligned}$$

where the last equality follows from the assumptions $\mathbb{E}[\tilde{\mathbf{v}}_c \tilde{\mathbf{v}}_c^T] = \mathbf{E}[\tilde{\mathbf{v}}_s \tilde{\mathbf{v}}_s^T] = \mathbf{K}$ and $\mathbb{E}[\tilde{\mathbf{v}}_c \tilde{\mathbf{v}}_s^T] = \mathbb{E}[\tilde{\mathbf{v}}_s \tilde{\mathbf{v}}_c^T] = 0$ that we have made in the statement of the lemma.

Similarly, we can calculate

$$\begin{aligned} \mathbf{K}_{ss} &\triangleq \mathbb{E}[\mathbf{v}_s \mathbf{v}_s^T] \\ &= \mathbb{E}[(\mathbf{U}_c \tilde{\mathbf{v}}_s + \mathbf{U}_s \tilde{\mathbf{v}}_c)(\mathbf{U}_c \tilde{\mathbf{v}}_s + \mathbf{U}_s \tilde{\mathbf{v}}_c)^T] \\ &= \mathbb{E}[\mathbf{U}_c \tilde{\mathbf{v}}_s \tilde{\mathbf{v}}_s^T \mathbf{U}_c^T + \mathbf{U}_s \tilde{\mathbf{v}}_c \tilde{\mathbf{v}}_s^T \mathbf{U}_c^T + \\ &\quad \mathbf{U}_c \tilde{\mathbf{v}}_s \tilde{\mathbf{v}}_c^T \mathbf{U}_s^T + \mathbf{U}_s \tilde{\mathbf{v}}_c \tilde{\mathbf{v}}_c^T \mathbf{U}_s^T] \\ &= \mathbf{K}_{cc}, \end{aligned}$$

and

$$\begin{aligned} \mathbf{K}_{cs} &\triangleq \mathbb{E}[\mathbf{v}_c \mathbf{v}_s^T] = \mathbf{U}_c \mathbf{K} \mathbf{U}_s^T - \mathbf{U}_s \mathbf{K} \mathbf{U}_c^T \\ &= \mathbb{E}[(\mathbf{U}_c \tilde{\mathbf{v}}_c - \mathbf{U}_s \tilde{\mathbf{v}}_s)(\mathbf{U}_c \tilde{\mathbf{v}}_s + \mathbf{U}_s \tilde{\mathbf{v}}_c)^T] \\ &= \mathbb{E}[\mathbf{U}_c \tilde{\mathbf{v}}_c \tilde{\mathbf{v}}_s^T \mathbf{U}_c^T - \mathbf{U}_s \tilde{\mathbf{v}}_s \tilde{\mathbf{v}}_s^T \mathbf{U}_c^T + \\ &\quad \mathbf{U}_c \tilde{\mathbf{v}}_c \tilde{\mathbf{v}}_c^T \mathbf{U}_s^T - \mathbf{U}_s \tilde{\mathbf{v}}_s \tilde{\mathbf{v}}_c^T \mathbf{U}_s^T] \\ &= \mathbf{U}_c \mathbf{K} \mathbf{U}_s^T - \mathbf{U}_s \mathbf{K} \mathbf{U}_c^T \end{aligned}$$

Therefore, $\mathbf{K}_{cc}$ and $\mathbf{K}_{ss}$ are symmetric matrices, and $\mathbf{K}_{cs} = -\mathbf{K}_{cs}^T$. Using the latter, for any $\mathbf{x} \in \mathbb{R}^K$, we have $\mathbf{x}^T \mathbf{K}_{cs} \mathbf{x} = (\mathbf{x}^T \mathbf{K}_{cs} \mathbf{x})^T = \mathbf{x}^T \mathbf{K}_{cs}^T \mathbf{x} = -\mathbf{x}^T \mathbf{K}_{cs} \mathbf{x}$, and thus $\mathbf{x}^T \mathbf{K}_{cs} \mathbf{x} = 0$. Using these facts, we now show that $\mathbf{K}_{cc} + j\mathbf{K}_{cs}$ is positive definite.

For any $\mathbf{x} \in \mathbb{C}^K - \{0\}$, we have

$$\begin{aligned} \mathbf{x}^H (\mathbf{K}_{cc} + j\mathbf{K}_{cs}) \mathbf{x} &= (\mathbf{x}_c^T - j\mathbf{x}_s^T) (\mathbf{K}_{cc} + j\mathbf{K}_{cs}) (\mathbf{x}_c + j\mathbf{x}_s) \\ &= \mathbf{x}_c^T \mathbf{K}_{cc} \mathbf{x}_c - \mathbf{x}_c^T \mathbf{K}_{cs} \mathbf{x}_s + \mathbf{x}_s^T \mathbf{K}_{cs} \mathbf{x}_c + \mathbf{x}_s^T \mathbf{K}_{cc} \mathbf{x}_s \\ &\quad + j(\mathbf{x}_c^T \mathbf{K}_{cs} \mathbf{x}_c + \mathbf{x}_c^T \mathbf{K}_{cc} \mathbf{x}_s - \mathbf{x}_s^T \mathbf{K}_{cc} \mathbf{x}_c + \mathbf{x}_s^T \mathbf{K}_{cs} \mathbf{x}_s) \\ &= \mathbf{x}_c^T \mathbf{K}_{cc} \mathbf{x}_c + 2\mathbf{x}_s^T \mathbf{K}_{cs} \mathbf{x}_c + \mathbf{x}_s^T \mathbf{K}_{cc} \mathbf{x}_s \\ &= \mathbf{x}_c^T \mathbb{E}[\mathbf{v}_s \mathbf{v}_s^T] \mathbf{x}_c + 2\mathbf{x}_s^T \mathbb{E}[\mathbf{v}_c \mathbf{v}_s^T] \mathbf{x}_c + \mathbf{x}_s^T \mathbb{E}[\mathbf{v}_c \mathbf{v}_c^T] \mathbf{x}_s \\ &= \mathbb{E}[(\mathbf{x}_c^T \mathbf{v}_s + \mathbf{x}_s^T \mathbf{v}_c)(\mathbf{x}_c^T \mathbf{v}_s + \mathbf{x}_s^T \mathbf{v}_c)^T] \\ &= \mathbb{E}[(\mathbf{x}_c^T \mathbf{U}_s + \mathbf{x}_s^T \mathbf{U}_c) \tilde{\mathbf{v}}_c + (\mathbf{x}_c^T \mathbf{U}_c - \mathbf{x}_s^T \mathbf{U}_s) \tilde{\mathbf{v}}_s] \\ &\quad ((\mathbf{x}_c^T \mathbf{U}_s + \mathbf{x}_s^T \mathbf{U}_c) \tilde{\mathbf{v}}_c + (\mathbf{x}_c^T \mathbf{U}_c - \mathbf{x}_s^T \mathbf{U}_s) \tilde{\mathbf{v}}_s)^T] \\ &= \mathbf{x}_1^T \mathbf{K} \mathbf{x}_1 + \mathbf{x}_2^T \mathbf{K} \mathbf{x}_2, \end{aligned} \quad (82)$$

where $\mathbf{x}_1 = \mathbf{U}_s^T \mathbf{x}_c + \mathbf{U}_c^T \mathbf{x}_s$, and $\mathbf{x}_2 = \mathbf{U}_c^T \mathbf{x}_c - \mathbf{U}_s^T \mathbf{x}_s$. On the other hand,

$$\mathbf{U} \mathbf{x} = (\mathbf{U}_c^T + j\mathbf{U}_s^T) (\mathbf{x}_c + j\mathbf{x}_s) = \mathbf{x}_2 + j\mathbf{x}_1. \quad (83)$$

Moreover, since $\mathbf{U}$ is a unitary matrix, it is positive definite. Thus, $\mathbf{U} \mathbf{x} \neq 0$, and as a result, either $\mathbf{x}_1 \neq 0$ or $\mathbf{x}_2 \neq 0$, according to (83). Also, as indicated in the statement of the lemma, $\mathbf{K}$ is a diagonal matrix with positive entries, and is thus positive definite. Since either $\mathbf{x}_1 \neq 0$ or $\mathbf{x}_2 \neq 0$, and $\mathbf{K}$ is positive definite, either $\mathbf{x}_1^T \mathbf{K} \mathbf{x}_1 > 0$, or $\mathbf{x}_2^T \mathbf{K} \mathbf{x}_2 > 0$. Using this observation in (82), we have $\mathbf{x}^H (\mathbf{K}_{cc} + j\mathbf{K}_{cs}) \mathbf{x} = \mathbf{x}_1^T \mathbf{K} \mathbf{x}_1 + \mathbf{x}_2^T \mathbf{K} \mathbf{x}_2 > 0$. Hence, $\mathbf{K}_{cc} + j\mathbf{K}_{cs}$ is positive definite.

Now, let $\mathbf{A} = \Re((\mathbf{K}_{cc} + j\mathbf{K}_{cs})^{-1})$, $\mathbf{B} = \Im((\mathbf{K}_{cc} + j\mathbf{K}_{cs})^{-1})$, and $\theta_{ij} = \tan^{-1}(B_{ij}/A_{ij})$. According to [73, Eq. (24)], the joint PDF of $X_i, i = 1, \dots, K$ can be expressed as

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{\det(\mathbf{D})}{(4\pi)^K} \int_{-\pi}^{\pi} \cdots \int_{-\pi}^{\pi} \exp \left(-\frac{1}{2} \chi(\boldsymbol{\phi}) \right) d\boldsymbol{\phi}, \quad (84)$$

where

$$\chi(\phi) = \sum_{i=1}^K A_{ii} x_i + 2 \sum_{\substack{i,j=1 \\ i < j}}^K (A_{ij}^2 + B_{ij}^2)^{\frac{1}{2}} \sqrt{x_i x_j} \cos(\phi_i - \phi_j + \theta_{ij}), \quad (85)$$

and $\mathbf{D} \triangleq \mathbf{A} + j\mathbf{B}$. Note that $\mathbf{D}$ is a Hermitian matrix [73, Eq. (21)], and thus $A = A^T$ and $B = -B^T$. Equivalently, we have $A_{ij} = A_{ji}$, $B_{ij} = -B_{ji}$, and $B_{ii} = 0$. Together with the definition of θ_{ij} , these equalities imply that $\theta_{ii} = \tan^{-1}(B_{ii}/A_{ii}) = \tan^{-1}(0/A_{ii}) = 0$ and $\theta_{ij} = \tan^{-1}(B_{ij}/A_{ij}) = \tan^{-1}(-B_{ji}/A_{ji}) = -\tan^{-1}(B_{ji}/A_{ji}) = -\theta_{ji}$. With these observations, we shall now find an alternate expression for $\chi(\phi)$. We have

$$\begin{aligned} \chi(\phi) &= \sum_{i=1}^K (A_{ii}^2 + B_{ii}^2)^{\frac{1}{2}} \sqrt{x_i x_i} \cos(\phi_i - \phi_i + \theta_{ii}) + \\ &\quad \sum_{\substack{i,j=1 \\ i < j}}^K (A_{ij}^2 + B_{ij}^2)^{\frac{1}{2}} \sqrt{x_i x_j} \cos(\phi_i - \phi_j + \theta_{ij}) + \\ &\quad \sum_{\substack{i,j=1 \\ j < i}}^K (A_{ji}^2 + B_{ji}^2)^{\frac{1}{2}} \sqrt{x_j x_i} \cos(\phi_j - \phi_i + \theta_{ji}), \quad (86) \end{aligned}$$

where the first summation is equal to the first summation in (85), and the second and the third summations are equal to half of the second summation in (85). Note that in (86), the second and the third summations are exactly the same expressions, if i is replaced by j and j is replaced by i in one of them.

For the third summation in (86), we have $A_{ji}^2 = A_{ij}^2$, $B_{ji} = (-B_{ij})^2 = B_{ij}^2$, and $\cos(\phi_j - \phi_i + \theta_{ji}) = \cos(\phi_j - \phi_i - \theta_{ij}) = \cos(\phi_i - \phi_j + \theta_{ij})$. Substituting these equalities to (86), we can observe that $\chi(\phi)$ can be expressed as

$$\chi(\phi) = \sum_{i=1}^K \sum_{j=1}^K (A_{ij}^2 + B_{ij}^2)^{\frac{1}{2}} \sqrt{x_i x_j} \cos(\phi_i - \phi_j + \theta_{ij}). \quad (87)$$

Now, using the facts that $\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$, and $\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$, we have

$$\begin{aligned} &\cos(\phi_i - \phi_j + \theta_{ij}) \\ &= \cos(\phi_i - \phi_j) \cos(\theta_{ij}) + \sin(\phi_i - \phi_j) \sin(\theta_{ij}) \\ &= \cos \phi_i \cos \phi_j \cos \theta_{ij} + \sin \phi_i \sin \phi_j \cos \theta_{ij} + \\ &\quad \sin \phi_i \cos \phi_j \sin \theta_{ij} - \cos \phi_i \sin \phi_j \sin \theta_{ij} \\ &= (A_{ij}^2 + B_{ij}^2)^{-\frac{1}{2}} (\cos \phi_i \cos \phi_j A_{ij} + \sin \phi_i \sin \phi_j A_{ij} + \\ &\quad \sin \phi_i \cos \phi_j B_{ij} - \cos \phi_i \sin \phi_j B_{ij}), \end{aligned}$$

where the last equality follows since $\sin(\theta_{ij}) = \sin \tan^{-1}(\frac{B_{ij}}{A_{ij}}) = (A_{ij}^2 + B_{ij}^2)^{-\frac{1}{2}} B_{ij}$, and similarly

$\cos(\theta_{ij}) = (A_{ij}^2 + B_{ij}^2)^{-\frac{1}{2}} A_{ij}$. Substituting to (87), and rearranging the terms, we obtain

$$\begin{aligned} \chi(\phi) &= \sum_{i=1}^K \sum_{j=1}^K \left(\sqrt{x_i} \cos \phi_i A_{ij} \sqrt{x_j} \cos \phi_j + \right. \\ &\quad \left. \sqrt{x_i} \sin \phi_i A_{ij} \sqrt{x_j} \sin \phi_j + \sqrt{x_i} \sin \phi_i B_{ij} \sqrt{x_j} \cos \phi_j - \right. \\ &\quad \left. \sqrt{x_i} \cos \phi_i B_{ij} \sqrt{x_j} \sin \phi_j \right) \\ &= [\Re(\mathbf{w})]^T \mathbf{A} \Re(\mathbf{w}) + [\Im(\mathbf{w})]^T \mathbf{A} \Im(\mathbf{w}) + \\ &\quad [\Im(\mathbf{w})]^T \mathbf{B} \Re(\mathbf{w}) - [\Re(\mathbf{w})]^T \mathbf{B} \Im(\mathbf{w}) \quad (88) \\ &= \mathbf{w}^H \mathbf{D} \mathbf{w} \quad (89) \end{aligned}$$

where $\mathbf{w}$ is a $K \times 1$ complex vector with i th component equal to $w_i = \sqrt{x_i} \cos \phi_i + j \sqrt{x_i} \sin \phi_i = \sqrt{x_i} e^{j\phi_i}$. The equality of (88) and (89) can be verified by expanding (89) as $\mathbf{w}^H \mathbf{D} \mathbf{w} = ([\Re(\mathbf{w})]^T - j[\Im(\mathbf{w})]^T)(\mathbf{A} + j\mathbf{B})(\Re(\mathbf{w}) + j\Im(\mathbf{w}))$.

We have previously shown that $\mathbf{K}_{cc} + j\mathbf{K}_{cs}$ is positive definite. It follows that $\mathbf{D} = (\mathbf{K}_{cc} + j\mathbf{K}_{cs})^{-1}$ is also positive definite, and thus has eigenvalues $\lambda_1, \dots, \lambda_K > 0$. Also, since $\mathbf{D}$ is a Hermitian matrix, it admits a spectral decomposition $\mathbf{D} = \sum_{k=1}^K \lambda_k \mathbf{u}_k \mathbf{u}_k^H$, where $\mathbf{u}_k$, $k = 1, \dots, K$ form an orthonormal basis for $\mathbb{C}^K$. It follows that

$$\begin{aligned} \chi(\phi) &= \mathbf{w}^H \mathbf{D} \mathbf{w} = \sum_{k=1}^K \lambda_k |\mathbf{u}_k^H \mathbf{w}|^2 \\ &> \min_i \lambda_i \sum_{k=1}^K |\mathbf{u}_k^H \mathbf{w}|^2 = \min_i \lambda_i \|\mathbf{w}\|^2 = \min_i \lambda_i \sum_{k=1}^K x_k, \end{aligned}$$

Substituting this lower bound to (84), we obtain the desired upper bound on the joint PDF of X_k , $k = 1, \dots, K$:

$$f_{\mathbf{X}}(\mathbf{x}) \leq 2^{-K} \det(\mathbf{D}) \exp \left(-\frac{\min_i \lambda_i}{2} \sum_{k=1}^K x_k \right). \quad (90)$$

Combining this upper bound with (81), we have

$$f_{X,W}(x, w) \leq \mathbf{1}_{x, w \geq 0} \frac{\det(\mathbf{D})}{4} \exp \left[-\frac{\min_i \lambda_i}{2} (x + w) \right] \quad (91)$$

for $K = 2$.

For $K \geq 3$, we need to find an upper bound for the integral in (79). Substituting (90) to (79), we obtain

$$\begin{aligned} f_{X,W}(x, w) &\leq \mathbf{1}_{x, w \geq 0} 2^{-K} \det(\mathbf{D}) \times \\ &\quad \exp \left[-\frac{\min_i \lambda_i}{2} (x + w) \right] \int \cdots \int_{\substack{x_3, \dots, x_K \geq 0 \\ x_3 + \cdots + x_K \leq w}} dx_K \cdots dx_3. \quad (92) \end{aligned}$$

We claim that

$$\int \cdots \int_{\substack{x_3, \dots, x_K \geq 0 \\ x_3 + \cdots + x_K \leq w}} dx_K \cdots dx_3 = \frac{w^{K-2}}{\Gamma(K-1)}. \quad (93)$$

We prove this by induction. For $K = 3$, it is obvious. Suppose that it is true $K = K' \geq 3$. Then,

$$\begin{aligned} & \int \cdots \int_{\substack{x_3, \dots, x_{K'+1} \geq 0 \\ x_3 + \cdots + x_{K'+1} \leq w}} dx_{K'+1} \cdots dx_3 \\ &= \int_0^w \int \cdots \int_{\substack{x_3, \dots, x_{K'+1} \geq 0 \\ x_3 + \cdots + x_{K'+1} \leq w - x_{K'+1}}} dx_{K'} \cdots dx_3 dx_{K'+1} \\ &= \int_0^w \frac{(w - x_{K'+1})^{K'-2}}{\Gamma(K' - 1)} dx_{K'+1} \\ &= \frac{w^{K'-1}}{\Gamma(K')}, \end{aligned}$$

which shows that the claim is also true for $K = K' + 1$. Substituting (93) to (92) we obtain

$$f_{X,W}(x, w) \leq \mathbf{1}_{x,w \geq 0} 2^{-K} \det(\mathbf{D}) \frac{w^{K-2}}{\Gamma(K-1)} \times \exp \left[-\frac{\min_i \lambda_i}{2} (x + w) \right], \quad K \geq 3. \quad (94)$$

Note that substituting $K = 2$ to (94) reduces it to (91).

Finally, setting $\lambda' = 2^{-K} \det(\mathbf{D})$ and $\lambda'' = \frac{\min_i \lambda_i}{2}$, we obtain the same upper bound as in the statement of the lemma for any $K \geq 2$. Since $\lambda_k > 0, \forall k$, we have $\det(\mathbf{D}) = \prod_{k=1}^K \lambda_k > 0$, and $\min_i \lambda_i > 0$. These imply that $\lambda', \lambda'' > 0$ as claimed. This concludes the proof. ■

Lemma 2: Let $X_1, \dots, X_n$ be n non-negative possibly dependent random variables, and $Z = \min_n X_n$. Then,

$$f_Z(z) \leq \sum_{i=1}^n f_{X_i}(z), \quad (95)$$

and

$$F_Z(z) \leq \sum_{i=1}^n F_{X_i}(z). \quad (96)$$

Proof: Let us recall Leibniz's integral rule: For functions of a single variable $a(z), b(z)$, and of two variables $f(x, z)$, we have

$$\frac{\partial}{\partial z} \int_{a(z)}^{b(z)} f(x, z) dx = \int_{a(z)}^{b(z)} \frac{\partial f}{\partial z} dx + f(b(z), z) \frac{\partial b}{\partial z} - f(a(z), z) \frac{\partial a}{\partial z}.$$

We first prove (95). Let $Z_k = \min\{X_1, \dots, X_k\}$. We will show that $f_{Z_k}(z) \leq \sum_{i=1}^k f_{X_i}(z)$, for any $1 \leq k \leq n$ by induction. For $k = 1$, it is obvious. Suppose it is true for $n > k \geq 1$. We have $f_{Z_k}(z) \leq \sum_{i=1}^k f_{X_i}(z)$. Noting that $Z_{k+1} = \min(Z_k, X_{k+1})$,

$$\begin{aligned} F_{Z_{k+1}}(z) &= P(Z_{k+1} \leq z) \\ &= P(Z_k \leq z \text{ or } X_{k+1} \leq z) \\ &= P(Z_k \leq z) + P(X_{k+1} \leq z) - \\ &\quad P(Z_k \leq z \text{ and } X_{k+1} \leq z) \\ &= F_{Z_k}(z) + F_{X_{k+1}}(z) - F_{Z_k, X_{k+1}}(z, z), \end{aligned}$$

$$\begin{aligned} & f_{Z_{k+1}}(z) \\ &= \frac{\partial}{\partial z} (F_{Z_k}(z) + F_{X_{k+1}}(z) - F_{Z_k, X_{k+1}}(z, z)) \\ &= f_{Z_k}(z) + f_{X_{k+1}}(z) - \frac{\partial}{\partial z} F_{Z_k, X_{k+1}}(z, z) \\ &\leq \sum_{i=1}^{k+1} f_{X_i}(z) - \frac{\partial}{\partial z} \int_0^z \int_0^z f_{Z_k, X_{k+1}}(u, v) du dv \\ &= \sum_{i=1}^{k+1} f_{X_i}(z) - \int_0^z \left\{ \frac{\partial}{\partial z} \int_0^z f_{Z_k, X_{k+1}}(u, v) dv \right\} du - \end{aligned} \quad (97)$$

$$\int_0^z f_{Z_k, X_{k+1}}(z, v) dv \quad (98)$$

$$\begin{aligned} &= \sum_{i=1}^{k+1} f_{X_i}(z) - \int_0^z f_{Z_k, X_{k+1}}(u, z) du - \\ &\quad \int_0^z f_{Z_k, X_{k+1}}(z, v) dv \quad (99) \end{aligned}$$

$$\leq \sum_{i=1}^{k+1} f_{X_i}(z),$$

where (97) follows from our assumption that $f_{Z_k}(z) \leq \sum_{i=1}^k f_{X_i}(z)$, and for both (98) and (99), we have used Leibniz's integral rule. This proves (95). Integrating both sides of (95) from 0 to z proves (96). ■

Lemma 3: Let $\alpha_1, \dots, \alpha_N$ be non-negative real constants. For any N non-negative real valued functions $f_n, n = 1, \dots, N$ of m variables $x_1, \dots, x_m$, let $f_n \triangleq f_n(x_1, \dots, x_m)$. Then, we have

$$\begin{aligned} \sum_{n=1}^N \alpha_n f_n &\leq \max_i \alpha_i \sum_{n=1}^N f_n, \\ \exp(-\sum_{n=1}^N \alpha_n f_n) &\leq \exp(-\min_i \alpha_i \sum_{n=1}^N f_n), \\ \sum_{n=1}^N \alpha_n f_n &\geq \min_i \alpha_i \sum_{n=1}^N f_n, \\ \exp(-\sum_{n=1}^N \alpha_n f_n) &\geq \exp(-\max_i \alpha_i \sum_{n=1}^N f_n). \end{aligned}$$

The proof is omitted since it is trivial. ■

Lemma 4: For $-\infty < a < b < \infty$, and $c \geq 0, c^x \leq \max\{c^a, c^b\}, \forall x \in (a, b)$, with the convention that $0^0 = 1$.

Proof: For $c \geq 1, c^x$ is an increasing function of x , and thus $c^x \leq c^b$. For $0 \leq c \leq 1, c^x$ is a decreasing function of x , and thus $c^x \leq c^a$. Combining the two bounds gives the desired result. ■

Lemma 5: For $0 < a < b < \infty, \forall x \in (a, b)$,

$$\Gamma(x) \leq \begin{cases} \Gamma(a), & b \leq \gamma_{\min} \\ \Gamma(b), & a \geq \gamma_{\min} \\ \max\{\Gamma(a), \Gamma(b)\}, & a < \gamma_{\min} < b \end{cases}, \quad (100)$$

where $\gamma_{\min} = 1.46163\dots$ is the unique positive real number at which the derivative of $\Gamma(\cdot)$ vanishes.

Proof: The proof is straightforward once we observe that the Gamma function is decreasing on $(0, \gamma_{\min}]$, and increasing on $[\gamma_{\min}, \infty)$ [74, Fig. 6.2 and Eq. 6.3.19]. ■

Lemma 6: Let $0 \leq \alpha \leq 1$, and $x_1, \dots, x_n \geq 0$. Then, $(\sum_{i=1}^n x_i)^\alpha \leq \sum_{i=1}^n x_i^\alpha$.

Proof: It is sufficient to prove the case for $n = 2$, as the extension to any $n \in \mathbb{N}$ is straightforward by induction. We need to show that for $x, y \geq 0$, we have $(x + y)^\alpha \leq x^\alpha + y^\alpha$. The proof is trivial if at least one of the following four conditions are satisfied: $\alpha = 0$, $\alpha = 1$, $x = 0$ or $y = 0$. We thus prove the case $0 < \alpha < 1$ with $x, y > 0$. Suppose that $0 < y \leq x$. Since $f(x) = \frac{x^\alpha}{x} = x^{\alpha-1}$ is decreasing on $(0, \alpha)$, $f(x + y) \leq f(x)$, or equivalently $\frac{(x+y)^\alpha}{x+y} \leq \frac{x^\alpha}{x}$. Multiplying both sides of the inequality with $x + y$, we have $(x + y)^\alpha \leq x^\alpha(1 + \frac{y}{x})$. Since $z \leq z^\alpha, \forall z \in (0, 1]$ and $0 < \frac{y}{x} \leq 1$, we have $\frac{y}{x} \leq (\frac{y}{x})^\alpha$. Therefore, $(x + y)^\alpha < x^\alpha(1 + (\frac{y}{x})^\alpha) = x^\alpha + y^\alpha$, concluding the proof for $0 < y \leq x$. The proof for $0 < x \leq y$ can be carried out in a similar manner. ■

Lemma 7: Let $\beta \geq 1$, and $x_1, \dots, x_n \geq 0$. Then, $(\sum_{i=1}^n x_i)^\beta \leq n^{\beta-1} \sum_{i=1}^n x_i^\beta$.

Proof: The proof is trivial if $\beta = 1$. We prove the case $\beta > 1$ using Hölder's inequality: For any $x_1, y_1, \dots, x_n, y_n \geq 0$, we have $\sum_{i=1}^n x_i y_i \leq (\sum_{i=1}^n x_i^p)^{\frac{1}{p}} (\sum_{i=1}^n y_i^q)^{\frac{1}{q}}$ with $p, q \in (1, \infty)$ and $\frac{1}{p} + \frac{1}{q} = 1$. Setting $y_1 = \dots = y_n = 1$, $p = \beta$, and $q = \frac{\beta}{\beta-1}$, we have $\sum_{i=1}^n x_i \leq (\sum_{i=1}^n x_i^\beta)^{\frac{1}{\beta}} n^{\frac{\beta-1}{\beta}}$. Raising both sides of the inequality to the power β gives the desired result. ■

B. Upper Bounds on the PDF and the CDF of Ω_r

We can now find the desired upper bounds on the PDF and CDF of $\Omega_r = \gamma^L(\mathbf{e}_r, \mathbf{h})$.

Proposition 3: For all $\omega > 0$, $y \geq 0$ and P sufficiently large,

1) If $K = 1$,

$$f_{\Omega_r}(\omega) \leq C_{25} \exp\left(-C_{26} \frac{\omega}{P}\right) \psi_0(\omega), \quad (101)$$

and

$$F_{\Omega_r}(y + \omega) - F_{\Omega_r}(y) \leq C_{25} \omega \psi_0(\omega), \quad (102)$$

where

$$\psi_0(\omega) \triangleq \frac{1}{P} \left(1 + \omega^{-\frac{1}{2}}\right), \quad (103)$$

and $0 < C_{25}, C_{26} < \infty$ are constants. Otherwise,

2) If $K > 1$,

$$f_{\Omega_r}(\omega) \leq C_{27} \exp\left(-C_{28} \frac{\omega}{P}\right) \psi(\omega), \quad (104)$$

$$F_{\Omega_r}(y + \omega) - F_{\Omega_r}(y) \leq C_{27} \left(\omega \psi(\omega) + \frac{\log P}{P^2} y^{2 - \frac{1}{\log P}} \right), \quad (105)$$

and in particular, for $y = 0$,

$$F_{\Omega_r}(\omega) \leq C_{27} \omega \psi(\omega), \quad (106)$$

where

$$\psi(\omega) = \frac{\log P}{P} \left(1 + \omega^{-\frac{1}{2 \log P}} + \omega^{1 - \frac{1}{\log P}} \frac{1}{P} \right), \quad (107)$$

and $0 < C_{27}, C_{28} < \infty$ are constants.

Proof: First we prove the case for $K > 1$. According to (16), we have

$$\Omega_r = \min_{\ell} \min_{\mathbf{s} \neq \hat{\mathbf{s}}} \Omega_{r, \ell, \mathbf{s}, \hat{\mathbf{s}}}, \quad (108)$$

where

$$\Omega_{r, \ell, \mathbf{s}, \hat{\mathbf{s}}} \triangleq \frac{1}{4\bar{C}} \frac{\left| \sum_{k=1}^K (s_k - \hat{s}_k) \sqrt{P_{S_k}} f_{kr} \right|^2 |g_{r\ell}|^2 P_{R_r}}{1 + \sum_{k=1}^K |f_{kr}|^2 P_{S_k} + |g_{r\ell}|^2 P_{R_r}}. \quad (109)$$

We have omitted the condition $\mathbf{s}, \hat{\mathbf{s}} \in \mathcal{S}$ in (108) as it is clear from the context.

First, we find an upper bound on the PDF and the CDF of $\Omega_{r, \ell, \mathbf{s}, \hat{\mathbf{s}}}$.

Consider a fixed r, ℓ , and $\mathbf{s} \neq \hat{\mathbf{s}}$. For notational convenience, let $Z \triangleq \Omega_{r, \ell, \mathbf{s}, \hat{\mathbf{s}}}$ is as defined in (109). Also, let

$$\mathbf{f}' \triangleq [\sqrt{p_{S_1}} f_{1r} \cdots \sqrt{p_{S_K}} f_{Kr}]^T, \quad (110)$$

$$\boldsymbol{\delta} \triangleq [s_1 - \hat{s}_1 \cdots s_K - \hat{s}_K]^H, \quad (111)$$

$$\tilde{\boldsymbol{\delta}} \triangleq \boldsymbol{\delta} / \|\boldsymbol{\delta}\|, \quad (112)$$

$$X \triangleq |\langle \mathbf{f}', \tilde{\boldsymbol{\delta}} \rangle|^2, \quad (113)$$

$$Y \triangleq |g_{r\ell}|^2 P_{R_r}, \quad (114)$$

and

$$W \triangleq \|\mathbf{f}'\|^2 - X. \quad (115)$$

Then, we have

$$Z = \frac{1}{4\bar{C}} \frac{\left| \sum_{k=1}^K (s_k - \hat{s}_k) \sqrt{p_{S_k}} P f_{kr} \right|^2 |g_{r\ell}|^2 P_{R_r} P}{1 + \sum_{k=1}^K |f_{kr}|^2 p_{S_k} P + |g_{r\ell}|^2 P_{R_r} P} \quad (116)$$

$$= \frac{1}{4\bar{C}} \frac{|\sum_{k=1}^K \delta_k^H f'_k|^2 Y P^2}{1 + \sum_{k=1}^K |f'_k|^2 P + Y P} \quad (117)$$

$$= \frac{1}{4\bar{C}} \frac{|\langle \mathbf{f}', \boldsymbol{\delta} \rangle|^2 Y P^2}{1 + \|\mathbf{f}'\|^2 P + Y P} \quad (118)$$

$$= \frac{\|\boldsymbol{\delta}\|^2}{4\bar{C}} \frac{|\langle \mathbf{f}', \tilde{\boldsymbol{\delta}} \rangle|^2 Y P^2}{1 + \|\mathbf{f}'\|^2 P + Y P} \quad (119)$$

$$= \frac{C_{29} X Y P^2}{1 + X P + W P + Y P}, \quad (120)$$

where $C_{29} = \frac{1}{4\bar{C}} \|\boldsymbol{\delta}\|^2$. The first equality follows from (109) and the identities $P_{S_k} = p_{S_k} P$ and $P_{R_r} = p_{R_r} P$ provided in Section I-A. For (117), we have used the definitions of $\mathbf{f}'$, $\boldsymbol{\delta}$,

and Y in (110), (111), and (114), respectively. The third equality follows since $\sum_{k=1}^K \delta_k^H f'_k = \langle \mathbf{f}', \tilde{\boldsymbol{\delta}} \rangle$, and $\sum_{k=1}^K |f'_k|^2 = \|\mathbf{f}'\|^2$. Also, (119) holds by the definition of $\tilde{\boldsymbol{\delta}}$, and (120) follows from (113) and (115).

Using a transformation of random variables [72], we have

$$f_{Z,X,W}(z, x, w) = f_{Y,X,W}(y, x, w) |\det(\mathbf{J}_0)| \quad (121)$$

$$= f_Y(y) f_{X,W}(x, w) |\det(\mathbf{J}_0)|, \quad (122)$$

where the Jacobian matrix $\mathbf{J}_0$ is given by

$$\mathbf{J}_0 = \begin{bmatrix} \frac{\partial y}{\partial z} & \frac{\partial y}{\partial x} & \frac{\partial y}{\partial w} \\ \frac{\partial x}{\partial z} & \frac{\partial x}{\partial x} & \frac{\partial x}{\partial w} \\ \frac{\partial w}{\partial z} & \frac{\partial w}{\partial x} & \frac{\partial w}{\partial w} \end{bmatrix}, \quad (123)$$

and (122) holds since Y is independent of X and W .

From (120), we have $z = \frac{C_{29}xyP^2}{1+xP+wP+yP}$. Solving for y , we obtain

$$y = \frac{z(1+xP+wP)}{C_{29}xP^2 - zP}, \quad (124)$$

Now, since $\frac{\partial x}{\partial z} = \frac{\partial x}{\partial w} = \frac{\partial w}{\partial z} = \frac{\partial w}{\partial x} = 0$, the matrix in (123) is a triangular matrix, and its determinant is equal to the product of its diagonal elements. Therefore,

$$\det(\mathbf{J}_0) = \frac{\partial y}{\partial z} \frac{\partial x}{\partial x} \frac{\partial w}{\partial w} = \frac{C_{29}xP^2(1+xP+wP)}{(C_{29}xP^2 - zP)^2}, \quad (125)$$

where the partial derivative $\frac{\partial y}{\partial z}$ is calculated using (124).

Substituting (124) and (125) to (122) and averaging out X and W , we obtain

$$f_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_Y\left(\frac{z(1+xP+wP)}{C_{29}xP^2 - zP}\right) \times \left| \frac{C_{29}xP^2(1+xP+wP)}{(C_{29}xP^2 - zP)^2} \right| f_{X,W}(x, w) dx dw. \quad (126)$$

Since X, Y and W are nonnegative random variables, we need to evaluate (126) only for $x, w, z, \frac{z(1+xP+wP)}{C_{29}xP^2 - zP} \geq 0$, or equivalently for $z, w \geq 0$ and $x \geq \frac{z}{C_{29}P}$. Also, with such conditions on x and w , the value of the absolute value in (126) is equal to its argument. Therefore,

$$f_Z(z) = \mathbf{1}_{z \geq 0} \int_0^{\infty} \int_{\frac{z}{C_{29}P}}^{\infty} f_Y\left(\frac{z(1+xP+wP)}{C_{29}xP^2 - zP}\right) \times \frac{C_{29}xP^2(1+xP+wP)}{(C_{29}xP^2 - zP)^2} f_{X,W}(x, w) dx dw. \quad (127)$$

In order to evaluate $f_Z(z)$, we need the PDF of Y and the joint PDF of X and W . First, we determine the PDF of Y . According to our definition in Section I-A, $g_{r\ell} \sim \text{CN}(0, \sigma_{g_{r\ell}}^2)$, and thus $|g_{r\ell}|^2 \sim \Gamma(1, \sigma_{g_{r\ell}}^2)$. Therefore, using (114), we have

$Y = |g_{r\ell}|^2 p_{R_r} \sim \Gamma(1, \sigma_{g_{r\ell}}^2 p_{R_r})$. In other words, letting $C_{30} \triangleq (\sigma_{g_{r\ell}}^2 p_{R_r})^{-1}$, we have

$$f_Y(y) = \mathbf{1}_{y \geq 0} C_{30} \exp(-C_{30}y). \quad (128)$$

Since our current goal is to find an upper bound on the PDF and the CDF of Z , according to (127), we can work with an upper bound on $f_{X,W}(x, w)$ rather than the exact $f_{X,W}(x, w)$. We now find such a bound on the joint PDF of X and W using Lemma 1. Note that $\Re(\mathbf{f}')$ and $\Im(\mathbf{f}')$ have equal diagonal covariance matrices, i.e., $\mathbb{E}[\Re(\mathbf{f}')\Re(\mathbf{f}')^T] = \mathbb{E}[\Im(\mathbf{f}')\Im(\mathbf{f}')^T] = \mathbf{K}'$, where $K'_{ii} = \sigma_{f_{ir}}^2$ and $K'_{ij} = 0 \iff i \neq j$. Also, let us recall from (113) and (115) that $X = \langle \mathbf{f}', \tilde{\boldsymbol{\delta}} \rangle^2$, and $W = \|\mathbf{f}'\|^2 - X$ where $\|\tilde{\boldsymbol{\delta}}\| = 1$. We can therefore apply Lemma 1 to obtain

$$f_{X,W}(x, w) \leq \mathbf{1}_{x, w \geq 0} C_{31} w^{K-2} \exp[-C_{32}(x+w)], \quad (129)$$

where $0 < C_{31}, C_{32} < \infty$ are constants.

Substituting (128) and (129) to (127), we have

$$f_Z(z) \leq \mathbf{1}_{z \geq 0} C_{30} C_{31} \int_0^{\infty} \int_{\frac{z}{C_{29}P}}^{\infty} \exp\left(-\frac{C_{30}z(1+xP+wP)}{C_{29}xP^2 - zP}\right) \times \frac{C_{29}xP^2(1+xP+wP)}{(C_{29}xP^2 - zP)^2} e^{-C_{32}x} dx w^{K-2} e^{-C_{32}w} dw. \quad (130)$$

From now on, we omit the factor $\mathbf{1}_{z \geq 0}$ as it is clear from the context.

Using a change of variables $u = C_{29}xP - z$,

$$\begin{aligned} f_Z(z) &\leq C_{30} C_{31} \int_0^{\infty} \int_0^{\infty} \exp\left(-\frac{C_{30}z(1 + \frac{u+z}{C_{29}} + wP)}{uP}\right) \times \\ &\quad \frac{(u+z)P(1 + \frac{u+z}{C_{29}} + wP)}{u^2 P^2} e^{-C_{32}\frac{u+z}{C_{29}P}} \frac{du}{C_{29}P} \frac{w^{K-2}}{e^{C_{32}w}} dw \\ &= C_{30} C_{32} \exp\left[-\underbrace{\frac{C_{30} + C_{32}}{C_{29}}}_{\triangleq C_{33}} \frac{z}{P}\right] \times \\ &\quad \int_0^{\infty} \int_0^{\infty} \exp\left[-\frac{C_{30}z(C_{29} + z + C_{29}wP)}{C_{29}P} \frac{1}{u} - \frac{C_{32}}{C_{29}P} u\right] \times \\ &\quad \left[\frac{1}{C_{29}^2 P^2} + \frac{C_{29} + 2z + C_{29}wP}{C_{29}^2 P^2} \frac{1}{u} + \frac{z(C_{29} + z + C_{29}wP)}{C_{29}^2 P^2} \frac{1}{u^2} \right] du w^{K-2} e^{-C_{32}w} dw \\ &\leq C_{30} C_{31} e^{-\frac{C_{33}z}{P}} \int_0^{\infty} \int_0^{\infty} \exp\left\{-\underbrace{\frac{\min\{C_{30}, C_{30}C_{29}, C_{32}\}}{C_{29}}}_{\triangleq C_{34}} \right. \\ &\quad \left. \left[\frac{z(1+z+wP)}{P} \frac{1}{u} + \frac{1}{P} u \right] \right\} \underbrace{\frac{\max\{2, C_{29}\}}{C_{29}^2}}_{\triangleq C_{35}} \times \end{aligned}$$

$$\left[\frac{1}{P^2} + \frac{1+z+wP}{P^2} \frac{1}{u} + \frac{z(1+z+wP)}{P^2} \frac{1}{u^2} \right] \times \int_0^\infty u^{\alpha-1} e^{-\frac{\beta}{u} - \gamma u} du = 2 \left(\frac{\beta}{\gamma} \right)^{\frac{\alpha}{2}} K_\alpha \left(\sqrt{4\beta\gamma} \right), \beta, \gamma > 0. \quad (131)$$

where (131) follows from two applications of Lemma 3; one for the exponential function, and the other for the last square-bracketed expression inside the integral.

The inner integral in (131) can be evaluated using [75, Eq. 3.417.9]:

$$\int_0^\infty u^{\alpha-1} e^{-\frac{\beta}{u} - \gamma u} du = 2 \left(\frac{\beta}{\gamma} \right)^{\frac{\alpha}{2}} K_\alpha \left(\sqrt{4\beta\gamma} \right), \beta, \gamma > 0. \quad (132)$$

This gives us

$$f_Z(z) \leq C_{30} C_{31} C_{35} e^{-\frac{C_{33}z}{P}} \int_0^\infty \left[\frac{\kappa K_1(\kappa)}{C_{34}P} + \frac{z^{-1} \kappa^2 K_0(\kappa)}{2C_{34}^2} + \frac{\kappa K_{-1}(\kappa)}{C_{34}P} \right] w^{K-2} e^{-C_{32}w} dw,$$

where $\kappa = \sqrt{4C_{34}^2 z(1+z+wP)/P^2}$. Now, using the fact that $K_{-1}(x) = K_1(x)$, $\forall x \in \mathbb{R}$ [74, Eq. 9.6.6], and then applying Lemma 3 to take out the constants inside the parentheses, we obtain

$$f_Z(z) \leq C_{30} C_{31} C_{35} \underbrace{\max \left\{ \frac{2}{C_{34}}, \frac{1}{2C_{34}^2} \right\}}_{\triangleq C_{36}} e^{-\frac{C_{33}z}{P}} \times \int_0^\infty [P^{-1} \kappa K_1(\kappa) + z^{-1} \kappa^2 K_0(\kappa)] w^{K-2} e^{-C_{32}w} dw. \quad (133)$$

Now, let us find an upper bound for $K_0(\kappa)$ in (133). According to [74, Eq. 9.6.24], we have $K_\nu(z) = \int_0^\infty e^{-z \cosh t} \cosh(\nu t) dt$, $t, \nu \in \mathbb{R}$. Moreover, since $\cosh(\nu t)$ is an increasing function of ν , $K_\nu(z)$ is also an increasing function of ν . It follows that

$$K_0(\kappa) \leq K_\nu(\kappa), \nu \geq 0. \quad (134)$$

Also, from [8, Eq. 25], we have

$$K_\nu(\kappa) \leq 2^{\nu-1} \Gamma(\nu) \kappa^{-\nu}, \nu > 0. \quad (135)$$

Now let us set $0 < \nu < 1$. In this case,

$$\begin{aligned} \Gamma(\nu) &= \nu^{-1} \Gamma(\nu+1) \\ &\leq \nu^{-1} \max_{0 \leq \nu' \leq 1} \Gamma(\nu'+1) \\ &= \nu^{-1} \max_{1 \leq \nu' \leq 2} \Gamma(\nu') \\ &= \nu^{-1} \max\{\Gamma(1), \Gamma(2)\} \\ &= \nu^{-1}, \end{aligned} \quad (136)$$

where the second last equality follows from the third condition in (100) of Lemma 5.

Combining (134), (135) and (136) yields

$$\begin{aligned} K_0(\kappa) &\leq 2^{\nu-1} \nu^{-1} \kappa^{-\nu} \\ &\leq \nu^{-1} \kappa^{-\nu}, \end{aligned} \quad (137)$$

where the last inequality follows since $2^{\nu-1} < 1$ for $0 < \nu < 1$ by Lemma 4.

Since $\kappa K_1(\kappa) \leq 1$ [8, Eq. 25], and $1 < \nu^{-1}$, we have $\kappa K_1(\kappa) < \nu^{-1}$. Using this inequality and (137), (133) can be further bounded as

$$\begin{aligned} f_Z(z) &\leq C_{36} e^{-\frac{C_{33}z}{P}} \frac{1}{\nu P} \int_0^\infty (1+z^{-1}P\kappa^{2-\nu}) w^{K-2} e^{-C_{32}w} dw \\ &= C_{36} e^{-\frac{C_{33}z}{P}} \frac{1}{\nu P} \left[\int_0^\infty w^{K-2} e^{-C_{32}w} dw + \right. \\ &\quad \left. z^{-1}P \int_0^\infty \kappa^{2-\nu} w^{K-2} e^{-C_{32}w} dw \right] \end{aligned} \quad (138)$$

$$\begin{aligned} &= C_{36} e^{-\frac{C_{33}z}{P}} \frac{1}{\nu P} \left(\frac{\Gamma(K-1)}{C_{32}^{K-1}} + (2C_{34})^{2-\nu} \frac{z^{-\frac{\nu}{2}}}{P^{1-\nu}} \times \right. \\ &\quad \left. \int_0^\infty (1+z+wP)^{1-\frac{\nu}{2}} w^{K-2} e^{-C_{32}w} dw \right) \\ &\leq C_{36} \underbrace{\max \left\{ \frac{\Gamma(K-1)}{C_{32}^{K-1}}, 2C_{34}, (2C_{34})^2 \right\}}_{\triangleq C_{37}} e^{-\frac{C_{33}z}{P}} \frac{1}{\nu P} \times \end{aligned} \quad (139)$$

$$\left(1 + \frac{z^{-\frac{\nu}{2}}}{P^{1-\nu}} \int_0^\infty (1+z^{1-\frac{\nu}{2}} + w^{1-\frac{\nu}{2}} P^{1-\frac{\nu}{2}}) w^{K-2} e^{-C_{32}w} dw \right) \quad (140)$$

$$\begin{aligned} &= C_{37} e^{-\frac{C_{33}z}{P}} \frac{1}{\nu P} \left\{ 1 + \frac{z^{-\frac{\nu}{2}}}{P^{1-\nu}} \left[(1+z^{1-\frac{\nu}{2}}) \frac{\Gamma(K-1)}{C_{32}^{K-1}} + \right. \right. \\ &\quad \left. \left. \frac{\Gamma(K-\frac{\nu}{2})}{C_{32}^{K-\frac{\nu}{2}}} P^{1-\frac{\nu}{2}} \right] \right\}. \end{aligned} \quad (141)$$

In the derivation above, (139) follows from the evaluation of the first integral in (138) and the substitution of the value of κ . For (140), we have used Lemmas 3, 4, and 6.

Now, $K - \frac{1}{2} < K - \frac{\nu}{2} < K$, and since $K - \frac{1}{2} > \frac{3}{2} > \gamma_0$, $\forall K \geq 2$, according to Lemma 5, we have $\Gamma(K - \frac{\nu}{2}) \leq \Gamma(K)$. Substituting this inequality to (141), then applying Lemmas 3 and 4, we have

$$\begin{aligned} f_Z(z) &\leq C_{37} \underbrace{\max \left\{ 1, \frac{\Gamma(K-1)}{C_{32}^{K-1}}, \frac{\Gamma(K)}{C_{32}^K}, \frac{\Gamma(K)}{C_{32}^{K-\frac{1}{2}}} \right\}}_{\triangleq C_{38}} \times \\ &\quad \frac{e^{-\frac{C_{33}z}{P}}}{\nu P} \left[1 + \frac{z^{-\frac{\nu}{2}}}{P^{1-\nu}} (1 + z^{1-\frac{\nu}{2}} + P^{1-\frac{\nu}{2}}) \right] \end{aligned}$$

$$= C_{38} e^{-\frac{C_{33} z}{P}} \frac{\log P}{P} \left[1 + z^{-\frac{1}{2 \log P}} P^{-1 + \frac{1}{\log P}} + z^{1 - \frac{1}{\log P}} P^{-1 + \frac{1}{\log P}} + z^{-\frac{1}{2 \log P}} P^{\frac{1}{2 \log P}} \right], \quad (142)$$

where (142) follows from the substitution $\nu = \frac{1}{\log P}$ and rearranging the terms. Note that for $\nu = \frac{1}{\log P}$, the condition $0 < \nu < 1$ is satisfied for $P > e$.

Regarding (142), note that $P^{\frac{1}{\log P}} = [\exp(\log P)]^{\frac{1}{\log P}} = \exp(\frac{\log P}{\log P}) = e$, and similarly $P^{\frac{1}{2 \log P}} = \sqrt{e}$. Substituting these equalities to (142), and applying Lemma 3, we obtain

$$\begin{aligned} f_Z(z) &\leq e C_{38} e^{-\frac{C_{33} z}{P}} \frac{\log P}{P} \left(1 + z^{-\frac{1}{2 \log P}} P^{-1} + z^{1 - \frac{1}{\log P}} P^{-1} + z^{-\frac{1}{2 \log P}} P^{\frac{1}{2 \log P}} \right) \\ &\leq \underbrace{2e C_{38}}_{\triangleq C_{39}} e^{-\frac{C_{33} z}{P}} \frac{\log P}{P} \left(1 + z^{-\frac{1}{2 \log P}} + z^{1 - \frac{1}{\log P}} P^{-1} \right). \end{aligned} \quad (143)$$

$$\triangleq C_{39} \quad (144)$$

For (144), we have applied the inequality $P^{-1} \leq 1$ for $P \geq 1$ on the second term in (143) and used Lemma 3.

In general, the constants C_{39} and C_{33} in (144) depend on r , ℓ , s , and $\hat{s}$. Let $C_{39,r,\ell,s,\hat{s}}$ and $C_{33,r,\ell,s,\hat{s}}$ denote the dependent versions of C_{39} and C_{33} , respectively. Using Lemma 2, we have

$$\begin{aligned} f_{\Omega_r}(\omega) &\leq \sum_{\ell} \sum_{s \neq \hat{s}} \Omega_{r,\ell,s,\hat{s}}(\omega) \\ &\leq \frac{C_{27} \log P}{2} e^{-C_{28} \frac{\omega}{P}} \left(1 + \omega^{-\frac{1}{2 \log P}} + \omega^{1 - \frac{1}{\log P}} P^{-1} \right), \end{aligned} \quad (145)$$

where $C_{27} = 2L|\mathcal{A}||\mathcal{A} - 1| \max_{r,\ell,s \neq \hat{s}} C_{39,r,\ell,s,\hat{s}}$, and $C_{28} = \min_{r,\ell,s \neq \hat{s}} C_{33,r,\ell,s,\hat{s}}$ are the constants in the statement of the proposition. This implies the upper bound on the PDF of Ω_r . Now,

$$\begin{aligned} F_{\Omega_r}(y + \omega) - F_{\Omega_r}(y) &= \int_y^{y+\omega} f_{\Omega_r}(\omega) d\omega \\ &\leq \frac{C_{27} \log P}{2} \frac{1}{P} \int_y^{y+\omega} \left(1 + x^{-\frac{1}{2 \log P}} + x^{1 - \frac{1}{\log P}} \frac{1}{P} \right) dx \\ &= \frac{C_{27} \log P}{2} \frac{1}{P} \left\{ \omega + \frac{2 \log P}{2 \log P - 1} \left[(y + \omega)^{1 - \frac{1}{2 \log P}} - y^{1 - \frac{1}{2 \log P}} \right] + \frac{\log P}{2 \log P - 1} \frac{(y + \omega)^{2 - \frac{1}{\log P}} - y^{2 - \frac{1}{\log P}}}{P} \right\} \end{aligned} \quad (146)$$

$$\quad (147)$$

where (146) follows from (145) and the fact that $\exp(-C_{26} \frac{\omega}{P}) \leq 1$.

Let us now find a more compact bound for (147). First, for the coefficients of the second and the third term inside the curly-bracketed expression, we have

$$\begin{aligned} P > e &\implies \log P > 1 \implies \log P < 2 \log P - 1 \\ &\implies \frac{\log P}{2 \log P - 1} < 1, \end{aligned} \quad (148)$$

and similarly

$$\frac{2 \log P}{2 \log P - 1} < 2. \quad (149)$$

Also, $\frac{1}{2} < 1 - \frac{2}{\log P} < 1$ for $P > e$, and thus we can apply Lemma 6 to obtain

$$(y + \omega)^{1 - \frac{2}{\log P}} \leq y^{1 - \frac{2}{\log P}} + \omega^{1 - \frac{2}{\log P}}. \quad (150)$$

Similarly, $1 < 2 - \frac{1}{\log P} < 2$ for $P > e$, and by Lemma 7,

$$\begin{aligned} (y + \omega)^{2 - \frac{1}{\log P}} &\leq 2^{1 - \frac{1}{\log P}} (y^{2 - \frac{1}{\log P}} + \omega^{2 - \frac{1}{\log P}}) \\ &< 2(y^{2 - \frac{1}{\log P}} + \omega^{2 - \frac{1}{\log P}}). \end{aligned} \quad (151)$$

The last inequality follows from Lemma 4.

Substituting (148), (149), (150), and (151) to (147), and after some simplifications, we have

$$\begin{aligned} F_{\Omega_r}(y + \omega) - F_{\Omega_r}(y) &\leq \frac{C_{27} \log P}{2} \frac{1}{P} \left\{ \omega + 2\omega^{1 - \frac{1}{2 \log P}} + \frac{y^{2 - \frac{1}{\log P}} + 2\omega^{2 - \frac{1}{\log P}}}{P} \right\} \\ &\leq C_{27} \frac{\log P}{P} \left\{ \omega + \omega^{1 - \frac{1}{2 \log P}} + \frac{y^{2 - \frac{1}{\log P}} + \omega^{2 - \frac{1}{\log P}}}{P} \right\} \\ &= C_{27} \left\{ \omega \psi(\omega) + y^{2 - \frac{1}{\log P}} \frac{\log P}{P^2} \right\}, \end{aligned}$$

which is the same upper bound as in the statement of the proposition. The second inequality follows from Lemma 3. This concludes the proof for $K > 1$.

For $K = 1$, we observe by (108) and (109) that Ω_r has the following form

$$\begin{aligned} \Omega_r &= \min_{\ell} \min_{s_1 \neq \hat{s}_1} \frac{1}{4} \frac{|(s_1 - \hat{s}_1) \sqrt{P_{S_1}} f_{1r}|^2 |g_{r\ell}|^2 P_{R_r}}{1 + |f_{1r}|^2 P_{S_1} + |g_{r\ell}|^2 P_{R_r}} \\ &= \underbrace{\min_{s_1 \neq \hat{s}_1} \frac{|s_1 - \hat{s}_1|^2}{4}}_{\triangleq C_{40}} \frac{|f_{1r}|^2 P_{S_1} \min_{\ell} |g_{r\ell}|^2 P_{R_r}}{1 + |f_{1r}|^2 P_{S_1} + \min_{\ell} |g_{r\ell}|^2 P_{R_r}} \\ &= C_{40} \frac{\bar{X}_r \bar{Y}_r P^2}{1 + \bar{X}_r P + \bar{Y}_r P}, \end{aligned}$$

where $\bar{X}_r \triangleq |f_{1r}|^2 P_{S_1}$, and $\bar{Y}_r \triangleq p_{R_r} \min_{\ell} |g_{r\ell}|^2$. It is straightforward to show that $\bar{X}_r \sim \Gamma(1, C_{41,r})$ and $\bar{Y}_r \sim \Gamma(1, C_{42,r})$, where $C_{41,r} = p_{S_1} \sigma_{f_{1r}}^2$ and $C_{42,r} = p_{R_r} (\sum_{\ell} \sigma_{g_{r\ell}}^2)^{-1}$. Moreover, $\bar{X}_r, \bar{Y}_r, r = 1, \dots, R$ are independent.

Now, let $\Upsilon_r \triangleq \frac{\bar{X}_r \bar{Y}_r P^2}{1 + \bar{X}_r P + \bar{Y}_r P}$ and note that $\Omega_r = C_{40} \Upsilon_r$. Combining this with the PDF of Υ_r given by [8, Eq. 25], we have

$$\begin{aligned} f_{\Omega_r}(\omega) &= \exp\left(-\frac{C_{41,r} + C_{42,r}}{C_{40}} \frac{\omega}{P}\right) \frac{1}{P^2} \times \\ &\quad \left[P(C_{41,r} + C_{42,r}) \bar{\kappa} K_1(\bar{\kappa}) + 2C_{41,r} C_{42,r} K_0(\bar{\kappa}) \left(1 + \frac{2\omega}{C_{40}}\right) \right], \end{aligned} \quad (152)$$

where

$$\bar{\kappa} \triangleq \sqrt{\frac{4C_{41,r}C_{42,r}\omega(1 + \frac{\omega}{C_{40}})}{C_{40}P^2}},$$

and we have similarly omitted the factor $\mathbf{1}_{\omega \geq 0}$ on the right hand side of (152) for brevity.

Using the inequalities $\bar{\kappa}K_1(\bar{\kappa}) \leq 1$ [8, Eq. 25], $K_0(\bar{\kappa}) \leq 2\bar{\kappa}^{-1}$ [8, Eq. 27], and substituting the value of $\bar{\kappa}$, we obtain

$$f_{\Omega_r}(\omega) \leq \exp\left(-\frac{C_{41,r} + C_{42,r}}{C_{40}} \frac{\omega}{P}\right) \frac{1}{P} \times \left[C_{41,r} + C_{42,r} + 2\sqrt{C_{41,r}C_{42,r}} \frac{C_{40} + 2\omega}{\sqrt{\omega(C_{40} + \omega)}} \right].$$

Since

$$\frac{C_{40} + 2\omega}{\sqrt{\omega(C_{40} + \omega)}} \leq 2 \frac{C_{40} + \omega}{\sqrt{\omega(C_{40} + \omega)}} = 2(\sqrt{C_{40}\omega}^{-\frac{1}{2}} + 1),$$

we have

$$f_{\Omega_r}(\omega) \leq \exp\left(-\frac{C_{41,r} + C_{42,r}}{C_{40}} \frac{\omega}{P}\right) \frac{1}{P} \times \left[C_{41,r} + C_{42,r} + 4\sqrt{C_{41,r}C_{42,r}}(1 + \sqrt{C_{40}\omega}^{-\frac{1}{2}}) \right] \leq \frac{C_{25}}{2} \exp\left(-C_{26} \frac{\omega}{P}\right) \frac{1}{P} (1 + \omega^{-\frac{1}{2}}), \quad (153)$$

where $C_{25} \triangleq 2 \max_r \{ \max \{ C_{41,r} + C_{42,r} + 4\sqrt{C_{41,r}C_{42,r}}, 4\sqrt{C_{41,r}C_{42,r}C_{40}} \} \}$ and $C_{26} \triangleq \min_r \frac{C_{41,r} + C_{42,r}}{C_{40}}$ are the constants in the statement of the proposition. The second inequality is a consequence of Lemma 3, and implies the upper bound in (101). Now,

$$\begin{aligned} F_{\Omega_r}(y + \omega) - F_{\Omega_r}(y) &= \int_y^{y+\omega} f_{\Omega_r}(x) dx \\ &\leq \frac{C_{25}}{2P} \int_y^{y+\omega} (1 + x^{-\frac{1}{2}}) dx \\ &= \frac{C_{25}}{2P} \left(\omega + 2(y + \omega)^{\frac{1}{2}} - 2y^{\frac{1}{2}} \right) \\ &\leq \frac{C_{25}}{2P} \left(\omega + 2\omega^{\frac{1}{2}} \right) \\ &\leq \frac{C_{25}}{P} \omega (1 + \omega^{-\frac{1}{2}}), \end{aligned}$$

where the first inequality follows from (153) and the fact that $\exp(-C_{26} \frac{\omega}{P}) \leq 1$. The second and third inequalities follow from Lemmas 6 and 3, respectively. This proves (102) and thus concludes the proof of the proposition. ■

C. Proof of (21) and (22)

Finally we can prove the desired upper bounds using the results in Appendices B-A and B-B. Here, we provide a proof of (22) that corresponds to the case $K > 1$. The proof of (21) regarding the case $K = 1$ is very similar. Thus, we skip it for brevity.

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Let $\Omega = \max_r \Omega_r$, where $\Omega_r = \gamma^L(\mathbf{e}_r, \mathbf{h})$ is the NSNR given that the r th relay is selected. Note that the NSNR provided by GQ_{C_8} is thus equal to Ω . Therefore, by Proposition 2, we have

$$\text{NER}(\text{GQ}_{C_8}) \quad (154)$$

$$\leq C_4 \mathbb{E}[\exp(-\Omega)] \quad (155)$$

$$\leq C_4 \int_0^\infty e^{-w} \sum_{r=1}^R f_{\Omega_r}(w) \prod_{\substack{q=1 \\ q \neq r}}^R F_{\Omega_q}(w) dw \quad (156)$$

$$\leq RC_4 C_{27}^R \frac{\log^R P}{P^R} \int_0^\infty \omega^{R-1} e^{-w} \exp\left(-C_{28} \frac{\omega}{P}\right) \times \left(1 + \omega^{-\frac{1}{2 \log P}} + \omega^{1 - \frac{1}{\log P}} \frac{1}{P}\right)^R d\omega \quad (157)$$

$$\leq R3^{R-1} C_4 C_{27}^R \frac{\log^R P}{P^R} \int_0^\infty \omega^{R-1} e^{-w} \times \left(1 + \omega^{-\frac{R}{2 \log P}} + \omega^{R - \frac{R}{\log P}} \frac{1}{P^R}\right) d\omega \quad (158)$$

$$\begin{aligned} &= R3^{R-1} C_4 C_{27}^R \frac{\log^R P}{P^R} \times \\ &\quad \left[\Gamma(R) + \Gamma\left(R - \frac{R}{2 \log P}\right) + \Gamma\left(2R - \frac{R}{\log P}\right) \frac{1}{P^R} \right] \\ &\leq R3^{R-1} C_4 C_{27}^R \frac{\log^R P}{P^R} [\Gamma(R) + \Gamma(R) + \Gamma(2R)] \\ &\leq C_{43} \frac{\log^R P}{P^R}, \end{aligned}$$

where $C_{43} = R3^R C_4 C_{27}^R \Gamma(2R)$, and (156) follows from the order statistics of independent RVs. For (157), we have used Proposition 3 and the fact that $\exp(-C_{28} \frac{\omega}{P}) \leq 1$. For (158), we have used Lemma 7. This concludes the proof. ■

APPENDIX C

PROOF OF THE LOWER BOUND FOR $K > 1$ IN THEOREM 2

We start with a lower bound on the CNER. Note that the CNER is the probability that at least one receiver incorrectly decodes its desired symbol(s). Using this definition, we have

$$\text{CNER}(\mathbf{x}, \mathbf{h}) \geq \text{SER}_1^{\text{IML}}(\mathbf{x}, \mathbf{h}). \quad (159)$$

In other words, the CNER is lower bounded by the SER at the first receiver. Suppose that, for some $k \in \mathcal{D}_1$, a genie reveals all the transmitted symbols but s_k to the first receiver. The error rate of this genie-aided scheme provides a lower bound on the SER at the first receiver. It follows from (159) that the error rate with the genie aided scheme is also a lower bound on the CNER. Without loss of generality assume that $1 \in \mathcal{D}_1$, i.e., the first transmitter is in the decoding set of the first receiver. Let us now fix some $\hat{s}_1, \hat{s}_2 \in \mathcal{S}_1$ with $\hat{s}_1 \neq \hat{s}_2$. We have

$$\text{CNER}(\mathbf{x}, \mathbf{h}) \geq \frac{1}{|\mathcal{S}_1|} \mathbb{Q}(\sqrt{2\gamma^U(\mathbf{x}, \mathbf{h})}), \quad (160)$$

where

$$\gamma^u(\mathbf{x}, \mathbf{h}) \triangleq \frac{|\sum_{r=1}^R f_{1r} \sqrt{\rho'_r} g_{r1} x_r|^2 |\dot{s}_1 - \dot{s}_2|^2 P_{S_1}}{4(1 + \sum_{r=1}^R \rho'_r |g_{r1}|^2 |x_r|^2)}. \quad (161)$$

Note that in (160), $Q(\sqrt{2\gamma^u(\mathbf{x}, \mathbf{h})})$ is the probability that the received signal y_1 lies closer to the signal point that corresponds to $\dot{s}'_2$ compared to the one that corresponds to $\dot{s}'_1$, given $\dot{s}'_1$ is sent.

Let us find an upper bound on $\gamma^u(\mathbf{x}, \mathbf{h})$ for any $\mathbf{h}$, and $\mathbf{x} \in \mathcal{X}$. Let $\tilde{\mathbf{x}} \triangleq \frac{\sqrt{R}}{\|\mathbf{x}\|} \mathbf{x}$ be the projection of $\mathbf{x}$ onto the hypersphere with radius $\sqrt{R}$. Then, we have

$$\gamma^u(\mathbf{x}, \mathbf{h}) \leq \underbrace{\frac{|\dot{s}_1 - \dot{s}_2|^2}{4}}_{\triangleq C_{44}} P_{S_1} \frac{\frac{R}{\|\mathbf{x}\|^2} \left| \sum_{r=1}^R f_{1r} \sqrt{\rho'_r} g_{r1} x_r \right|^2}{1 + \frac{R}{\|\mathbf{x}\|^2} \sum_{r=1}^R \rho'_r |g_{r1}|^2 |x_r|^2} \quad (162)$$

$$= C_{44} P_{S_1} \frac{\left| \sum_{r=1}^R f_{1r} \sqrt{\rho'_r} g_{r1} \tilde{x}_r \right|^2}{1 + \sum_{r=1}^R \rho'_r |g_{r1}|^2 |\tilde{x}_r|^2} \quad (163)$$

$$= C_{44} P_{S_1} \frac{\left| \sum_{r=1}^R f_{1r} \sqrt{\rho'_r} \sqrt{R} g_{r1} \tilde{x}_r \right|^2}{R + \sum_{r=1}^R \rho'_r R |g_{r1}|^2 |\tilde{x}_r|^2} \quad (164)$$

The first inequality follows since for any arbitrary non-negative real numbers a, b , we have $\frac{a}{1+b} \leq \frac{ca}{1+cb}$ for $c \geq 1$, and $|x_r| \leq 1$, $\forall r \implies \frac{R}{\|\mathbf{x}\|^2} \geq 1$. The equality in (162) is by the definition of $\tilde{\mathbf{x}}$, (163) follows by multiplying both the numerator and the denominator of (162) by R , and (164) is a consequence of $R = \sum_{r=1}^R |\tilde{x}_r|^2$.

Applying the Cauchy-Schwarz inequality to (164),

$$\begin{aligned} \gamma^u(\mathbf{x}, \mathbf{h}) &\leq C_{44} P_{S_1} \sum_{r=1}^R \frac{|f_{1r}|^2 \rho'_r R |g_{r1}|^2}{1 + \rho'_r R |g_{r1}|^2} \\ &\leq C_{44} P_{S_1} \sum_{r=1}^R \frac{|f_{1r}|^2 \rho_r R |g_{r1}|^2}{1 + \rho_r R |g_{r1}|^2} \\ &= C_{44} P_{S_1} \sum_{r=1}^R \frac{|f_{1r}|^2 R |g_{r1}|^2 P_{R_r}}{1 + \mathbb{E}_s \left[\left| \sum_{k=1}^K f_{kr} s_k \sqrt{P_{S_k}} \right|^2 \right] + R |g_{r1}|^2 P_{R_r}}, \end{aligned} \quad (165)$$

where for (165), we have used the fact that $\rho'_r \leq \rho_r$. The equality follows by the substitution of the value of ρ_r in (3).

For the expectation in the denominator of (165), we have

$$\mathbb{E}_s \left[\left| \sum_{k=1}^K f_{kr} s_k \sqrt{P_{S_k}} \right|^2 \right]$$

$$\begin{aligned} &= \sum_{k=1}^K |f_{kr}|^2 (1 - |\mathbb{E}[s_k]|^2) P_{S_k} + \left| \sum_{k=1}^K f_{kr} \mathbb{E}[s_k] \sqrt{P_{S_k}} \right|^2 \\ &\geq \sum_{k=1}^K |f_{kr}|^2 (1 - |\mathbb{E}[s_k]|^2) P_{S_k} \\ &\geq \epsilon_s \sum_{k=1}^K |f_{kr}|^2 P_{S_k}, \end{aligned} \quad (166)$$

where the second inequality is by our assumption in Section II-B that $|\mathbb{E}[s_k]|^2 \leq 1 - \epsilon_s, \forall k$. Substituting (166) to (165), we obtain

$$\begin{aligned} \gamma^u(\mathbf{x}, \mathbf{h}) &\leq R^2 C_{44} \max_r \frac{|f_{1r}|^2 |g_{r1}|^2 P_{S_1} P_{R_r}}{1 + \epsilon_s \sum_k |f_{kr}|^2 P_{S_k} + R |g_{r1}|^2 P_{R_r}} \\ &\leq \frac{R^2 C_{44}}{\epsilon_s} \max_r \frac{|f_{1r}|^2 |g_{r1}|^2 P_{S_1} P_{R_r}}{1 + |f_{1r}|^2 P_{S_1} + |f_{2r}|^2 P_{S_2} + |g_{r1}|^2 P_{R_r}} \\ &\leq \underbrace{\frac{R^2 C_{44} \max_r \{\sigma_{f_{1r}}^2, \sigma_{g_{r1}}^2, \sigma_{P_{R_r}}\} P_{S_1}}{\epsilon_s \min_r \{1, \sigma_{f_{1r}}^2 P_{S_1}, \sigma_{f_{2r}}^2 P_{S_2}, \sigma_{g_{r1}}^2 P_{R_r}\}}}_{\triangleq C_{45}} \times \\ &\quad \max_r \frac{X_r Y_r P^2}{1 + X_r P + W_r P + Y_r P}, \end{aligned} \quad (167)$$

where $X_r = \sigma_{f_{1r}}^2 P_{S_1}$, $Y_r = \sigma_{g_{r1}}^2 P_{R_r}$, and $W_r = \sigma_{f_{2r}}^2 P_{S_2}$. Note that $X_r, Y_r, W_r \sim \Gamma(1, 1)$ and they are independent. Let $Z_r^u \triangleq (X_r Y_r P^2) / (1 + X_r P + W_r P + Y_r P)$. We can rewrite (167) as $\gamma^u(\mathbf{x}, \mathbf{h}) \leq C_{45} \max_r Z_r^u$. Now, let

$$Z^u \triangleq \max_r Z_r^u, \quad (168)$$

$$\text{NER}^L(Q) \triangleq \frac{1}{|S_1|} \mathbb{E}[Q(\sqrt{2C_{45} Z^u})]. \quad (169)$$

According to (160), $\text{NER}(Q) \geq \text{NER}^L(Q), \forall Q$. It is therefore sufficient to find a lower bound on $\text{NER}^L(Q)$. Using the fact that $Q(x) \geq \frac{1}{\sqrt{2\pi}} \frac{x}{1+x^2} e^{-\frac{x^2}{2}}$, we have

$$\text{NER}^L(Q) \geq \frac{1}{|S_1| \sqrt{\pi}} \int_0^\infty \frac{\sqrt{z}}{1+2z} \exp(-z C_{45}) f_{Z^u}(z) dz. \quad (170)$$

We thus need to find a lower bound for the PDF of Z^u . Using order statistics, we have

$$f_{Z^u}(z) = \sum_{r=1}^R f_{Z_r^u}(z) \prod_{\substack{q=1 \\ q \neq r}}^R F_{Z_q^u}(z). \quad (171)$$

In the following, we find a lower bound on the PDF and CDF of Z_r^u , for any r . We first evaluate the PDF of Z_r^u . Using a transformation of random variables [72], the PDF of Z_r^u can be expressed as

$$\begin{aligned} f_{Z_r^u}(z) &= \int_0^\infty \int_0^\infty f_{Y_r} \left(\frac{z(1+xP+wP)}{xP^2 - zP} \right) \times \\ &\quad \frac{xP^2(1+xP+wP)}{(xP^2 - zP)^2} e^{-x} dx e^{-w} dw. \end{aligned}$$

This PDF is in the same form as (120) in Proposition 3, and can be evaluated using the same methods discussed therein. We have

$$\begin{aligned} f_{Z_r^v}(z) &= e^{-\frac{2z}{P}} \int_0^\infty \left[\frac{1}{P} \sqrt{\frac{4z(1+wP+z)}{P^2}} K_1 \left(\sqrt{\frac{4z(1+wP+z)}{P^2}} \right) + \frac{2(1+2z+wP)}{P^2} K_0 \left(\sqrt{\frac{4z(1+wP+z)}{P^2}} \right) \right] e^{-w} dw \\ &\geq \frac{e^{-\frac{2z}{P}}}{P^2} \int_0^\infty 2(1+wP+z) K_0 \left(\sqrt{\frac{4z(1+wP+z)}{P^2}} \right) e^{-w} dw. \end{aligned}$$

Using the fact that for any $z > 0$, $K_0(z) = -\log(\frac{z}{2}) - \gamma_e + (1 - \gamma_e)\frac{1}{4z^2} + (1 + 1/2 - \gamma_e)\frac{z^4}{32} + \dots$ [75], we have $K_0(z) \geq -\log(\frac{z}{2}) - \gamma_e = \log(\frac{2}{z}) - \gamma_e$, and thus

$$\begin{aligned} f_{Z_r^v}(z) &\geq \frac{e^{-\frac{2z}{P}}}{P^2} \int_0^\infty (1+z+wP) \left[\log \left(\frac{P^2}{z(1+z+wP)} \right) - 2\gamma_e \right] e^{-w} dw \\ &= \frac{e^{-\frac{2z}{P}}}{P^2} \left\{ (-2\gamma_e + 2\log P - \log z) \int_0^\infty (1+z+wP) e^{-w} dw - \int_0^\infty (1+z+wP) \log(1+z+wP) e^{-w} dw \right\}. \quad (172) \end{aligned}$$

It is straightforward to evaluate the first integral in (172) as

$$\int_0^\infty (1+z+wP) e^{-w} dw = 1+z+P. \quad (173)$$

For the second integral in (172), we recall the rule of integration by parts: For functions $u = f(x)$ and $v = g(x)$, and differentials $du = \frac{\partial f(x)}{\partial x} dx$ and $dv = \frac{\partial g(x)}{\partial x} dx$, we have $\int_0^\infty u dv = uv|_0^\infty - \int_0^\infty v du$. We first set $u = (1+z+wP) \log(1+z+wP)$ and $dv = e^{-w} dw$. This gives us

$$\begin{aligned} \int_0^\infty (1+z+wP) \log(1+z+wP) e^{-w} dw &= (1+z) \log(1+z) + P + P \int_0^\infty \log(1+z+wP) e^{-w} dw. \quad (174) \end{aligned}$$

We now apply the rule of integration by parts one more time with $u = \log(1+z+wP)$ and $dv = e^{-w} dw$ to obtain

$$\int_0^\infty \log(1+z+wP) e^{-w} dw \quad (175)$$

$$= \log(1+z) + \int_0^\infty \frac{P}{1+z+wP} e^{-w} dw \quad (176)$$

$$= \log(1+z) + e^{\frac{1+z}{P}} \int_1^\infty \frac{1}{u} e^{-\frac{1+z}{P}u} du \quad (177)$$

$$= \log(1+z) + e^{\frac{1+z}{P}} E_1 \left(\frac{1+z}{P} \right). \quad (178)$$

The second equality follows from a change of variables $u = \frac{wP}{1+z} + 1$, and the last inequality is by the definition of the exponential integral in Section I-H. Substituting (173) and the combination of (174) and (178) to (172), we obtain

$$\begin{aligned} f_{Z_r^v}(z) &= \frac{e^{-\frac{2z}{P}}}{P^2} \left\{ (-2\gamma_e + 2\log P - \log z)(1+z+P) - \left[(1+z) \log(1+z) + P \log(1+z) + P + P e^{\frac{1+z}{P}} E_1 \left(\frac{1+z}{P} \right) \right] \right\}. \end{aligned}$$

Now, note that

$$\log z \leq \log(1+z) \leq z, \quad (179)$$

and

$$\begin{aligned} e^{\frac{1+z}{P}} E_1 \left(\frac{1+z}{P} \right) &\leq \log \left(1 + \frac{P}{1+z} \right) \leq \log(1+z+P) \\ &\leq \log(1+z+P+zP) = \log(1+z) + \log(1+P) \\ &\leq z + \log(2P) \leq z + 1 + \log P. \quad (180) \end{aligned}$$

The first inequality in (180) follows since $e^x E_1(x) \leq \log(1 + \frac{1}{x})$, $x > 0$ [74, Eq. 5.1.20]. Using (179) and (180), we have

$$\begin{aligned} f_{Z_r^v}(z) &\geq \frac{e^{-\frac{2z}{P}}}{P^2} \left\{ (-2\gamma_e + 2\log P - z)(1+z+P) - \left[(1+z)z + Pz + P + P(z+1+\log P) \right] \right\} \\ &= \frac{e^{-\frac{2z}{P}}}{P^2} \left\{ P \log P + 2\log P(1+z) - \left[(2\gamma_e + z)(1+z+P) + z(1+z) + 2(z+1)P \right] \right\} \\ &\geq \frac{e^{-\frac{2z}{P}}}{P^2} \left\{ P \log P - P[(2+z)(2+z) + z(1+z) + 2(z+1)] \right\} \\ &= \frac{e^{-\frac{2z}{P}}}{P} \left[\log P - (2z^2 + 7z + 6) \right] \quad (181) \end{aligned}$$

Since $2z^2 + 7z + 6 < 2z^2 + 7(1+z^2) + 6 < 13(1+z^2)$, $\forall z \geq 0$, (181) leads to a more compact lower bound

$$f_{Z_r^v}(z) \geq \phi(z), \quad (182)$$

where

$$\phi(z) \triangleq e^{-\frac{2z}{P}} \frac{1}{P} [\log P - 13(1+z^2)]. \quad (183)$$

Now suppose that $P \geq e^{13}$, and let $z_0 \triangleq (\frac{\log P}{13} - 1)^{\frac{1}{2}}$. Then, $\phi(z) \geq 0$ for $z \leq z_0$, and $\phi(z) < 0$, otherwise. For $z \leq z_0$, the CDF $F_{Z_r^v}(z)$ can be lower bounded by

$$F_{Z_r^v}(z) \geq \int_0^z \exp\left(-\frac{2x}{P}\right) \frac{1}{P} [\log P - 13(1+x^2)] dx \quad (184)$$

$$\geq e^{-\frac{2z}{P}} \frac{1}{P} \int_0^z [\log P - 13(1+x^2)] dx \quad (185)$$

$$= e^{-\frac{2z}{P}} \frac{z}{P} \left[\log P - 13 \left(1 + \frac{z^2}{3} \right) \right] \quad (186)$$

$$\geq e^{-\frac{2z}{P}} \frac{z}{P} [\log P - 13(1+z^2)] \quad (187)$$

$$= z\phi(z). \quad (188)$$

Since $F_{Z_r^v}$ is a CDF, $f_{Z_r^v}(z) \geq 0, \forall z$. Therefore, for $z > z_0$, we can choose any negative function as a lower bound on $f_{Z_r^v}(z)$. But, (188) is negative for $z > z_0$. Thus, it is a lower bound on $F_{Z_r^v}(z)$ that holds for all z .

We can now find a lower bound for the PDF of Z^v by substituting (182) and (188) to (171). For $z \leq z_0$, the PDF $f_{Z^v}(z)$ can be lower bounded by

$$\begin{aligned} f_{Z^v}(z) &\geq Rz^{R-1}\phi^R(z) \\ &= Rz^{R-1} \frac{e^{-\frac{2Rz}{P}}}{P^R} [\log P - 13(1+z^2)]^R \end{aligned} \quad (189)$$

$$\geq Rz^{R-1} \frac{e^{-2Rz}}{P^R} [\log P - 13(1+z^2)]^R \quad (190)$$

$$= Rz^{R-1} \frac{e^{-2Rz}}{P^R} \sum_{i=0}^R \binom{R}{i} \log^{R-i} P (-13)^i (1+z^2)^i \quad (191)$$

$$\begin{aligned} &\geq Rz^{R-1} \frac{e^{-2Rz}}{P^R} \left[\log^R P - \sum_{i=1}^R \binom{R}{i} \log^{R-i} P 13^i (1+z^2)^i \right] \\ &\geq Rz^{R-1} \frac{e^{-2Rz}}{P^R} \left[\log^R P - \frac{R13^R}{2^{1-2R}} \log^{R-1} P (1+z^{2R}) \right] \end{aligned} \quad (192)$$

Inequality (190) follows since $P \geq 1$ for sufficiently large P , and (191) is by the binomial expansion. The last inequality follows since $\sum_{i=0}^R \binom{R}{i} = 2^R$, and thus $\binom{R}{i} \leq 2^R, \forall i \in \{1, \dots, R\}$; $\log^{R-i} P \leq \log^{R-1} P, \forall P \geq e, \forall i \in \{1, \dots, R-1\}$; and $(1+z^2)^i \leq (1+z^2)^R \leq 2^{R-1}(1+z^{2R}), \forall z \in \mathbb{R}, \forall i \in \{1, \dots, R-1\}$. The last inequality in the previous sentence is a consequence of Lemma 7.

Now, similarly, since $f_{Z^v}(z)$ is a PDF, $f_{Z^v}(z) \geq 0, \forall z$. Therefore, for $z > z_0$, we can choose any negative function as a lower bound on $f_{Z^v}(z)$. But, it is straightforward to verify that (192) is negative for $z > z_0$. Thus, it is a lower bound on $f_{Z^v}(z)$ that holds for all z . We can therefore use it to bound (170) as

$$\begin{aligned} \text{NER}^L(\mathbf{Q}) &\geq \frac{R}{|\mathcal{S}_1|\sqrt{\pi}} \frac{\log^R P}{P^R} \int_0^\infty \frac{z^{R-\frac{1}{2}}}{1+2z} e^{-z(\mathcal{C}_{45}+2R)} dz - \\ &\frac{R^2 2^{2R-1} 13^R \log^{R-1} P}{|\mathcal{S}_1|\sqrt{\pi}} \frac{1}{P^R} \int_0^\infty \frac{z^{R-\frac{1}{2}}(1+z^{2R})}{1+2z} e^{-z(\mathcal{C}_{45}+2R)} dz. \end{aligned} \quad (193)$$

The first integral in (193) can be lower bounded by

$$\begin{aligned} \int_0^\infty \frac{z^{R-\frac{1}{2}}}{1+2z} e^{-z(\mathcal{C}_{45}+2R)} dz &\geq \int_0^1 \frac{z^{R-\frac{1}{2}}}{1+2z} e^{-z(\mathcal{C}_{45}+2R)} dz \\ &\geq \frac{e^{-(\mathcal{C}_{45}+2R)}}{3} \int_0^1 z^{R-\frac{1}{2}} dz \\ &= \frac{2e^{-(\mathcal{C}_{45}+2R)}}{3(2R+1)}. \end{aligned} \quad (194)$$

For the second integral in (193), we have

$$\begin{aligned} &\int_0^\infty \frac{z^{R-\frac{1}{2}}(1+z^{2R})}{1+2z} e^{-z(\mathcal{C}_{45}+2R)} dz \\ &\leq \int_0^\infty z^{R-\frac{1}{2}}(1+z^{2R}) e^{-z(\mathcal{C}_{45}+2R)} dz \\ &= \frac{\Gamma(R+\frac{1}{2})}{(\mathcal{C}_{45}+2R)^{R+\frac{1}{2}}} + \frac{\Gamma(3R+\frac{1}{2})}{(\mathcal{C}_{45}+2R)^{3R+\frac{1}{2}}}. \end{aligned} \quad (195)$$

Substituting (194) and (195) to (193), it follows that

$$\text{NER}^L(\mathbf{Q}) \geq 2\mathcal{C}_{46} P^{-R} (\log^R P - \mathcal{C}_{47} \log^{R-1} P),$$

for some constants $\infty > \mathcal{C}_{46}, \mathcal{C}_{47} > 0$ independent of P .

Finally, $\mathcal{C}_{47} \log^{R-1} P \leq \frac{1}{2} \log^R P, \forall P \geq \exp(2\mathcal{C}_{47})$, and thus

$$\text{NER}^L(\mathbf{Q}) \geq \mathcal{C}_{46} P^{-R} \log^R P,$$

for all $P \geq \exp(\max\{13, 2\mathcal{C}_{47}\})$. This concludes the proof. ■

APPENDIX D

PROOF OF THE UPPER BOUND IN THEOREM 3

For notational convenience, let us first define the following:

$$\begin{aligned} \Upsilon_{r,1} &\triangleq \frac{\min_{\ell} |g_{r\ell}|^2 p_{R_r}}{2(\varsigma-1)^2}, \\ \Upsilon_{r,2,s,\hat{s}} &\triangleq \frac{1}{4} \left| \sum_{k=1}^K (s_k - \hat{s}_k) \sqrt{p_{S_k}} f_{kr} \right|^2, \\ \Upsilon_{r,2} &\triangleq \min_{\substack{s,\hat{s} \in \mathcal{S} \\ s \neq \hat{s}}} \Upsilon_{r,2,s,\hat{s}}, \\ \Upsilon_r &\triangleq \min\{\Upsilon_{r,1}, \Upsilon_{r,2}\}, \\ \Upsilon &\triangleq \max_r \Upsilon_r. \end{aligned}$$

According to (15) and (18), $\gamma^L(\mathbf{e}_r, \mathbf{h}) = P\Upsilon_r$, and thus $P\Upsilon_r$ is indeed the NSNR given that the r th relay is selected. Since our GQ chooses the relay selection vector that maximizes the NSNR, the unconditional NSNR is $P \max_r \Upsilon_r = P\Upsilon$. It follows by Proposition 2 that the NER with the GQ can be upper bounded as

$$\text{NER}(\text{GQ}_{\mathcal{C}_s}) \leq \mathcal{C}_4 \int_0^\infty e^{-vP} f_{\Upsilon}(v) dv. \quad (196)$$

In the following, we find an upper bound on the PDF of Υ . Since Υ is the maximum of R independent random variables Υ_r , $r = 1, \dots, R$, by order statistics, we have

$$f_{\Upsilon}(v) = \sum_{r=1}^R f_{\Upsilon_r}(v) \prod_{\substack{q=1 \\ q \neq r}}^R F_{\Upsilon_q}(v). \quad (197)$$

We thus need to find upper bounds on the PDF and the CDF of Υ_r for any r . First, using Lemma 2, we obtain

$$\begin{aligned} f_{\Upsilon_r}(v) &\leq f_{\Upsilon_{r,1}}(v) + f_{\Upsilon_{r,2}}(v) \\ &\leq f_{\Upsilon_{r,1}}(v) + \sum_{\mathbf{s} \neq \hat{\mathbf{s}}} f_{\Upsilon_{r,2,\mathbf{s},\hat{\mathbf{s}}}}(v), \end{aligned} \quad (198)$$

where we have omitted the condition $\mathbf{s}, \hat{\mathbf{s}} \in \mathcal{S}$ as it is clear from the context.

Note that $f_{\Upsilon_{r,1}}(v) = \mathbf{1}_{v \geq 0} C_{48,r} \exp(-C_{48,r}v)$, where $C_{48,r} \triangleq \sum_{\ell=1}^L \frac{2(\zeta-1)^2}{\sigma_{\theta_{r,\ell}}^2 p_{Rr}}$. From now on, we omit the term $\mathbf{1}_{v \geq 0}$ as it is clear from the context. Also, it is straightforward to show using Lemma 1 that $f_{\Upsilon_{r,2,\mathbf{s},\hat{\mathbf{s}}}}(v) \leq C_{49,r,\mathbf{s},\hat{\mathbf{s}}} \exp(-C_{50,r,\mathbf{s},\hat{\mathbf{s}}}v)$ for constants $0 < C_{49,r,\mathbf{s},\hat{\mathbf{s}}}, C_{50,r,\mathbf{s},\hat{\mathbf{s}}} < \infty$. Substituting these inequalities to (198), we have

$$\begin{aligned} f_{\Upsilon_r}(v) &\leq C_{48,r} \exp(-C_{48,r}v) + \sum_{\mathbf{s} \neq \hat{\mathbf{s}}} C_{49,r,\mathbf{s},\hat{\mathbf{s}}} \exp(-C_{50,r,\mathbf{s},\hat{\mathbf{s}}}v) \\ &\leq \max_r C_{48,r} \exp(-\min_r C_{48,r}v) + \\ &\quad |\mathcal{A}|(|\mathcal{A}| - 1) \max_{r,\mathbf{s} \neq \hat{\mathbf{s}}} C_{49,r,\mathbf{s},\hat{\mathbf{s}}} \exp(-\min_{r,\mathbf{s} \neq \hat{\mathbf{s}}} C_{50,r,\mathbf{s},\hat{\mathbf{s}}}v) \\ &\leq C_{51} \exp(-C_{52}v), \end{aligned} \quad (199)$$

where $C_{51} \triangleq 2 \max\{\max_r C_{48,r}, \max_{r,\mathbf{s} \neq \hat{\mathbf{s}}} C_{49,r,\mathbf{s},\hat{\mathbf{s}}}\}$, and $C_{52} \triangleq \min\{\min_r C_{48,r}, \min_{r,\mathbf{s} \neq \hat{\mathbf{s}}} C_{50,r,\mathbf{s},\hat{\mathbf{s}}}\}$. The second inequality follows from Lemma 3. We have thus obtained an r -independent upper bound on the PDF of Υ_r . Now, using this upper bound, we have

$$\begin{aligned} F_{\Upsilon_r}(v) &= \int_0^v f_{\Upsilon_r}(\omega) d\omega \leq \int_0^v C_{51} \exp(-C_{52}\omega) d\omega \\ &\leq \int_0^v C_{51} d\omega \\ &= v C_{51}. \end{aligned} \quad (200)$$

Substituting (199) and (200) to (197), we have

$$f_{\Upsilon}(v) \leq R C_{51}^R v^{R-1} \exp(-C_{52}v). \quad (201)$$

The upper bound on the NER in the statement of the theorem follows by substituting (201) to (196) and performing the integration. This concludes the proof. ■

APPENDIX E PROOF OF THEOREM 4

We need the following lemma to prove the theorem.

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Lemma 8: For any local quantizer $\text{LQ}_{\xi,N}^{(\mathbf{f}|\mathbf{v})}$, let $\text{LD}(\xi, N)$ denote its localization distortion, as defined in (28). Then, for any $\xi > 0$ and $N \geq 1$,

$$\text{LD}(\xi, N) \leq \text{LD}^{\text{U}}(\xi, N), \quad (202)$$

where

$$\text{LD}^{\text{U}}(\xi, N) \triangleq \mathbb{E}_{\mathbf{h}} [\text{CNER}(\text{LQ}_{\xi,N}^{(\mathbf{f}|\mathbf{v})}(\mathbf{h}), \mathbf{h}) | |\mathcal{R}_l| \geq 2] \mathbb{P}(|\mathcal{R}_l| \geq 2),$$

and $\mathcal{R}_l$ is as defined in Section V-B-VI.

Therefore, combining (202) and (28), we have

$$\text{NER}(\text{LQ}_{\xi,N}^{(\mathbf{f}|\mathbf{v})}) \leq \text{NER}(\text{GQ}_{C_s}) + \text{LD}^{\text{U}}(\xi, N). \quad (203)$$

Proof: The NER with the quantizer $\text{LQ}_{\xi,N}^{(\mathbf{f}|\mathbf{v})}$ can be expressed as

$$\begin{aligned} \text{NER}(\text{LQ}_{\xi,N}^{(\mathbf{f}|\mathbf{v})}) &= \mathbb{E}_{\mathbf{h}} [\text{CNER}(\text{LQ}_{\xi,N}^{(\mathbf{f}|\mathbf{v})}(\mathbf{h}), \mathbf{h}) | \mathcal{R}_l = \mathcal{R}_g] \mathbb{P}(\mathcal{R}_l = \mathcal{R}_g) + \\ &\quad \mathbb{E}_{\mathbf{h}} [\text{CNER}(\text{LQ}_{\xi,N}^{(\mathbf{f}|\mathbf{v})}(\mathbf{h}), \mathbf{h}) | \mathcal{R}_l \neq \mathcal{R}_g] \mathbb{P}(\mathcal{R}_l \neq \mathcal{R}_g), \end{aligned} \quad (204)$$

where $\mathcal{R}_g$ is as defined in Section V-B-VI. Let us call the first and the second terms in the right hand side of (204) as NER_1^{LQ} and NER_2^{LQ} , respectively. For the first term, we have

$$\text{NER}_1^{\text{LQ}} = \mathbb{E}_{\mathbf{h}} [\text{CNER}(\text{GQ}_{C_s}(\mathbf{h}), \mathbf{h}) | \mathcal{R}_l = \mathcal{R}_g] \mathbb{P}(\mathcal{R}_l = \mathcal{R}_g) \quad (205)$$

$$\begin{aligned} &= \mathbb{E}_{\mathbf{h}} [\text{CNER}(\text{GQ}_{C_s}(\mathbf{h}), \mathbf{h})] - \\ &\quad \mathbb{E}_{\mathbf{h}} [\text{CNER}(\text{GQ}_{C_s}(\mathbf{h}), \mathbf{h}) | \mathcal{R}_l \neq \mathcal{R}_g] \mathbb{P}(\mathcal{R}_l \neq \mathcal{R}_g) \end{aligned} \quad (206)$$

$$\leq \mathbb{E}_{\mathbf{h}} [\text{CNER}(\text{GQ}_{C_s}(\mathbf{h}), \mathbf{h})] \quad (207)$$

$$= \text{NER}(\text{GQ}_{C_s}), \quad (208)$$

where (205) follows from the fact that the LQ provides the same beamforming vector (and thus the same CNER) as the GQ whenever $\mathcal{R}_l = \mathcal{R}_g$, and (206) follows from the law of total probability. The inequality in (208) holds since the last term in (206) is non-negative, and (208) follows from the definition of NER in (10).

We now find an upper bound for NER_2^{LQ} . First, note that

$$\{\mathbf{h} : |\mathcal{R}_l| \neq |\mathcal{R}_g|\} = \{\mathbf{h} : |\mathcal{R}_l| > |\mathcal{R}_g|\} \quad (209)$$

$$= \{\mathbf{h} : |\mathcal{R}_l| > |\mathcal{R}_g| \text{ and } |\mathcal{R}_g| \geq 1\} \quad (210)$$

$$\subset \{\mathbf{h} : |\mathcal{R}_l| \geq 2\}, \quad (211)$$

where (209) follows since $\mathcal{R}_g \subset \mathcal{R}_l$ as discussed in Section V-B-VI, and (210) holds since $\mathcal{R}_g$ is non-empty. The inclusion in (211) holds since $(|\mathcal{R}_l| > |\mathcal{R}_g| \text{ and } |\mathcal{R}_g| \geq 1) \implies |\mathcal{R}_l| \geq 2$. Hence, $\{\mathbf{h} : |\mathcal{R}_l| \neq |\mathcal{R}_g|\} \subset \{\mathbf{h} : |\mathcal{R}_l| \geq 2\}$, using which we obtain

$$\text{NER}_2^{\text{LQ}} \leq \mathbb{E}_{\mathbf{h}} [\text{CNER}(\text{LQ}_{\xi,N}^{(\mathbf{f}|\mathbf{v})}(\mathbf{h}), \mathbf{h}) | |\mathcal{R}_l| \geq 2] \mathbb{P}(|\mathcal{R}_l| \geq 2). \quad (212)$$

Combining (212), (208), and (204), we obtain (203). It is now straightforward to show, by the definition in (28), that $\text{LD}(\xi, N) \leq \text{LD}^U(\xi, N)$ as claimed. ■

We now prove the theorem for $K > 1$. The proof for $K = 1$ is very similar. It is thus omitted. Let $\Omega = \max_r \Omega_r$, where $\Omega_r = \gamma^L(\mathbf{e}_r, \mathbf{h})$ is the NSNR given that the r th relay is selected. We prove the theorem by finding an upper bound on the localization distortion defined in (28). According to the Lemma 8, it is sufficient to calculate the CNER given $|\mathcal{R}_l| \geq 2$. Note that $|\mathcal{R}_l| \geq 2$ if and only if there exists $r, q \in \{1, \dots, R\}$, $r \neq q$ such that $\mathcal{N}(\Omega_r) = \mathcal{N}(\Omega_q) = \mathcal{N}(\Omega)$. Also, since $N_{\mathbf{f}} = 2$, $\mathcal{N}(\Omega)$ is either 0 or 1. Depending on $\mathcal{N}(\Omega)$, we divide the calculation of $\text{LD}^U(\xi_{\mathbf{f}}, 2)$ to two separate parts as

$$\text{LD}^U(\xi_{\mathbf{f}}, 2) = \sum_{i=1}^2 \text{LD}_i^U(\xi_{\mathbf{f}}, 2). \quad (213)$$

The first part is concerned with the case $\mathcal{N}(\Omega) = 0$. Since $\Omega = \max_r \Omega_r$, $\mathcal{N}(\Omega) = 0$ if and only if $\Omega_r \in [0, \xi_{\mathbf{f}})$, $\forall r$. Given that $\mathcal{N}(\Omega) = 0$, the decoder chooses one of the R relay selection vectors, and thus the NSNR is at least $\min_r \Omega_r$. Therefore, using Proposition 1, we have

$$\begin{aligned} \text{LD}_1^U(\xi_{\mathbf{f}}, 2) &\leq \mathcal{C}_4 \int_0^{\xi_{\mathbf{f}}} \cdots \int_0^{\xi_{\mathbf{f}}} \exp\left(-\min_r \omega_r\right) \prod_r f_{\Omega_r}(\omega_r) \prod_r d\omega_r \\ &\leq \mathcal{C}_4 \int_0^{\xi_{\mathbf{f}}} \cdots \int_0^{\xi_{\mathbf{f}}} \prod_r f_{\Omega_r}(\omega_r) \prod_r d\omega_r \\ &= \mathcal{C}_4 [F_{\Omega_r}(\xi_{\mathbf{f}})]^R \\ &\leq \mathcal{C}_4 \mathcal{C}_{27}^R \xi_{\mathbf{f}}^R \left[\frac{\log P}{P} \left(1 + \xi_{\mathbf{f}}^{-\frac{1}{2 \log P}} + \xi_{\mathbf{f}}^{1 - \frac{1}{\log P}} \frac{1}{P} \right) \right]^R \\ &= \mathcal{C}_4 (RC_{27})^R \frac{\log^{2R} P}{P^R} \left(1 + (R \log P)^{-\frac{1}{2 \log P}} + \right. \\ &\quad \left. (R \log P)^{1 - \frac{1}{\log P}} \frac{1}{P} \right)^R \quad (215) \\ &\leq \mathcal{C}_4 (RC_{27})^{R3^{R-1}} \frac{\log^{2R} P}{P^R} \left(1 + (R \log P)^{-\frac{R}{2 \log P}} + \right. \\ &\quad \left. (R \log P)^{R - \frac{R}{\log P}} \frac{1}{P^R} \right), \quad (216) \end{aligned}$$

where (214) follows from Proposition 3, (215) follows from the substitution $\xi_{\mathbf{f}} = R \log P$, and (216) is a consequence of Lemma 7. Now, for the terms inside the parenthesis in (216), note that for P sufficiently large $R \log P \geq 1$, $\frac{R}{2 \log P} \geq 0$, and $\frac{R}{\log P} \geq 0$. Therefore, for P sufficiently large, $(R \log P)^{-\frac{R}{2 \log P}} \leq 1$, and $(R \log P)^{-\frac{R}{\log P}} \leq 1$. Substituting these inequalities to (216), we have

$$\text{LD}_1^U(\xi_{\mathbf{f}}, 2) \leq \mathcal{C}_4 (RC_{27})^{R3^{R-1}} \frac{\log^{2R} P}{P^R} \left(2 + R^R \frac{\log^R P}{P^R} \right).$$

But, $\frac{\log^R P}{P^R} \leq 1$ for $P \geq e$, and thus

$$\text{LD}_1^U(\xi_{\mathbf{f}}, 2) \leq \underbrace{\mathcal{C}_4 (RC_{27})^{R3^{R-1}} (2 + R^R)}_{\triangleq \mathcal{C}_{53}} \frac{\log^{2R} P}{P^R}. \quad (217)$$

For the second part, we consider the case $\mathcal{N}(\Omega) = 1 \iff \exists r \in \mathcal{R}, \Omega_r \in [\xi_{\mathbf{f}}, \infty)$. In this case, the minimum NSNR is $\xi_{\mathbf{f}} = R \log P$, and we simply have

$$\text{LD}_2^U(\xi_{\mathbf{f}}, 2) \leq \mathcal{C}_4 \exp(-R \log P) = \mathcal{C}_4 P^{-R}. \quad (218)$$

Substituting (217) and (218) to (213), we have

$$\text{LD}^U(\xi_{\mathbf{f}}, 2) \leq \mathcal{C}_{53} \frac{\log^{2R} P}{P^R} + \mathcal{C}_4 \frac{1}{P^R} \leq (\mathcal{C}_{53} + \mathcal{C}_4) \frac{\log^{2R} P}{P^R},$$

and the last inequality follows since $1 \leq \log^{2R} P$ whenever $P \geq e$. This concludes the proof. ■

APPENDIX F PROOF OF THEOREM 6

We prove the theorem for $K > 1$. The proof for $K = 1$ is very similar and thus has been skipped for brevity.

Let $\Omega = \max_r \Omega_r$, where $\Omega_r = \gamma^L(\mathbf{e}_r, \mathbf{h})$ is the NSNR given that the r th relay is selected. We need to find an upper bound on the localization distortion defined in (28). According to Lemma 8, it is sufficient to calculate the CNER given $|\mathcal{R}_l| \geq 2$. Depending on $\mathcal{N}(\Omega)$, we divide the calculation of $\text{LD}^U(\xi, N)$ to three separate parts as

$$\text{LD}^U(\xi, N) = \sum_{i=1}^3 \text{LD}_i^U(\xi, N). \quad (219)$$

The first part is concerned with the case where $\mathcal{N}(\Omega) = 0 \iff \Omega_r \in [0, \xi)$, $\forall r$. The NSNR is obviously at least 0. Using Propositions 2 and 3, we have

$$\begin{aligned} \text{LD}_1^U(\xi, N) &\leq \mathcal{C}_4 \int_0^{\xi} \cdots \int_0^{\xi} \prod_r f_{\Omega_r}(\omega_r) \prod_r d\omega_r \\ &\leq \mathcal{C}_4 \mathcal{C}_{27}^R \xi^R \psi^R(\xi). \end{aligned} \quad (220)$$

Now we consider the term $\psi(\xi)$ in (220). For future reference, we shall calculate an upper bound for the more general quantity $\psi(n\xi)$, for any $n \in \{1, \dots, N-1\}$. We have

$$\begin{aligned} \psi(n\xi) &= \frac{\log P}{P} \left[1 + (n\xi)^{-\frac{1}{2 \log P}} + (n\xi)^{1 - \frac{1}{\log P}} \frac{1}{P} \right] \\ &= \frac{\log P}{P} \left(1 + n^{-\frac{1}{2 \log P}} \Lambda^{\frac{1}{2 \log P}} + \frac{n^{1 - \frac{1}{\log P}} \xi \Lambda^{\frac{1}{\log P}}}{P} \right). \end{aligned} \quad (221)$$

Now, note that for $P \geq e$, $\frac{1}{2 \log P} \leq \frac{1}{2}$, and similarly $\frac{1}{\log P} \leq 1$. As a result, $n^{-\frac{1}{2 \log P}} \leq 1$ and $n^{1 - \frac{1}{\log P}} \leq n$. Moreover, as discussed in Section V-D, $\epsilon \leq \Lambda \leq \Lambda_0 P$ for sufficiently large P , and thus $\Lambda^{\frac{1}{2 \log P}} \leq (\Lambda_0 P)^{\frac{1}{2 \log P}} = \Lambda_0^{\frac{1}{2 \log P}} P^{\frac{1}{2 \log P}} \leq \max\{1, \Lambda_0\} \sqrt{e}$. Similarly, $\Lambda^{\frac{1}{\log P}} \leq \max\{1, \Lambda_0\} e$. Substituting these inequalities to (221), we obtain

$$\psi(n\xi) \leq \frac{\log P}{P} \left[1 + \max\{1, \Lambda_0\} \left(\sqrt{e} + \frac{en\xi}{P} \right) \right]. \quad (222)$$

Moreover, for all $n \in \{1, \dots, N-1\}$, $n\xi < N\xi = \frac{1}{\Lambda}[\Lambda \log \Lambda + R\Lambda \log P] \leq \log \Lambda + R \log P \leq \log \Lambda_0 + (R+1) \log P$. Combining with (222), we can argue that there is a constant $0 < C_{55} < \infty$ such that

$$\psi(n\xi) \leq C_{55} \frac{\log P}{P}, \quad (223)$$

for all P sufficiently large. Using (220), it follows that

$$\begin{aligned} \text{LD}_1^U(\xi, N) &\leq \frac{C_4 C_{27}^R C_{55}^R \log^R P}{\Lambda^R P^R} \\ &\leq \frac{C_4 C_{27}^R C_{55}^R \log^R P}{\epsilon^{R-1} \Lambda P^R}. \end{aligned}$$

The second inequality follows since $\Lambda \geq \epsilon \implies \Lambda^{-1} \leq \epsilon^{-1} \implies \Lambda^{-(R-1)} \leq \epsilon^{-(R-1)}$.

For the second part, we evaluate the cases for which $\mathcal{N}(\Omega) \in \{1, \dots, N-2\}$. For each $n \in \{1, \dots, N-2\}$, suppose that $i \geq 2$ of Ω_r are in the interval $[n\xi, (n+1)\xi)$, and the rest $R-i$ of them are in $[0, n\xi)$. The minimum NSNR is at least $n\xi$. Also, there are $\binom{R}{i}$ possible ways to choose which Ω_r will be in $[n\xi, (n+1)\xi)$. Therefore,

$$\begin{aligned} \text{LD}_2^U(\xi, N) &\leq C_4 \sum_{i=2}^R \sum_{\mathcal{K} \in \mathcal{R}_i^R} \sum_{n=1}^{N-2} \underbrace{\int_0^{n\xi} \dots \int_0^{n\xi}}_{R-i \text{ integrals}} \underbrace{\int_{n\xi}^{(n+1)\xi} \dots \int_{n\xi}^{(n+1)\xi}}_{i \text{ integrals}} \\ &\quad \exp(-\min_{r \in \mathcal{K}} \omega_r) \prod_{r=1}^R f_{\Omega_r}(\omega_r) \prod_{r \in \mathcal{K}} d\omega_r \prod_{r' \in \mathcal{K}^c} d\omega_{r'}, \quad (224) \end{aligned}$$

where $\mathcal{R}_i^R$ is the collection of all possible i -combinations of the set $\{1, \dots, R\}$ (e.g., $\mathcal{R}_2^3 = \{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$), and $\mathcal{K}^c = \{1, \dots, R\} - \mathcal{K}$. Then, similarly, we can use Propositions 2 and 3 to arrive at

$$\begin{aligned} \text{LD}_2^U(\xi, N) &\leq C_4 C_{27}^R \sum_{i=2}^R \sum_{\mathcal{K} \in \mathcal{R}_i^R} \sum_{n=1}^{N-2} \left\{ e^{-n\xi} [n\xi \psi(n\xi)]^{R-i} \times \right. \\ &\quad \left. \left[\xi \psi(\xi) + \frac{\log P}{P^2} (n\xi)^{2-\frac{1}{\log P}} \right]^i \right\} \\ &\leq C_4 C_{27}^R \sum_{i=2}^R \binom{R}{i} 2^{i-1} \sum_{n=1}^{N-2} \left\{ n^{R-i} e^{-n\xi} \psi^{R-i}(n\xi) \times \right. \\ &\quad \left. \left[\xi^R \psi^i(\xi) + \xi^{R-i} (1 + n^{2i} \xi^{2i}) \frac{\log^i P}{P^{2i}} \right] \right\}, \quad (225) \end{aligned}$$

where (225) follows from Lemma 7, and the fact that $(n\xi)^{2i-\frac{1}{\log P}} \leq (1 + n^{2i} \xi^{2i})$, $i \geq 1$. Now, note that for any $i \in \{2, \dots, R\}$, we have $\binom{R}{i} < 2^R$ and $2^{i-1} < 2^R$. Also, by (223), $\psi^i(\xi) \leq C_{55}^i \log^i P P^{-i}$, and $\psi^{R-i}(n\xi) \leq C_{55}^{R-i} \log^{R-i} P P^{-(R-i)}$. Substituting these inequalities to (225) and rearranging the terms, we have

$$\text{LD}_2^U(\xi, N) \leq C_4 (4C_{27})^R \frac{\log^R P}{P^R} \sum_{i=2}^R C_{55}^{R-i} \left[\left(C_{55}^i \xi^R + \frac{\xi^{R-i}}{P^i} \right) \times \right.$$

$$\left. \sum_{n=1}^{N-2} n^{R-i} e^{-n\xi} + \frac{\xi^{R+i}}{P^i} \sum_{n=1}^{N-2} n^{R+i} e^{-n\xi} \right]. \quad (226)$$

Using Lemma 4 and Lemma 3 to take out the C_{55} -dependent terms in (226), we obtain

$$\begin{aligned} \text{LD}_2^U(\xi, N) &\leq C_{56} \frac{\log^R P}{P^R} \sum_{i=2}^R \left[\left(\xi^R + \frac{\xi^{R-i}}{P^i} \right) \times \right. \\ &\quad \left. \sum_{n=1}^{N-2} n^{R-i} e^{-n\xi} + \frac{\xi^{R+i}}{P^i} \sum_{n=1}^{N-2} n^{R+i} e^{-n\xi} \right], \quad (227) \end{aligned}$$

where $C_{56} \triangleq C_4 (4C_{27})^R \max\{1, C_{55}^R\} \max\{1, C_{55}^{R-2}\}$.

Next, we evaluate the summations with respect to n in (227). The following lemma provides a useful upper bound:

Lemma 9: Let f be a non-negative real valued Riemann integrable function with $f(x) < \infty$, $\forall x \in \mathbb{R}$ that is increasing on the interval $(-\infty, b)$, and decreasing on (b, ∞) . Then

$$\sum_{n=n_1}^{n_2} f(n) \leq \int_{n_1}^{n_2} f(x) dx + 2b. \quad (228)$$

Proof: Let $n_b = \lfloor b \rfloor$ be the largest integer less than b . Assume that $n_1 < n_b < n_2$. Then,

$$\begin{aligned} \sum_{n=n_1}^{n_b-1} f(n) &= \sum_{n=n_1}^{n_b-1} \int_n^{n+1} f(n) dx \leq \sum_{n=n_1}^{n_b-1} \int_n^{n+1} f(x) dx \\ &= \int_{n_1}^{n_b} f(x) dx, \quad (229) \end{aligned}$$

where the inequality follows from the fact that f is increasing in the range of integration. Also,

$$\begin{aligned} \sum_{n=n_b+2}^{n_2} f(n) &= \sum_{n=n_b+2}^{n_2} \int_n^{n+1} f(n) dx \leq \sum_{n=n_b+2}^{n_2} \int_n^{n+1} f(x-1) dx \\ &= \int_{n_b+1}^{n_2} f(x) dx, \quad (230) \end{aligned}$$

where the inequality follows since f is decreasing on (n_b+1, ∞) , and thus for $n_b+2 \leq n \leq x \leq n+1$, $f(n) \leq f(x-1)$. Finally, combining (229) and (230),

$$\begin{aligned} \sum_{n=n_1}^{n_2} f(n) &= \sum_{n=n_1}^{n_b-1} f(n) + f(n_b) + f(n_b+1) + \sum_{n=n_b+2}^{n_2} f(n) \\ &\leq \int_{n_1}^{n_b} f(x) dx + b + b + \int_{n_b+1}^{n_2} f(x) dx \\ &\leq \int_{n_1}^{n_2} f(x) dx + 2b, \end{aligned}$$

which is the desired inequality for $n_1 < n_b < n_2$. The other cases can be proved similarly. We skip them for brevity. ■

Note that the function $f(x) = x^\alpha \exp(-x\xi)$ has a global maximum at $x = \alpha/\xi$ with $f(x) = \alpha^\alpha \xi^{-\alpha} \exp(-\alpha)$. Moreover, for any $0 \leq a < b < \infty$, $\int_a^b f(x) dx \leq \int_0^\infty f(x) dx =$

$\Gamma(\alpha + 1)\xi^{-(\alpha+1)}$. Therefore, using Lemma 9, for any $2 \leq i \leq R$, we have

$$\begin{aligned} & \sum_{n=1}^{N-2} n^{R-i} e^{-n\xi} \\ & \leq \Gamma(R-i+1)\xi^{-(R-i+1)} + 2(R-i)^{R-i}\xi^{-(R-i)} \\ & \leq \Gamma(R-i+1)\xi^{-(R-i+1)} + 2\epsilon^{-1}(R-i)^{R-i}\xi^{-(R-i+1)} \\ & \leq \underbrace{\max\{\Gamma(R-1), 2\epsilon^{-1}(R-2)^{R-2}\}}_{\triangleq \mathcal{C}_{57}} \xi^{-(R-i+1)}. \end{aligned} \quad (231)$$

where the second inequality follows from the assumption that $\Lambda \geq \epsilon \implies \xi \leq \epsilon^{-1}$ for sufficiently large P . For the third inequality we have used the facts that $\Gamma(R-i+1) \leq \Gamma(R-1)$ and $(R-i)^{R-i} \leq (R-2)^{R-2}$.

Similarly, we have

$$\begin{aligned} & \sum_{n=1}^{N-2} n^{R+i} e^{-n\xi} \\ & \leq \Gamma(R+i+1)\xi^{-(R+i+1)} + 2(R+i)^{R+i}\xi^{-(R+i)} \\ & \leq \Gamma(R+i+1)\xi^{-(R+i+1)} + 2\epsilon^{-1}(R+i)^{R+i}\xi^{-(R+i+1)} \\ & \leq \underbrace{\max\{\Gamma(2R+1), 2\epsilon^{-1}(2R)^{2R}\}}_{\triangleq \mathcal{C}_{58}} \xi^{-(R+i+1)}. \end{aligned} \quad (232)$$

Using (231) and (232), (227) can be bounded as:

$$\text{LD}_2^{\text{U}}(\xi, N) \leq \mathcal{C}_{56} \frac{\log^R P}{P^R} \sum_{i=2}^R \left[\mathcal{C}_{57} \left(\xi^{i-1} + \frac{\xi^{-1}}{P^i} \right) + \mathcal{C}_{58} \frac{\xi^{-1}}{P^i} \right].$$

On the other hand, $\Lambda \leq \Lambda_0 P \implies P \geq \Lambda_0^{-1} \xi^{-1} \implies P^{-i} \leq \Lambda_0^i \xi^i$, $i = 2, \dots, R$, and thus

$$\begin{aligned} & \text{LD}_2^{\text{U}}(\xi, N) \\ & \leq \mathcal{C}_{56} \frac{\log^R P}{P^R} \sum_{i=2}^R [\mathcal{C}_{57}(1 + \Lambda_0^i) + \mathcal{C}_{58}\Lambda_0^i] \xi^{i-1} \\ & \leq \mathcal{C}_{56} \underbrace{\max_{j \in \{2, R\}} [\mathcal{C}_{57}(1 + \Lambda_0^j) + \mathcal{C}_{58}\Lambda_0^j]}_{\triangleq \mathcal{C}_{59}} \frac{\log^R P}{P^R} \sum_{i=2}^R \xi^{i-1} \\ & = \mathcal{C}_{59} \frac{\log^R P}{P^R} \xi \sum_{i=2}^R \xi^{i-2} \\ & = \mathcal{C}_{59} \frac{\log^R P}{\Lambda P^R} \sum_{i=0}^{R-2} \xi^i. \end{aligned}$$

Similarly, $\Lambda \geq \epsilon \implies \frac{1}{\xi} \geq \epsilon \implies \xi \leq \epsilon^{-1} \implies \xi^i \leq \epsilon^{-i}$, $i = 0, \dots, R-2$. Therefore,

$$\text{LD}_2^{\text{U}}(\xi, N) \leq \underbrace{\mathcal{C}_{59} \sum_{i=0}^{R-2} \epsilon^{-i}}_{\triangleq \mathcal{C}_{60}} \frac{\log^R P}{\Lambda P^R}.$$

For the last part, we consider the case for which $\mathcal{N}(\Omega) = N-1$. In this case, the minimum NSNR is $(N-1)\xi$, which can be lower bounded by

$$(N-1)\xi = (\lceil \Lambda \log \Lambda + R\Lambda \log P \rceil - 1) \frac{1}{\Lambda}$$

$$\begin{aligned} & \geq (\Lambda \log \Lambda + R\Lambda \log P - 1) \frac{1}{\Lambda} \\ & = \log \Lambda + R \log P - \frac{1}{\Lambda}. \end{aligned}$$

Therefore, the contribution to the localization distortion is upper bounded by

$$\begin{aligned} \text{LD}_3^{\text{U}}(\xi, N) & \leq \mathcal{C}_4 \exp[-(N-1)\xi] \\ & \leq \mathcal{C}_4 \exp\left(-\log \Lambda - R \log P + \frac{1}{\Lambda}\right) \\ & \leq \mathcal{C}_4 \exp(\epsilon^{-1}) \frac{1}{\Lambda P^R} \\ & \leq \mathcal{C}_4 \exp(\epsilon^{-1}) \frac{\log^R P}{\Lambda P^R} \end{aligned}$$

Combining the final upper bounds for $\text{LD}_i^{\text{U}}(\xi, N)$, $i = 1, 2, 3$, we have $\text{LD}^{\text{U}}(\xi, N) \leq \mathcal{C}_{61} \frac{\log^R P}{\Lambda P^R}$ for all P sufficiently large, and a constant $0 < \mathcal{C}_{61} < \infty$ that is independent of P and Λ . This proves the upper bound on the localization distortion.

Now, let $\Omega_{r\ell} = \gamma_{\ell}(\mathbf{e}_r, \mathbf{h})$ denote the local NSNR at the ℓ th receiver given that the r th relay is selected. By the definition of our compressor in Section V-B-III, the ℓ th receiver uses $\lceil \log_2(N^R - 1) \rceil$ feedback bits if there exists a relay index r for which $\Omega_{r\ell} < (N-1)\xi$. Also, the ℓ th receiver uses 0 bits if $\forall r$, $\Omega_{r\ell} \geq (N-1)\xi$. Therefore,

$$\begin{aligned} \mathbf{R}_{\ell}(\text{LQ}_{\xi_v, N_v}^v) & = \lceil \log_2(N^R - 1) \rceil \mathbb{P}(\exists r, \Omega_{r\ell} < (N-1)\xi) \\ & \leq R \log_2 N \mathbb{P}(\exists r, \Omega_{r\ell} < (N-1)\xi) \\ & \leq R \log_2 N \mathbb{P}(\exists r, \min_{\ell} \Omega_{r\ell} < (N-1)\xi) \\ & = R \log_2 N \mathbb{P}(\exists r, \Omega_r < (N-1)\xi) \\ & \leq R \log_2 N \sum_{r=1}^R F_{\Omega_r}[(N-1)\xi], \end{aligned} \quad (233)$$

where the second inequality follows since $\exists r, \Omega_{r\ell} < (N-1)\xi \implies \exists r, \min_{\ell} \Omega_{r\ell} < (N-1)\xi$. The fourth inequality is a consequence of the union bound.

Now, according to Proposition 3,

$$\begin{aligned} F_{\Omega_r}[(N-1)\xi] & \leq \mathcal{C}_{27}(N-1)\xi \psi[(N-1)\xi] \\ & \leq \mathcal{C}_{27}\mathcal{C}_{55}(N-1)\xi \frac{\log P}{P} \\ & = \mathcal{C}_{27}\mathcal{C}_{55}(\lceil \Lambda \log \Lambda + R\Lambda \log P \rceil - 1) \frac{\log P}{\Lambda P} \\ & \leq \mathcal{C}_{27}\mathcal{C}_{55}[\log \Lambda + R \log P] \frac{\log P}{P} \\ & \leq \mathcal{C}_{27}\mathcal{C}_{55}[\log \Lambda_0 + (R+1) \log P] \frac{\log P}{P} \\ & \leq \mathcal{C}_{27}\mathcal{C}_{55}(R+1 + \Lambda_0) \frac{\log^2 P}{P}, \end{aligned} \quad (234)$$

where the second inequality follows from (223). For the fourth inequality, we have used the fact that $\Lambda \leq \Lambda_0 P$ for sufficiently large P . The last inequality follows since $\log \Lambda_0 < \Lambda_0$, $\forall \Lambda_0 > 0$. Using a similar chain of upper bounds,

$$\begin{aligned} \log_2 N & = \log_2 \lceil \Lambda \log \Lambda + R\Lambda \log P \rceil \\ & \leq \log_2(\Lambda \log \Lambda + R\Lambda \log P) + 1 \\ & < \log_2(\Lambda^2 + R\Lambda \log P) + 1 \end{aligned}$$

$$\begin{aligned}
&\leq \log_2(\Lambda_0^2 P^2 + R\Lambda_0 P \log P) + 1 \\
&< \log_2(\Lambda_0^2 P^2 + R\Lambda_0 P^2) + 1 \\
&= \log_2(\Lambda_0^2 + R\Lambda_0) + 2\log_2 P + 1. \quad (235)
\end{aligned}$$

Substituting (234) and (235) to (233), and after some straightforward manipulations, we obtain the upper bound on the rate of the ℓ th receiver in the statement of the theorem. This concludes the proof. ■

APPENDIX G

PROOF OF THE ARGUMENTS REGARDING NETWORKS WITH DIRECT LINKS

In this appendix, we provide the formal proofs of some of the arguments in Section VII regarding networks with direct links.

A. Maximal Diversity Gains for Networks With Direct Links

The DF strategy is a special case of a general coding scheme for the relay-interference network and thus the maximal diversity gain with the DF scheme is $(R + 1, 0)$. With a similar argument, we can obtain the diversity upper bound $(R + 1, 0)$ for AF relays with $K = 1$. For $K > 1$ and AF relays, similar to what we have done for the proof of (24) in Appendix C, we can assume without loss of generality that the first receiver is interested in the messages of the first transmitter and a genie reveals all the transmitted symbols but s_1 to the first receiver. Since the NER is lower bounded by the SER at the first receiver, it is sufficient to find a lower bound on the symbol error rate of a system with input-output relationship given by

$$\begin{bmatrix} \bar{y}_{01} \\ \bar{y}_{11} \end{bmatrix} = \begin{bmatrix} \beta_1 d_{11} + \sum_{r=1}^R \alpha_1 d_{11} x_r \sqrt{\bar{\rho}'_r} \alpha_1 f_{1r} g_{r1} \\ \sum_{r=1}^R x_r g_{r1} \sqrt{\bar{\rho}'_r} \eta_{0r} + \eta_{2\ell} \end{bmatrix} s_1 \sqrt{P_{S_1}} + \begin{bmatrix} \eta_{1\ell} \\ \eta_{2\ell} \end{bmatrix}. \quad (236)$$

In the light of the discussion above, in order to obtain (236), we have concatenated the received signals of the first receiver during the two transmission steps. We have then removed the terms that have one of $s_2, \dots, s_K$ as a factor. We have also allowed an arbitrary relay normalization factor $\bar{\rho}'_r \leq \bar{\rho}_r$.

The optimal decoder that minimizes the error probability for the system model in (236) is a maximum ratio combiner, which yields an SNR given by

$$\begin{aligned}
\gamma_{\text{DL}}^{\text{U}} &\triangleq |\alpha_1 d_{11}|^2 P_{S_1} + \frac{|\beta_1 d_{11} + \sum_{r=1}^R x_r \sqrt{\bar{\rho}'_r} \alpha_1 f_{1r} g_{r1}|^2 P_{S_1}}{1 + \sum_{r=1}^R |x_r|^2 |g_{r1}|^2 \bar{\rho}'_r} \\
&\leq |d_{11}|^2 P_{S_1} + \frac{2 \left[|\beta_1 d_{11}|^2 + \left| \sum_{r=1}^R x_r \sqrt{\bar{\rho}'_r} \alpha_1 f_{1r} g_{r1} \right|^2 \right] P_{S_1}}{1 + \sum_{r=1}^R |x_r|^2 |g_{r1}|^2 \bar{\rho}'_r} \\
&\leq 3|d_{11}|^2 P_{S_1} + \frac{2 \left| \sum_{r=1}^R x_r \sqrt{\bar{\rho}'_r} f_{1r} g_{r1} \right|^2 |\alpha_1|^2 P_{S_1}}{1 + \sum_{r=1}^R |x_r|^2 |g_{r1}|^2 \bar{\rho}'_r}. \quad (237)
\end{aligned}$$

The second inequality follows from Hölder's inequality and the fact that $|\alpha_1|^2 + |\beta_1|^2 \leq 1 \implies |\alpha_1|^2 \leq 1$. The third inequality follows since $|\beta_1|^2 \leq 1$.

In (237), the first term provides a diversity gain of $(1, 0)$. The second term has the same form as $\gamma^{\text{U}}(\mathbf{x}, \mathbf{h})$ in (161) apart from

some constant multipliers. Using exactly the same methods in Appendix C, we can show that it provides a diversity gain of at most $(R, -R)$. The two terms represent independent random variables, and therefore $\gamma_{\text{DL}}^{\text{U}}$ yields a diversity gain of at most $(R + 1, -R)$. We skip the details of the derivation since they are straightforward.

B. Achievable NER Using Direct Links Only

At the ℓ th receiver, instead of an individual ML decoder, suppose that we employ a joint ML decoder that attempts to decode all the transmitted symbols. The error probability with the joint ML decoder provides an upper bound on the error probability with the individual ML decoder.

Let $\text{SER}_{\ell}^{\text{DL}}(\mathbf{d}_{\ell})$ denote the conditional error probability with the joint ML decoder. Also, let $\text{CNER}^{\text{DL}}(\mathbf{d})$ denote the conditional NER, where $\mathbf{d} \triangleq [\mathbf{d}_1 \dots \mathbf{d}_K]$. Since the NER is the probability that at least one receiver incorrectly decodes its desired symbols, by a union bound over all the receivers, we have

$$\begin{aligned}
\text{CNER}^{\text{DL}}(\mathbf{d}) &\leq \sum_{\ell=1}^L \text{SER}^{\text{DL}}(\mathbf{d}_{\ell}) \\
&\leq L \max_{\ell} \text{SER}^{\text{DL}}(\mathbf{d}_{\ell}). \quad (238)
\end{aligned}$$

Now, using a union bound over all the transmitted symbols, we have

$$\begin{aligned}
\text{SER}^{\text{DL}}(\mathbf{d}_{\ell}) &\leq \frac{1}{|\mathcal{S}|} \sum_{s \neq \hat{s}} Q \left(\sqrt{\frac{1}{2} \left| \sum_{k=1}^K (s_k - \hat{s}_k) \sqrt{P_{S_k}} d_{k\ell} \right|^2} \right) \\
&\leq (|\mathcal{S}| - 1) Q \left(\sqrt{2\gamma_{\ell, \text{DL}}(\mathbf{d}_{\ell})} \right) \\
&\leq \frac{|\mathcal{S}| - 1}{2} \exp(-\gamma_{\ell, \text{DL}}(\mathbf{d}_{\ell})), \quad (239)
\end{aligned}$$

where $\gamma_{\ell, \text{DL}}(\mathbf{d}_{\ell})$ is as defined in (57). Substituting (239) to (238), we obtain (56) with $C_{21} = \frac{L(|\mathcal{S}|-1)}{2}$.

We now find an upper bound on $\text{NER}^{\text{DL}} \triangleq \mathbb{E}_{\mathbf{d}}[\text{CNER}^{\text{DL}}(\mathbf{d})]$. Let $Z \triangleq \min_{\ell} \min_{s \neq \hat{s}} Y_{\ell, s, \hat{s}}$ with $Y_{\ell, s, \hat{s}} \triangleq \frac{1}{4} \left| \sum_{k=1}^K (s_k - \hat{s}_k) \sqrt{P_{S_k}} d_{k\ell} \right|^2$. Note that

$$\begin{aligned}
\text{NER}^{\text{DL}} &\leq C_{21} \mathbb{E}[\exp(-ZP)] \\
&= C_{21} \int_0^{\infty} \exp(-zP) f_Z(z) dz. \quad (240)
\end{aligned}$$

It is straightforward to show that $Y_{\ell, s, \hat{s}}$ follows an exponential distribution with mean $\sum_k |s_k - \hat{s}_k|^2 P_{S_k} \sigma_{d_{k\ell}}^2$. Using Lemma 2, we can then show that $f_Z(z) \leq C_{62} \exp(-zC_{63})$ for finite and positive constants C_{62} and C_{63} that are independent of P . Substituting this upper bound on the PDF of Z to (240) and performing the integration, we can finally obtain $\text{NER}^{\text{DL}} \leq C_{22} P^{-1}$ for a constant $0 < C_{22} < \infty$.

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